
A Greedy PDE Router for Blending Neural Operators and Classical Methods

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1 **Revision note.** All text, tables and figures added in this revision are typeset in red (new figures carry
2 a red frame); everything in black is unchanged from the submitted version.

Abstract

3 When solving PDEs, classical numerical solvers are often computationally ex-
4 pensive, while machine learning methods can suffer from spectral bias, failing to
5 capture high-frequency components. Designing an optimal hybrid iterative solver-
6 where, at each iteration, a solver is selected from an ensemble of solvers to leverage
7 their complementary strengths—poses a challenging combinatorial problem. While
8 greedy selection is desirable for its constant-factor approximation guarantee to the
9 optimal solution under Lipschitz assumptions, it requires knowledge of the true
10 error at each step, which is unavailable in practice. We address this by proposing
11 an approximate greedy router that efficiently mimics a greedy approach to solver
12 selection. Empirical results on the Poisson and convection–diffusion equations
13 show that our method consistently reduces final error and area-under-the-curve
14 (AUC) of the error trajectory relative to single-solver baselines and existing hybrid
15 approaches such as HINTS. In particular, our method reaches comparable error lev-
16 els in substantially fewer iterations while exhibiting more stable error decay. With a
17 cost-aware instantiation of the greedy rule, a lightweight router, and a band-limited
18 scale-equivariant corrector, these gains carry over to end-to-end wall-clock time on
19 a 128×128 grid: for every one of the 15 solver–equation pairings (five classical
20 solvers; Poisson, convection–diffusion and anisotropic diffusion) the learned router
21 reaches truncation-level accuracy $238\times$ – $1983\times$ faster than the classical solver
22 alone (which never reaches it on the anisotropic equation) and $2.68\times$ – $11.8\times$ faster
23 than HINTS, and it matches or beats HINTS even when the latter’s period is tuned
24 per setting.

25 1 Introduction

26 Natural phenomena and engineered systems are often governed by ordinary and partial differential
27 equations (PDEs). Solving these equations enables a variety of tasks - predicting the evolution of the
28 system through simulation (e.g., forecasting weather using the Navier-Stokes equations) [Kalnay,
29 2003, Bauer et al., 2015, Staniforth and Côté, 1991], addressing control problems (e.g., optimizing
30 heat shield design for spacecrafts using heat transfer equations) [Anderson, 1989, Tröltzsch, 2010],
31 and tackling inverse problems based on external measurements (e.g., reconstructing brain activity
32 from EEG data using electrophysiological models) [Baillet et al., 2001, Grech et al., 2008].

33 Traditionally, PDEs are solved using finite difference methods [Smith, 1985, LeVeque, 2007], which
34 discretize the spatial and/or temporal domain and approximate the solution by solving a system
35 of equations at the discrete grid points. Similarly, finite element methods [Hughes, 2003, Bathe,
36 2006, Brenner, 2008] construct a system of equations based on local basis functions, and then

rely on numerical linear algebra techniques—such as the Jacobi and Gauss-Seidel method [Saad, 2003]—to compute the solution. These iterative methods, however, lack generalization across different initial conditions, boundary conditions, or forcing functions, as even minor changes to any of these parameters require re-solving the entire system of equations from scratch.

These challenges have motivated the development of neural operators [Kovachki et al., 2023]—a class of machine learning models that aim to learn the solution operator directly, enabling fast inference across a range of parameters. Neural operators lift the classical linear layer in neural networks to an operator—typically a kernel integral operator—between function spaces. Despite strong approximation guarantees [Chen and Chen, 1995], neural operators are prone to spectral bias [Liu and Cai, 2021, Liu et al., 2024, You et al., 2024, Khodakarami et al., 2025, Xu et al., 2025], similar to traditional neural networks [Rahaman et al., 2019, Xu et al., 2019, Luo et al., 2019]. As a result, they favor learning low-frequency components of the solution function and often struggle to capture high-frequency features that iterative solvers excel at.

Recognizing this complementarity, Zhang et al. [2024] proposed HINTS, a hybrid solver that interleaves a neural operator pass every τ^{th} iteration of a classical solver, with τ fixed a priori. While HINTS reports significant empirical gains in convergence and accuracy over the standalone neural operator and Jacobi solver, this fixed schedule can be detrimental: a poorly timed neural correction may increase the error and undo recent progress.

Although state-dependent solver schedules are desirable, this naturally suggests a reinforcement learning formulation for learning routing policies. However, in our setting, the structure of iterative PDE solvers allows for a simpler approach: each solver induces a known update, and its effect can be evaluated through error-based signals. Leveraging this, we adopt a greedy rule that selects, at each iteration, the solver with the largest immediate error reduction. Such a rule can be near-optimal when the final error satisfies (weak) supermodularity—so that applying a beneficial solver earlier is at least as valuable as applying it later. To support this approach, we make the following contributions:

- We present a general hybrid PDE solver in which a routing rule chooses, at each iteration, from a set of classical solvers and neural operators. With oracle access to the true error at every step, we show that a greedy routing rule achieves a constant-factor approximation to the optimal strategy for linear PDEs, provided the updates are error-reducing (Lipschitz with constant < 1) and zero-preserving—conditions under which weak supermodularity holds.
- Given that the true error is not observed at test time, a convex surrogate loss is introduced, which, when minimized, enables the learned model to imitate the greedy router without access to true error information. This alignment is guaranteed through Bayes consistency.
- We empirically demonstrate that our approximate greedy solver achieves faster convergence than HINTS and individual solvers, often reaching comparable error levels in fewer iterations on the Poisson and convection-diffusion equations.

2 Background

Consider the following linear PDE:

$$\mathcal{L}_x^a(u) = f, x \in D; \quad \mathcal{B}_x(u) = b, x \in \partial D \quad (1)$$

where $\mathcal{L}_x^a$ is a differential operator with respect to the spatial variable $x \in D$ parameterized by the coefficient function a , f is a forcing function, u is the PDE solution, $\mathcal{B}_x$ is the boundary operator, and b is the boundary term. Such PDEs can be solved numerically using various discretization strategies, such as finite differences [Smith, 1985, LeVeque, 2007], finite element, and spectral methods [Boyd, 2001, Canuto et al., 2006]. Here, we focus on finite differences, which replace derivatives with difference quotients (e.x., $\partial_x u(x) \approx (u(x+h) - u(x))/h$) on a uniform grid.

Formally, let h be the grid size and $G_h(D)$ be a uniform grid with spacing h over D . For a function $g : D \rightarrow \mathbb{R}$, we denote its restriction to $G_h(D)$ by $g_h \in \mathbb{R}^N$, where $N = |G_h(D)|$. If $\mathcal{L}_h^a \in \mathbb{R}^{N \times N}$ is the discretized operator, the PDE reduces to a linear system of equations

$$\mathcal{L}_h^a u_h = f_h, \quad (2)$$

with the boundary conditions incorporated in $\mathcal{L}_h^a$. Unlike much of the neural operator literature, which retains function space formulations for discretization invariance, we follow the conventions of classical

numerical analysis and represent functions and operators as vectors and matrices. Discretization invariance is not essential here, since we focus on fixed-grid problems.

2.1 Iterative PDE Solvers

Direct methods like Gauss Elimination and Thomas Algorithm can be computationally expensive in high-dimensional domains when solving systems like Equation (2). In contrast, iterative methods like Jacobi and Gauss-Seidel offer computational speedups by iteratively updating the solution, gradually converging to the true solution for the PDE. The general iterative update is

$$u^{(t+1)} = u^{(t)} + C \left(f_h - \mathcal{L}_h^a u^{(t)} \right), \quad (3)$$

where $u^{(t)}$ is the t^{th} iterate of the solution and C is a preconditioning matrix. Jacobi uses $C = D^{-1}$, where D is the diagonal of $\mathcal{L}_h^a$, and Gauss-Seidel uses $C = (D + L)^{-1}$, where L denotes the strict lower triangular part. However, these methods tend to damp low frequency parts of the error slowly.

2.2 Neural Operators

Suppose we observe samples $\{(a_i, f_i, u_i)\}_{i=1}^N$ such that $(a_i, f_i) \sim \mathcal{P}$ are i.i.d. samples, and $u_i = \mathcal{G}^*(a_i, f_i)$ be generated by a deterministic solution operator $\mathcal{G}^*$. A neural operator (NO) $\mathcal{G}_\theta$ seeks to approximate $\mathcal{G}^*$ by minimizing the expected squared $L^2(D)$ error:

$$\mathcal{R}_{\text{NO}}(\mathcal{G}_\theta) = \mathbb{E}_{(a,f) \sim \mathcal{P}} \left[\int_D \|u_i(x) - \mathcal{G}_\theta(a_i, f_i)(x)\|_2^2 dx \right]$$

Neural operators use discretization-invariant layers. Prominent instantiations include Fourier Neural Operators (FNO), which use spectral transforms [Li et al., 2020a], Graph Neural Operators, which perform graph-based aggregation over sampled points [Li et al., 2020b], and DeepONets, which employ a trunk-branch decomposition to map input functions to output functions [Lu et al., 2021].

2.3 Greedy Optimization

Consider maximizing $g(S)$ over $S \subseteq \Omega$ with $|S| \leq T$ where $g : 2^\Omega \rightarrow \mathbb{R}$ is a set function defined on subsets of a set Ω . An exhaustive search is infeasible, so an efficient alternative is the greedy algorithm. It builds the solution subset iteratively by adding ω^* that yields the largest marginal gain, i.e. $\omega^* = \arg \max_{\omega \in \Omega \setminus S} g(S \cup \omega)$. When g is non-negative, monotone (adding elements never decreases the value), and submodular (diminishing returns), the greedy algorithm achieves a $(1 - e^{-1})$ approximation to the optimal solution [Nemhauser et al., 1978]. Formally, a function g is submodular if $g(A \cup \{\omega\}) - g(A) \geq g(B \cup \{\omega\}) - g(B)$ for all $A \subseteq B \subseteq \Omega$ and $\omega \in \Omega \setminus B$; intuitively, delaying an addition cannot increase its benefit. For set function minimization problems, supermodularity, where the inequality flips, plays an analogous role, and the greedy rule achieves constant-factor approximation guarantees [Liberty and Sviridenko, 2017].

The suboptimality of the greedy algorithm has been extensively studied for both set-function minimization [Bounia and Koriche, 2023] and maximization [Das and Kempe, 2018, Feige et al., 2011, Bian et al., 2017, Harshaw et al., 2019] when standard assumptions—non-negativity, monotonicity, and sub/supermodularity—are weakened. In contrast, results on greedy sequence maximization [Streeter and Golovin, 2008, Alaei et al., 2021, Zhang et al., 2015, Bernardini et al., 2020, Van Over et al., 2024, Tschitschek et al., 2017]—where the ordering of the elements affects the function—have led to sequential analogues of submodularity and monotonicity. We leverage these tools to analyze the suboptimality of the greedy solution to minimizing the final error.

3 General framework for Hybrid Solvers

To solve Equation (2), consider the following hybrid iterative update:

$$u_h^{(t+1)} = u_h^{(t)} + C_{S_t} \left(f_h - \mathcal{L}_h^a u_h^{(t)} \right) \quad (4)$$

where $\mathcal{C} = \{C_j\}_{j=1}^K$ is a set of preconditioning functions and $S_t \in [K] = \{1, \dots, K\}$ indexes the function chosen at step t . Here, we use “preconditioning function” broadly to include classical linear

updates ($C_j(x) = C_j x$ where C_j is a matrix), and learned models, such as neural operators. This update generalizes the classical iterative update in Equation (3), by allowing C_j to be non-linear and enabling the preconditioning function to be adaptively chosen at every step. $\mathcal{C}$ can also accommodate parameterized solver families (e.g., different Jacobi relaxation weights), enabling adaptive selection of solver parameters. As the number of solvers K grows, the chance of selecting a more effective update increases. Furthermore, HINTS [Zhang et al., 2024] is a special case of this hybrid solver with $K = 2$, where C_1 is a neural operator and C_2 is a classical preconditioner. Its routing rule is given by $S_t = \mathbf{1}_{t \bmod \tau > 0} + 1$, which selects C_1 every τ steps and C_2 otherwise.

Let $e_h^{(t)} = u_h - u_h^{(t)}$ denote the error at step t . Then,

$$e_h^{(t+1)} = u_h - u_h^{(t)} - C_{S_t} \left(\mathcal{L}_h^a u_h - \mathcal{L}_h^a u_h^{(t)} \right) = (I - C_{S_t} \circ \mathcal{L}_h^a) (e_h^{(t)})$$

where I is the identity map and $I - C_j \circ \mathcal{L}_h^a$ is the error propagation function for solver j . Here, “ $\circ$ ” denotes function composition (matrix multiplication for linear updates). The objective is to select a sequence $S = (S_1, \dots, S_T)$ that minimizes the error norm after T steps or $\|e_h^{(T)}\|_2^2$:

$$\min_{S_t \in [K], |S| \leq T} h(S) := \left\| (I - C_{S_{|S|}} \circ \mathcal{L}_h^a) \circ \dots \circ (I - C_{S_1} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \quad (5)$$

with compositions applied from right to left, so that the S_1 update acts on the initial error.

4 Greedy Algorithm

Equation (5) defines a combinatorial optimization problem with a search space exponential in T , making exact optimization intractable for large T or K . As a starting point, we consider an “omniscient” greedy algorithm that assumes access to the true initial error $e_h^{(0)}$. This is unrealistic since knowledge of $e_h^{(0)}$ could directly recover the solution via $u_h = u_h^{(0)} + e_h^{(0)}$, but it provides a useful benchmark. In Section 5, we relax this assumption using a practical learning strategy that is Bayes consistent with this omniscient rule, thereby recovering the guarantees shown below.

As discussed in Section 2.3, when supermodularity and monotonicity hold, the greedy rule—such as the one described in Algorithm 1—enjoys constant-factor approximation guarantees. However, classical results focus on *set* functions, with only recent extensions made to sequences. Building on this line, we introduce a sequence-based notion of weak supermodularity.

Algorithm 1 Greedy Algorithm for a Hybrid PDE solver

Require: $\{C_j\}_{j=1}^K, T, \mathcal{L}_h^a, e_h^{(0)}$
 $S^0 \leftarrow \emptyset$
for $t < T$ **do**
 $S^{t+1} \leftarrow S^t \oplus \arg \min_{j \in [K]} \|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$
 $e_h^{(t+1)} \leftarrow (I - C_{S^{t+1}} \circ \mathcal{L}_h^a)(e_h^{(t)})$
end for
return S^T

Let Ω^* denote the space of sequences with elements in Ω . For $S, S' \in \Omega^*$, we denote their concatenation as $S \oplus S'$. S is a prefix of S' or $S \preceq S'$ if $S' = S \oplus L$ for some $L \in \Omega^*$. A function $g : \Omega^* \rightarrow \mathbb{R}$ is prefix monotonically non-increasing if $g(S \oplus S') \leq g(S)$ for all $S, S' \in \Omega^*$, and postfix monotonically non-increasing if $g(S' \oplus S) \leq g(S)$ for all $S, S' \in \Omega^*$. A prefix non-increasing function g is sequence supermodular if, for all $S', S \in \Omega^* : S \preceq S'$, it holds that

$$g(S) - g(S \oplus \omega) \geq g(S') - g(S' \oplus \omega), \quad \forall \omega \in \Omega \quad (6)$$

However, $h(S)$ (defined in Equation (5)) may not satisfy this property in general. Therefore, we introduce weak sequence supermodularity. A prefix non-increasing function g is weakly supermodular with respect to $S' \in \Omega^*$ if, for any $S \in \Omega^{|S'|}$, there exists $\alpha(S') \geq 1$ such that

$$g(S) - g(S \oplus S') \leq \alpha(S') \sum_{i \in [|S'|]} g(S) - g(S \oplus S'_i) \quad (7)$$

The parameter $\alpha(S')$ or the supermodularity ratio quantifies deviation from exact sequence supermodularity. Expanding the $g(S) - g(S \oplus S')$ as a telescoping sum $\sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) - g(S \oplus (S'_1, \dots, S'_i))$ shows that the marginal decrease from appending S'_i after its predecessors is controlled by the effect of appending S'_i directly to S . Thus, postponing the inclusion of S'_i cannot yield a significantly larger benefit than adding it earlier. The supermodularity ratio $\alpha(S)$ quantifies the extent to which future gains from delays may exceed immediate gains.

Having introduced these notions, we now characterize the suboptimality of greedy solutions of weakly supermodular and postfix monotonic sequence functions in Theorem 4.1.

Theorem 4.1. *Let $g : \Omega^* \rightarrow \mathbb{R}$ be a weakly supermodular function with respect to the optimal solution $O = \arg \min_{S \in \Omega^T} h(S)$ with a supermodularity ratio of $\alpha(O)$ and postfix monotonicity. Let the greedy solution of length T be S^T . If $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$, then*

$$g(\emptyset) - g(S^T) \geq (1 - \phi_T(\alpha(O))) (g(\emptyset) - g(O))$$

The proof of Theorem 4.1 appears in Section A.2. As $T \rightarrow \infty$, the factor $1 - \phi_T(\alpha(O))$ decreases to $1 - e^{-1/\alpha(O)}$, so the worst-case performance of the greedy rule saturates rather than degrades with horizon. Additionally, larger $\alpha(O)$, which indicates higher reward for delayed inclusions, loosens the suboptimality bound. Theorem 4.1 requires that the sequence objective h be weakly supermodular with respect to the optimal solution and postfix monotone—properties established in Proposition 4.2. We defer the proof to Section A.3.

Proposition 4.2. *Suppose that for all $j \in [K]$, the error propagation function $I - C_j \circ \mathcal{L}_h^a$ is ρ_j -Lipschitz continuous with $\rho_j < 1$, and that $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$. Then, the function h is weakly supermodular with respect to the optimal solution O , with*

$$\alpha(O) = \max \left\{ \frac{4}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

Furthermore, if $I - C_j \circ \mathcal{L}_h^a$ is invertible for all $j \in [K]$, h is also postfix non-increasing.

The conditions of Proposition 4.2 are natural. For classical solvers, the Lipschitz constant is $\|I_N - C_j \mathcal{L}_h^a\|$, which is typically less than 1 for well-posed linear elliptic PDEs (i.e., the update is damping errors). For neural networks, Lipschitz continuity can be enforced via weight regularization [Gouk et al., 2021] and a sufficiently trained model should approximate $(\mathcal{L}_h^a)^{-1}$ well enough to make the Lipschitz constant small. The requirement $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$ is both natural and desirable: it precludes spurious updates when the residual $f_h - \mathcal{L}_h^a u_h^{(t)}$ is 0. This holds by design for classical schemes, and it can be enforced for any learned model by excluding bias terms.

The form of $\alpha(O)$ in Proposition 4.2 highlights that the suboptimality factor in Theorem 4.1 is governed by the collective contraction factors of the solvers chosen by the optimal solution: as $\sum_{i=1}^T \rho_{O_i}^2 \rightarrow T$, $\alpha(O)$ grows, resulting in a weaker bound. Invertibility of the error propagation functions is often satisfied with Jacobi and Gauss-Seidel updates. However, neural networks with dimension changes or ReLU activations may yield non-invertible error propagation maps. Nevertheless, our experiments show that greedy routers remain effective even when invertibility is not met.

Theorem 4.1 thus indicates that the approximation guarantee is strongest when the sequence is nearly supermodular ($\alpha(O) \approx 1$). Generally, if all error propagation maps share an eigenbasis, supermodularity is established. This occurs, for example, for linear, constant-coefficient PDEs with periodic boundary conditions where the solver ensemble includes Jacobi, Gauss-Seidel, and a single-layer linear Fourier Neural Operator. The proof of Theorem 4.3 is deferred to Section A.5.

Proposition 4.3. *Let $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$ for all $j \in [K]$ and $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$. Then, h is supermodular.*

Cost-aware greedy routing. Algorithm 1 treats every update as equally expensive, whereas in practice a neural-operator evaluation and a classical sweep have very different costs, and the quantity of interest is the error reached per unit of wall-clock time. We make the greedy rule cost-aware without changing the analysis: let c_j denote the (measured) cost of one application of C_j , take the cost u of one neural-operator call as the unit, and replace each preconditioner by the *macro-action*

205 $C_j^{(m_j)}$ that applies C_j for $m_j = \max\{1, \text{round}(u/c_j)\}$ consecutive iterations, so that all actions
 206 cheaper than the unit cost approximately u ; an action dearer than the unit (a multigrid cycle, say)
 207 is applied once and its error is compared per unit of cost, $\|e\|^{u/(m_j c_j)}$, which reduces to the plain
 208 comparison when the costs match. For a linear update, m consecutive sweeps are again a stationary
 209 iteration, $(I - C_j \circ \mathcal{L}_h^a)^m = I - C_j^{(m)} \mathcal{L}_h^a$ with $C_j^{(m)} = \sum_{i=0}^{m-1} (I - C_j \mathcal{L}_h^a)^i C_j$, and for a nonlinear
 210 C_j the macro-action is the m -fold composition of its error propagation map. Hence $\{C_j^{(m_j)}\}_{j=1}^K$ is
 211 again a set of preconditioning functions in the sense of Section 4: if $I - C_j \circ \mathcal{L}_h^a$ is ρ_j -Lipschitz,
 212 zero-preserving and invertible, then so is its m_j -fold composition (with constant $\rho_j^{m_j} < 1$), and
 213 Theorem 4.1, Propositions 4.2 and 4.3, and the consistency result of Section 5 apply verbatim to the
 214 macro-action ensemble. Running Algorithm 1 on macro-actions selects, at every decision epoch, the
 215 action with the largest error reduction per unit cost, and a horizon of T macro-actions corresponds to
 216 a wall-clock budget of $\approx Tu$; Section E uses this cost-aware instantiation throughout.

217 5 Approximate Greedy Router

218 The results in Section 4 indicate that the error reduction from the greedy solution closely matches that
 219 of the optimal sequence, but this is predicated on having access to the initial error, $e_h^{(0)}$. Inaccurate
 220 error estimates can result in poor solver decisions, causing errors to amplify. To remedy this, we learn
 221 a router r that selects solvers myopically, as in Algorithm 1, without access to true errors.

222 We adopt the following learning setup. Let $\mathcal{A}, \mathcal{F}, \mathcal{U} \subseteq \mathbb{R}^N$ denote the spaces of coefficient, forcing,
 223 and solution functions on the grid $G_h(D)$. We assume an application-specific data distribution $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$
 224 over $\mathcal{A} \times \mathcal{F}$ that reflects test time conditions. During training, (a_h, f_h) is drawn from $\mathcal{P}_{\mathcal{A} \times \mathcal{F}}$, and a
 225 high-accuracy reference solution u_h is computed, providing the true per-step error. The router r is
 226 learned offline on this data; at test time, it operates without access to true errors.

227 At iteration t , router r selects a solver using the coefficient $a_h \in \mathcal{A}$, forcing $f_h \in \mathcal{F}$, and the current
 228 iterate $u_h^{(t)} \in \mathcal{U}$. If $r(a_h, f_h, u_h^{(t)}) = j$, the next iterate is computed with solver j , resulting in an
 229 error of $\|(I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)})\|_2^2$. Learning such a router requires minimizing the following loss:

$$l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h) = \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)})=j} \quad (8)$$

230 with risk $\mathcal{R}_{\text{route}}(r) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} [l_{\text{route}}(r, a_h, f_h, u_h^{(t)}, u_h)]$. For analysis, we fix the iterate genera-
 231 tion via teacher-forcing [Williams and Zipser, 1989, Lamb et al., 2016]: during training, the iterates
 232 fed to the router are produced by an oracle greedy rollout, not by the router’s own past choices.
 233 Formally, with $u_h^{(0)} = 0_N$, for $t \geq 1$,

$$j_t^* \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) (e_h^{(t-1)}) \right\|_2^2, \quad u_h^{(t)} = u_h^{(t-1)} + C_{j_t^*} (f_h - \mathcal{L}_h^a u_h^{(t-1)})$$

234 Thus, $\{u_h^{(t)}\}$ is a deterministic function of (a_h, f_h) . At each t , l_{route} is evaluated on $r(a_h, f_h, u_h^{(t)})$
 235 which takes in the teacher-forced iterate. Finally, the greedy choice j_{t+1}^* is used to advance $u_h^{(t+1)}$.
 236 Thus the input distribution seen by r depends on (a_h, f_h) , not on the router’s predictions. However,
 237 full teacher forcing induces a distributional mismatch at test time (exposure bias). In experiments, we
 238 mitigate this with scheduled sampling [Bengio et al., 2015]; see Section C for details.

239 Minimizing $\mathcal{R}_{\text{route}}(r)$ yields the following Bayes-Optimal Router:

$$r^*(a_h, f_h, u_h^{(t)}) \in \operatorname{argmin}_{j \in [K]} \left\| (I - C_j \circ \mathcal{L}_h^a) (e_h^{(t)}) \right\|_2^2 \quad (9)$$

240 which aligns with the update in Algorithm 1. Hence, l_{route} is consistent with learning the greedy rule.

241 5.1 Surrogate Loss

242 Since Equation (8) is discontinuous and non-convex, direct minimization is intractable in practice.
 243 We instead introduce a surrogate loss that (1) is convex, enabling efficient optimization; (2) upper
 244 bounds the original loss; and (3) is Bayes consistent, ensuring the Bayes optimal decision of the

original loss is preserved upon minimization. Formally, a surrogate ϕ is considered to be Bayes consistent with respect to the loss l if

$$\lim_{n \rightarrow \infty} \mathcal{R}_\phi(f_n) - \mathcal{R}_\phi^* \implies \lim_{n \rightarrow \infty} \mathcal{R}_l(f_n) - \mathcal{R}_l^*$$

where $\mathcal{R}_l(f) = \mathbb{E}[l(f(X), Y)]$ and $\mathcal{R}_l^* = \inf_f \mathcal{R}_l(f)$. In other words, in the limit of infinite data, if the risk of a sequence of learned hypotheses $\{f_n\}$ converges to the optimal risk under ϕ , it also converges to the optimal risk with respect to the original loss l .

To define a surrogate loss for the routing problem, consider a set of scoring functions $\mathbf{g} = \{g_j\}_{j=1}^K$ with $g_j : \mathcal{A} \times \mathcal{F} \times \mathcal{U} \rightarrow \mathbb{R}$ and we define the router as $r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$. For example, $\mathbf{g}$ can be a neural network with K outputs. Then, minimizing the following surrogate loss yields the same decision as minimizing Equation (8):

$$\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h) = - \sum_{j=1}^K \sum_{k=1}^K \tilde{c}_k(a_h, u_h^{(t)}, u_h) \mathbf{1}_{k \neq j} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{m=1}^K \exp(g_m(a_h, f_h, u_h^{(t)}))} \right) \quad (10)$$

where $\tilde{c}_j(a_h, u_h^{(t)}, u_h) = \|(I - C_j \circ \mathcal{L}_h^a)(u_h - u_h^{(t)})\|_2^2$. The Ψ -risk is denoted by $\mathcal{R}_\Psi(\mathbf{g}) = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}}[\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)]$. The convexity of Ψ with respect to $\mathbf{g}$ follows from the convexity of log-softmax function in its inputs. Moreover, Ψ upper bounds l_{route} up to a constant factor, and we refer the reader to Section B.2 for the proof. Finally, Theorem 5.1 shows that Ψ achieves Bayes consistency with respect to l_{route} ; the proof can be found in Section B.3.

Theorem 5.1. *Let $\tilde{c}_j(a_h, u_h^{(t)}, u_h) < \bar{E} < \infty$ for all $j \in [K]$. If there exists $j \in [K]$ such that $\tilde{c}_j(a_h, u_h^{(t)}, u_h) > E_{\min} > 0$, then, for any collection of solvers $\{C_j\}_{j=1}^K$ and linear discrete operator $\mathcal{L}_h^a$, Ψ is Bayes consistent surrogate for l_{route} .*

Theorem 5.1 is a cost-sensitive analogue of the classical Bayes-consistency of multiclass cross-entropy for 0-1 loss: in the infinite-sample limit, minimizing cross-entropy recovers the true conditional class probabilities, so the induced decision is Bayes optimal. Our result extends this to cross-entropy with instance-dependent weights $\sum_{k \neq j} \tilde{c}_k(a_h, u_h^{(t)}, u_h)$. The uniform upper bound holds when all preconditioning functions are error-damping. It is also reasonable to assume at least one solver cannot annihilate the error in one step, yielding the lower bound. Under these conditions, minimizing Ψ recovers the Bayes-optimal router of Equation (9), i.e., the greedy solution of Equation (5). Following the work of Mao et al. [2024], the proof uses standard conditional-risk calibration: we relate the excess risks of l_{route} and Ψ and take the infinite-sample limit.

6 Related Works

Hybrid PDE Solvers: Early data-driven solvers sought convergence guarantees by predicting parameters—e.g., preconditioning matrices, multi-grid smoothers or restriction matrices—within iterative schemes [Taghibakhshi et al., 2021, Caldana et al., 2024, Kopaničáková and Karniadakis, 2025, Huang et al., 2022, Katrutsa et al., 2020]. These works, however, do not leverage the neural surrogates’ ability to generalize across varying coefficients and forcings highlighted in Section 2.2 and thus, offer only modest speedups over classical solvers. HINTS [Zhang et al., 2024] introduced hybrid solvers that interleave a classical method with a pre-trained DeepONet on a fixed schedule (e.g., 24 Jacobi steps, then one DeepONet correction), achieving faster convergence than the standalone numerical solver. Subsequent works extend this idea to new geometries [Kahana et al., 2023] and analyze error mode damping, replacing DeepONet with alternatives such as MIONet [Jin et al., 2022] [Hu and Jin, 2025] or FNO [Li et al., 2020a, Cui et al., 2022]. However, these hybrid solvers are limited in two ways: they, firstly, only combine the trained surrogate with a *single* numerical solver, and secondly rely on a fixed, heuristic schedule. Our proposed method addresses these shortcomings, resulting in significant empirical improvements (Section 7).

Model Routing: Routing [Shnitzer et al., 2023, Hu et al., 2024, Ding et al., 2024, Huang et al., 2025] selects, for each input, a model from a fixed set to optimize a task metric under cost or latency constraints. Simple heuristics—e.g., thresholding a cheap model’s uncertainty estimates [Chuang et al.,

289 2024, 2025]—often poorly balance the cost-accuracy tradeoff, motivating many systems to instead
 290 learn a router that maps inputs to the best model [Hari and Thomson, 2023, Mohammadshahi et al.,
 291 2024, Šakota et al., 2024]. We similarly learn a router, but over numerical and neural solvers for
 292 PDEs at the iteration level. Our work is the first to apply routing to the problem of learning hybrid
 293 PDE solvers, in contrast to prior approaches that use fixed schedules. Additionally, by exploiting
 294 the algebraic structure of PDE solvers, we derive theoretical guarantees for our routing strategy
 295 (Section 4), unlike typical model-routing settings.

296 **Mixture of Experts:** Classical mixture-of-experts (MoE) [Jacobs et al., 1991, Jordan and Jacobs,
 297 1994] uses a gating network to combine the outputs of multiple experts, trained jointly with the gate.
 298 In modern LLMs, MoE instead performs sparse, token-level routing to a subset of in-layer experts,
 299 enabling large capacity without proportional compute, typically with load-balancing mechanisms
 300 [Shazeer et al., 2017, Lepikhin et al., 2020, Fedus et al., 2022, Jiang et al., 2024]. Our method can be
 301 regarded as gating of a different kind: “experts” (solvers) are not jointly trained with the router, and a
 302 single router is reused across iterations, unlike MoE which commonly employs layer-specific gates.

303 7 Experiments

304 We empirically demonstrate the fast, uniform convergence of the approximate greedy router on the
 305 Poisson and Convection-Diffusion (ConvDiff) equations posed on the unit domain $D = [0, 1]^2$:

$$\forall \mathbf{x} \in D \quad \textbf{Poisson: } -\Delta u(\mathbf{x}) = f(\mathbf{x}) \quad \textbf{ConvDiff: } -\Delta u(\mathbf{x}) + 20 \frac{\partial}{\partial x_1} u(\mathbf{x}) + 20 \frac{\partial}{\partial x_2} u(\mathbf{x}) = f(\mathbf{x})$$

306 We impose periodic boundary conditions for both equations, and center f to satisfy the compatibility
 307 condition $\int_D f(\mathbf{x}) d\mathbf{x} = 0$; Dirichlet results are deferred to Section D.8. The domain D is discretized
 308 on a 31×31 uniform grid; results on a finer grid are provided in Section D.7. Data is sampled from a
 309 zero-mean Hierarchical Gaussian Random Field on the periodic domain with covariance operator
 310 $\alpha(-\Delta + \beta I)^{-\gamma}$, where α, β, γ vary across samples (See Section C). We run two experimental settings.
 311 In the first, we restrict to solver pairs ($K = 2$) and compare against HINTS and standalone classical
 312 solvers. In the second, we assess performance as the solver ensemble grows; since HINTS does not
 313 support this setting, we simply compare against individual solvers. In both cases, we consider a range
 314 of iterative methods described below. Performance is reported over 128 test samples via the mean
 315 final error $\|e_h^{(T)}\|_2$ and the mean area under the curve (AUC), defined as $\sum_{t=1}^T \|e_h^{(t)}\|_2$; the “curve”
 316 refers to the error over iterations. AUC characterizes the full convergence behavior, which is critical
 317 in practice where solutions must be reached in minimal time.

318 **Paired Solver Experiments:** We seek to optimally solve the aforementioned PDEs with access
 319 to a single numerical scheme and a trained DeepONet. In particular, we consider settings coupling
 320 the DeepONet with Jacobi, Gauss-Seidel (GS), and Symmetric Gauss-Seidel (SymGS) solvers. For
 321 each setup, we compare our learned router against the single-solver schedules (Jacobi only, GS only,
 322 and SymGS only) and HINTS variants (HINTS-Jacobi, HINTS-GS, HINTS-SymGS), where the
 323 DeepONet correction are applied every 24 iterations. We train LSTM-based routers using the loss
 324 in Equation (10) separately for each pairwise settings. The LSTM captures the sequential nature of
 325 routing decisions where the benefit of each action depends on the error trajectory and past choices.
 326 Training details of the DeepONet and the routers can be found in Section C. To ensure fairness,
 327 comparisons are restricted to methods with identical solver access, and are grouped by solver family
 328 (e.g., Jacobi-related methods). In addition, we include a *true greedy* oracle, which selects at each
 329 iteration the solver that yields the largest immediate error reduction. This oracle has access to the
 330 ground-truth error and is therefore not implementable in practice, but helps evaluate how well the
 331 learned router approximates the optimal policy. Each method is run for 300 iterations.

332 As shown in Table 1, the greedy router achieves the lowest final error and AUC across all solver
 333 families, outperforming their single-solver and HINTS counterparts. Moreover, its performance
 334 closely tracks that of the true greedy oracle, indicating that our surrogate-based training procedure
 335 effectively approximates the greedy strategy. This can be further seen in the routing visualizations in
 336 Section D.6, which demonstrate that the router learns qualitatively similar strategies to the oracle.
 337 The improvements over HINTS are particularly notable: while HINTS can improve over single-
 338 solver schedules, it exhibits oscillatory error behavior (See Figure 1) due to invoking the DeepONet

Table 1: Final error and AUC of L_2^2 error (lower is better). Values are mean ($\pm$ standard error) over 128 test instances; Error is reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Bold indicates the best method within each solver family. True-Greedy denotes an oracle policy with access to exact error information

Equation	Poisson		ConvDiff	
Methods	$\ e_h^{(T)}\ \times 10^3$	AUC	$\ e_h^{(T)}\ \times 10^3$	AUC
Jacobi-related Solvers				
Jacobi Only	0.383 (1.029)	0.821 (2.154)	0.136 (0.364)	0.312 (0.788)
HINTS-Jacobi	0.759 (0.016)	0.393 (0.590)	0.447 (0.013)	0.202 (0.256)
Learned Greedy-Jacobi	0.054 (0.142)	0.165 (0.368)	0.033 (0.097)	0.098 (0.239)
True-Greedy-Jacobi	0.021 (0.018)	0.094 (0.170)	0.012 (0.012)	0.049 (0.091)
GS-related Solvers				
GS only	0.019 (0.051)	0.435 (1.141)	$< 10^{-3}$	0.088 (0.219)
HINTS-GS	0.724 (0.000)	0.325 (0.488)	0.456 (0.000)	0.130 (0.154)
Learned Greedy-GS	0.002 (0.004)	0.083 (0.152)	$< 10^{-3}$	0.031 (0.078)
True-Greedy-GS	0.001 (0.001)	0.057 (0.107)	$< 10^{-3}$	0.019 (0.038)
SymGS-related Solvers				
SymGS only	$< 10^{-3}$	0.227 (0.593)	$< 10^{-3}$	0.071 (0.178)
HINTS-SymGS	0.709 (0.000)	0.252 (0.380)	0.463 (0.000)	0.116 (0.131)
Learned Greedy-SymGS	$< 10^{-3}$	0.052 (0.102)	$< 10^{-3}$	0.022 (0.039)
True-Greedy-SymGS	$< 10^{-3}$	0.034 (0.066)	$< 10^{-3}$	0.016 (0.032)

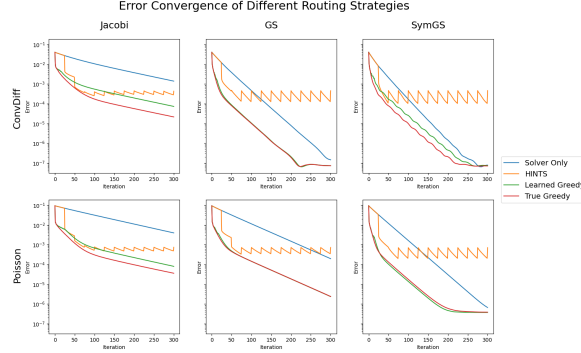


Figure 1: Convergence histories for representative test instances. Rows: Convection Diffusion (top) and Poisson (bottom). Columns: Jacobi, Gauss–Seidel (GS), and Symmetric Gauss–Seidel (SymGS). Greedy yields near-monotone decay and the lowest errors, whereas HINTS shows sawtooth behaviors.

at suboptimal times. In contrast, the greedy routers selectively applies updates that reduce error, resulting in near-monotone convergence. These gains arise from adapting routing decisions across both samples for a given PDE and PDEs. For instance, the neural operator is used more frequently in the convection–diffusion setting than in Poisson (Figure 2), a behavior that must be manually specified in HINTS but is learned automatically by our method. This shows that improvements stem from selective use of the learned component, rather than frequent invocation as in HINTS.

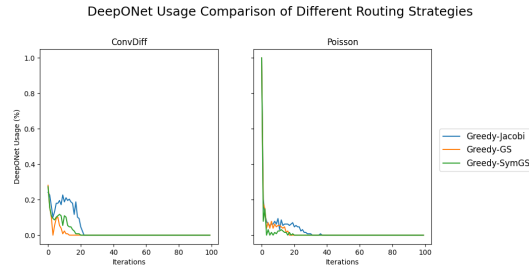


Figure 2: DeepONet usage across iterations for different greedy routing strategies on convection–diffusion (left) and Poisson (right). The variability across iterations highlights the non-uniform, sample-dependent nature of the learned routing strategy. See Section D.6 for detailed routing trajectories. For readability, we display the first 100 iterations (out of 300 total).

Solver Ensembles Experiments: We consider routing over a collection of more than two solvers. Specifically, we study ensembles $\mathcal{W}$ of increasing size, consisting of a trained DeepONet and a diverse set of numerical solvers, including Jacobi, GS, SymGS, Damped Jacobi ($\omega = 0.67$) and Successive Over-relaxation (SOR) with $\omega = 1.5$. Our goal is to evaluate whether enlarging the solver set yields gains beyond pairwise configurations. To this end, we compare Router($\text{NO} \cup \mathcal{W}$) against the best pairwise version of our method, Router($\text{NO} \cup \{\text{solver}\}$) for solver $\in \mathcal{W}$. As shown in Table 3, larger ensembles often yield improvements over both final error and AUC relative to the best pairwise configuration, indicating that additional solver diversity can be leveraged. The router achieves these gains by learning to route to each member of the ensemble for a non-trivial proportion of the iterates, as measured in Section D.6. Finally, despite the increased size of the action space, the learned router continues to produce improvements, demonstrating robustness of both the training procedure and the convex surrogate of Section 5.1; true greedy comparisons are deferred to Section D.1. These results show that our method can scale to richer solver sets, enabling improved performance without sacrificing learnability.

Wall-clock time-to-solution: Iteration counts favor methods whose iterations are expensive, so we complement the results above with an end-to-end wall-clock study on a 128×128 grid (128^2 unknowns) for all five classical solvers and both PDEs, plus a third, harder PDE—anisotropic diffusion $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$ with $\epsilon = 10^{-2}$ on the same periodic domain and data, the standard case in which point smoothers and multigrid with point smoothing degrade; Section E details the protocol. Three ingredients make the comparison meaningful and the router deployable: (i) all classical solvers are implemented as $O(N^2)$ stencil or cached sparse-triangular sweeps (matching the dense implementations used elsewhere to machine precision), so the classical baseline is not artificially slowed; (ii) the DeepONet corrector acts on the residual through a 64×64 sensor grid with a Fourier trunk basis and is applied scale-equivariantly, so one call solves the smooth half of the spectrum to $\approx 10^{-6}$ at the cost of about 12 Jacobi or 3 Gauss–Seidel sweeps and never injects high-frequency error; and (iii) the router is made *cost-aware* through the cost-equalised macro-actions of Section 4, and the LSTM is replaced by a router over $6+K$ scalar features (log residual, its recent decay rate, the previous action, and call counts) with a two-layer MLP whose decision costs $\approx 5 \mu\text{s}$, trained with the surrogate of Section 5.1 to imitate the cost-aware greedy oracle. Table 2 reports the median wall-clock time to reach truncation-level relative error $\varepsilon = h^2$, the accuracy beyond which further algebraic refinement is not meaningful on an $N \times N$ grid. The learned router reaches it $238\times$ – $1983\times$ faster than the classical solver alone on the two isotropic equations (on the anisotropic equation no point smoother reaches it at all within 60,000 iterations) and $2.68\times$ – $11.8\times$ faster than HINTS with the published period $\tau=25$, in every one of the 15 solver–equation pairings (Table 2); it is also never slower than HINTS with the best period $\tau \in \{5, 10, 25, 50\}$ chosen *a posteriori* on the test set for that pairing ($1.21\times$ – $3.22\times$; Table 17), whereas the best τ itself varies across pairings and tolerances. The router tracks the cost-aware oracle within $1.06\times$ in median time, and the iteration-based metrics of Table 1 carry over to the finer grid (Table 18). Figure 3 shows where the gain comes from: the router calls the corrector once at the very start (where the error is smooth and the corrector removes four orders of magnitude at once), spends a solver-dependent number of sweeps on the high-frequency band, and calls it again only if the smooth error re-emerges—between one and three calls in total—while HINTS spends $\tau-1$ sweeps before its first call and keeps calling the corrector when it no longer helps. At tolerances far below h^2 the undamped-Jacobi pairing is limited by the near-Nyquist modes that neither component damps (Table 21), but the router degrades gracefully there because it stops selecting the corrector, whereas fixed schedules keep paying for it (Section E). Against the classical methods a practitioner would use on these problems (Table 26), the router with its best stationary pairing is $3.62\times$ – $3.71\times$ faster than geometric multigrid alone, $3.85\times$ – $5.31\times$ faster than the best multigrid-preconditioned Krylov method, and only the FFT direct solve—a method specific to constant-coefficient periodic problems, which we use to compute ground truth—is faster, by $2.87\times$ – $2.91\times$; routing over $\{\text{NO}, \text{multigrid}\}$ is itself $1.93\times$ – $2.03\times$ faster than multigrid alone. The gains persist on 256^2 grids for every pairing and on 512^2 grids for the GS, SOR and multigrid pairings (Table 27); the one exception is undamped Jacobi at 512^2 , where the corrector’s fixed band leaves the smoother a stalled high-frequency band, every schedule needs the same ≈ 80 sweeps, and the router only matches (Poisson) or, on convection–diffusion, falls short of a fixed schedule (Section E.4). All 128^2 and 256^2 comparisons are significant at $p < 0.01$ (paired Wilcoxon over instances) and stable across five independent router-training seeds (Section E.6). The same construction extends to solver ensembles (Table 24): over the four ensembles of Table 3 and both PDEs, the router over $\text{NO} \cup \mathcal{W}$ matches the best pairwise router of its members to within

Table 2: Median wall-clock time to truncation-level relative error $\varepsilon = h^2$ on the 128×128 grid, 64 test instances per cell (paired median speedup over the solver-only baseline in parentheses). “HINTS (best τ)” is the best of $\tau \in \{5, 10, 25, 50\}$ chosen per cell on the test set itself. The greedy oracle is the cost-agnostic Algorithm 1, the cost-aware oracle is Algorithm 1 on cost-equalised macro-actions; both use the true error and are not deployable. Bold marks the fastest deployable method in each row; the learned router pays for all of its features and decisions.

Equation	Solver	Solver only	HINTS ($\tau=25$)	HINTS (best τ)	Greedy oracle	Cost-aware oracle	Learned router (ours)
Poisson	Jacobi	1.15 s	2.65 ms (412 $\times$)	1.10 ms (984 $\times$; $\tau=5$)	854 μ s (1283 $\times$)	942 μ s (1199 $\times$)	987 μs (1154$\times$)
	Jacobi (0.67)	1.62 s	2.71 ms (621 $\times$)	1.13 ms (1464 $\times$; $\tau=5$)	915 μ s (1904 $\times$)	922 μ s (1836 $\times$)	912 μs (1863$\times$)
	GS	1.83 s	6.65 ms (277 $\times$)	1.99 ms (906 $\times$; $\tau=5$)	960 μ s (1872 $\times$)	953 μ s (1892 $\times$)	1.01 ms (1809$\times$)
	SymGS	1.52 s	10.3 ms (149 $\times$)	2.60 ms (585 $\times$; $\tau=5$)	1.09 ms (1395 $\times$)	1.13 ms (1330$\times$)	1.13 ms (1331$\times$)
	SOR (1.5)	600.9 ms	12.2 ms (49.4 $\times$)	3.37 ms (180 $\times$; $\tau=5$)	1.03 ms (583 $\times$)	1.02 ms (582 $\times$)	1.02 ms (576$\times$)
ConvDiff	Jacobi	1.51 s	3.44 ms (436 $\times$)	1.28 ms (1145 $\times$; $\tau=5$)	992 μ s (1532 $\times$)	1.15 ms (1319 $\times$)	1.14 ms (1317$\times$)
	Jacobi (0.67)	2.22 s	3.40 ms (656 $\times$)	1.31 ms (1670 $\times$; $\tau=5$)	1.10 ms (2036 $\times$)	1.08 ms (1974 $\times$)	1.10 ms (1983$\times$)
	GS	1.63 s	8.27 ms (198 $\times$)	3.60 ms (453 $\times$; $\tau=5$)	1.23 ms (1341 $\times$)	1.32 ms (1222 $\times$)	1.35 ms (1243$\times$)
	SymGS	1.56 s	11.1 ms (141 $\times$)	2.96 ms (529 $\times$; $\tau=5$)	1.25 ms (1254 $\times$)	1.27 ms (1236 $\times$)	1.28 ms (1211$\times$)
	SOR (1.5)	383.6 ms	13.6 ms (28.0 $\times$)	3.64 ms (104 $\times$; $\tau=5$)	1.39 ms (273 $\times$)	1.63 ms (240 $\times$)	1.59 ms (238$\times$)
AnisoDiff	Jacobi	>4.62 s ¹⁶⁴	2.90 ms (> $10^3 \times$)	1.15 ms (> $10^3 \times$; $\tau=5$)	826 μ s (> $10^3 \times$)	892 μ s (> $10^3 \times$)	911 μs (> $10^3 \times$)
	Jacobi (0.67)	>4.52 s ¹⁶⁴	2.85 ms (> $10^3 \times$)	1.18 ms (> $10^3 \times$; $\tau=5$)	810 μ s (> $10^3 \times$)	823 μ s (> $10^3 \times$)	850 μs (> $10^3 \times$)
	GS	>13.63 s ¹⁶⁴	6.48 ms (> $10^3 \times$)	1.74 ms (> $10^3 \times$; $\tau=5$)	831 μ s (> $10^3 \times$)	833 μ s (> $10^3 \times$)	852 μs (> $10^3 \times$)
	SymGS	>23.15 s ¹⁶⁴	10.3 ms (> $10^3 \times$)	2.40 ms (> $10^3 \times$; $\tau=5$)	832 μ s (> $10^3 \times$)	844 μ s (> $10^3 \times$)	871 μs (> $10^3 \times$)
	SOR (1.5)	>13.80 s ¹⁶⁴	6.62 ms (> $10^3 \times$)	1.80 ms (> $10^3 \times$; $\tau=5$)	843 μ s (> $10^3 \times$)	862 μ s (> $10^3 \times$)	869 μs (> $10^3 \times$)

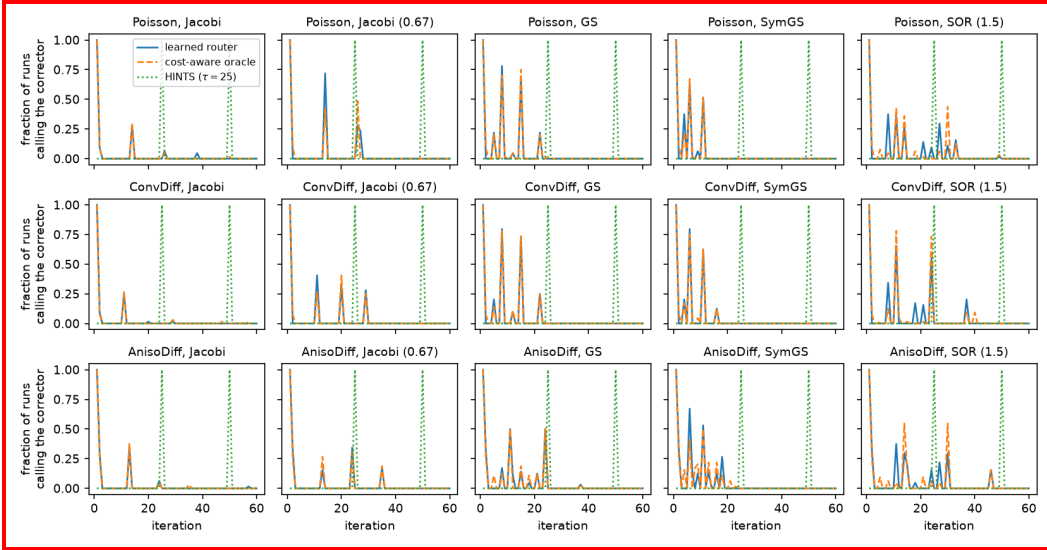


Figure 3: Corrector usage on the 128×128 grid: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ($\tau=25$), for every solver pairing (columns) and both equations (rows).

403 $\pm 15\%$ (ratio of medians $0.82 \times - 1.14 \times$) without knowing in advance which member is fastest, and is
404 $380 \times \rightarrow 10^3 \times$ faster than the best member solver alone; the remaining gap to the ensemble oracle,
405 which is faster than every pairwise router, is imitation error in choosing among near-equivalent
406 smoothers (Section E).

407 Additional experiments are provided in Section D.

408 8 Discussion

409 We have introduced an adaptive method for selecting solvers that efficiently minimizes the final error
410 of a PDE in iterative methods, opening many directions for future work. Our current DeepONet is
411 trained without accounting for its downstream role in correction term prediction; jointly training the
412 ML solver and router could yield larger gains. Another avenue is to use reinforcement learning to
413 learn a cost-aware routing strategy that optimizes error under compute budgets; Sections 4 and E show

Table 3: Final error and AUC of squared L^2 error for ensembles of increasing size. $\mathcal{W}$ denotes the set of numerical solvers and NO the neural operator (DeepONet). Each Router($\text{NO} \cup \mathcal{W}$) is compared against the best pairwise routing using our method, i.e., Router($\text{NO} \cup \{\text{solver}\}$) for solver $\in \mathcal{W}$. Values are mean ($\pm$ s.e.) over 128 test instances; Error is reported in $\times 10^{-3}$. p -values from paired one-sided t -tests comparing the AUC of the ensemble to that of the best pairwise router.

$\mathcal{W}$	Method	Poisson			ConvDiff		
		$\ e_h^{(T)}\ \times 10^3$	AUC	p -value (vs pairwise)	$\ e_h^{(T)}\ \times 10^3$	AUC	p -value (vs pairwise)
{Jacobi, GS}	Best Router($\text{NO} \cup \{\text{solver}\}$)	0.002 (0.004)	0.083 (0.152)	-	$< 10^{-3}$	0.031 (0.078)	-
	Router($\text{NO} \cup \mathcal{W}$)	0.002 (0.002)	0.078 (0.132)	$< 10^{-3}$	$< 10^{-3}$	0.021 (0.040)	0.004
{Jacobi, GS, SymGS}	Best Router($\text{NO} \cup \{\text{solver}\}$)	$< 10^{-3}$	0.052 (0.102)	-	$< 10^{-3}$	0.022 (0.039)	-
	Router($\text{NO} \cup \mathcal{W}$)	$< 10^{-3}$	0.041 (0.091)	$< 10^{-3}$	$< 10^{-3}$	0.018 (0.035)	$< 10^{-3}$
{Jacobi, GS, SymGS, Jacobi (0.67)}	Best Router($\text{NO} \cup \{\text{solver}\}$)	$< 10^{-3}$	0.052 (0.102)	-	$< 10^{-3}$	0.022 (0.039)	-
	Router($\text{NO} \cup \mathcal{W}$)	0.002 (0.003)	0.046 (0.083)	0.004	$< 10^{-3}$	0.019 (0.037)	0.002
{Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	Best Router($\text{NO} \cup \{\text{solver}\}$)	$< 10^{-3}$	0.052 (0.143)	-	$< 10^{-3}$	0.011 (0.030)	-
	Router($\text{NO} \cup \mathcal{W}$)	$< 10^{-3}$	0.043 (0.082)	0.03	$< 10^{-3}$	0.008 (0.019)	0.004

that a myopic cost-aware variant of the greedy rule—Algorithm 1 on cost-equalised macro-actions—already delivers uniform wall-clock gains, and RL could extend it to non-myopic budgeted routing. RL would additionally enable extending the ensemble routing to *continuous* parameterization of solvers. Finally, framing routing as an online learning problem could enable adaptation to unknown deployment conditions.

References

- Eugenia Kalnay. *Atmospheric modeling, data assimilation and predictability*. Cambridge university press, 2003.
- Peter Bauer, Alan Thorpe, and Gilbert Brunet. The quiet revolution of numerical weather prediction. *Nature*, 525(7567):47–55, 2015.
- Andrew Staniforth and Jean Côté. Semi-lagrangian integration schemes for atmospheric models—a review. *Monthly weather review*, 119(9):2206–2223, 1991.
- John David Anderson. *Hypersonic and high temperature gas dynamics*. Aiaa, 1989.
- Fredi Tröltzsch. *Optimal control of partial differential equations: theory, methods, and applications*, volume 112. American Mathematical Soc., 2010.
- Sylvain Baillet, John C Mosher, and Richard M Leahy. Electromagnetic brain mapping. *IEEE Signal processing magazine*, 18(6):14–30, 2001.
- Roberta Grech, Tracey Cassar, Joseph Muscat, Kenneth P Camilleri, Simon G Fabri, Michalis Zervakis, Petros Xanthopoulos, Vangelis Sakkalis, and Bart Vanrumste. Review on solving the inverse problem in eeg source analysis. *Journal of neuroengineering and rehabilitation*, 5:1–33, 2008.
- Gordon D Smith. *Numerical solution of partial differential equations: finite difference methods*. Oxford university press, 1985.
- Randall J LeVeque. *Finite difference methods for ordinary and partial differential equations: steady-state and time-dependent problems*. SIAM, 2007.
- Thomas JR Hughes. *The finite element method: linear static and dynamic finite element analysis*. Courier Corporation, 2003.
- Klaus-Jürgen Bathe. *Finite element procedures*. Klaus-Jurgen Bathe, 2006.
- Susanne C Brenner. *The mathematical theory of finite element methods*. Springer, 2008.
- Yousef Saad. *Iterative methods for sparse linear systems*. SIAM, 2003.
- Nikola Kovachki, Zongyi Li, Burigede Liu, Kamyar Azizzadenesheli, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Neural operator: Learning maps between function spaces with applications to pdes. *Journal of Machine Learning Research*, 24(89):1–97, 2023.

447 Tianping Chen and Hong Chen. Universal approximation to nonlinear operators by neural networks
448 with arbitrary activation functions and its application to dynamical systems. *IEEE transactions on*
449 *neural networks*, 6(4):911–917, 1995.

450 Lizuo Liu and Wei Cai. Multiscale deepnet for nonlinear operators in oscillatory function spaces for
451 building seismic wave responses. *arXiv preprint arXiv:2111.04860*, 2021.

452 Xinliang Liu, Bo Xu, Shuhao Cao, and Lei Zhang. Mitigating spectral bias for the multiscale operator
453 learning. *Journal of Computational Physics*, 506:112944, 2024.

454 Zhilin You, Zhenli Xu, and Wei Cai. Mscalefno: Multi-scale fourier neural operator learning for
455 oscillatory function spaces. *arXiv preprint arXiv:2412.20183*, 2024.

456 Siavash Khodakarami, Vivek Oommen, Aniruddha Bora, and George Em Karniadakis. Mitigating
457 spectral bias in neural operators via high-frequency scaling for physical systems. *arXiv preprint*
458 *arXiv:2503.13695*, 2025.

459 Zhi-Qin John Xu, Lulu Zhang, and Wei Cai. On understanding and overcoming spectral biases of
460 deep neural network learning methods for solving pdes. *arXiv preprint arXiv:2501.09987*, 2025.

461 Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred Hamprecht, Yoshua
462 Bengio, and Aaron Courville. On the spectral bias of neural networks. In *International conference*
463 *on machine learning*, pages 5301–5310. PMLR, 2019.

464 Zhi-Qin John Xu, Yaoyu Zhang, Tao Luo, Yanyang Xiao, and Zheng Ma. Frequency principle:
465 Fourier analysis sheds light on deep neural networks. *arXiv preprint arXiv:1901.06523*, 2019.

466 Tao Luo, Zheng Ma, Zhi-Qin John Xu, and Yaoyu Zhang. Theory of the frequency principle for
467 general deep neural networks. *arXiv preprint arXiv:1906.09235*, 2019.

468 Enrui Zhang, Adar Kahana, Alena Kopaničáková, Eli Turkel, Rishikesh Ranade, Jay Pathak, and
469 George Em Karniadakis. Blending neural operators and relaxation methods in pde numerical
470 solvers. *Nature Machine Intelligence*, pages 1–11, 2024.

471 John P Boyd. *Chebyshev and Fourier spectral methods*. Courier Corporation, 2001.

472 Claudio Canuto, M. Yousuff Hussaini, Alfio Quarteroni, and Thomas A. Zang. *Spectral Methods:*
473 *Fundamentals in Single Domains*. Springer, 2006. ISBN 9783540307264.

474 Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew
475 Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations.
476 *arXiv preprint arXiv:2010.08895*, 2020a.

477 Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Andrew Stuart, Kaushik
478 Bhattacharya, and Anima Anandkumar. Multipole graph neural operator for parametric partial
479 differential equations. *Advances in Neural Information Processing Systems*, 33:6755–6766, 2020b.

480 Lu Lu, Pengzhan Jin, Guofei Pang, Zhongqiang Zhang, and George Em Karniadakis. Learning
481 nonlinear operators via deepnet based on the universal approximation theorem of operators.
482 *Nature machine intelligence*, 3(3):218–229, 2021.

483 George L Nemhauser, Laurence A Wolsey, and Marshall L Fisher. An analysis of approximations for
484 maximizing submodular set functions—i. *Mathematical programming*, 14(1):265–294, 1978.

485 Edo Liberty and Maxim Sviridenko. Greedy minimization of weakly supermodular set functions.
486 In *Approximation, Randomization, and Combinatorial Optimization. Algorithms and Techniques*
487 *(APPROX/RANDOM 2017)*, pages 19–1. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2017.

488 Louenas Bounia and Frederic Koriche. Approximating probabilistic explanations via supermodular
489 minimization. In *Uncertainty in Artificial Intelligence*, pages 216–225. PMLR, 2023.

490 Abhimanyu Das and David Kempe. Approximate submodularity and its applications: Subset selection,
491 sparse approximation and dictionary selection. *Journal of Machine Learning Research*, 19(3):
492 1–34, 2018.

493 Uriel Feige, Vahab S Mirrokni, and Jan Vondrák. Maximizing non-monotone submodular functions.
494 *SIAM Journal on Computing*, 40(4):1133–1153, 2011.

495 Andrew An Bian, Joachim M Buhmann, Andreas Krause, and Sebastian Tschiatschek. Guarantees for
496 greedy maximization of non-submodular functions with applications. In *International conference*
497 *on machine learning*, pages 498–507. PMLR, 2017.

498 Chris Harshaw, Moran Feldman, Justin Ward, and Amin Karbasi. Submodular maximization beyond
499 non-negativity: Guarantees, fast algorithms, and applications. In *International Conference on*
500 *Machine Learning*, pages 2634–2643. PMLR, 2019.

501 Matthew Streeter and Daniel Golovin. An online algorithm for maximizing submodular functions.
502 *Advances in Neural Information Processing Systems*, 21, 2008.

503 Saeed Alaei, Ali Makhdoumi, and Azarakhsh Malekian. Maximizing sequence-submodular functions
504 and its application to online advertising. *Management Science*, 67(10):6030–6054, 2021.

505 Zhenliang Zhang, Edwin KP Chong, Ali Pezeshki, and William Moran. String submodular functions
506 with curvature constraints. *IEEE Transactions on Automatic Control*, 61(3):601–616, 2015.

507 Sara Bernardini, Fabio Fagnani, and Chiara Piacentini. Through the lens of sequence submodularity.
508 In *Proceedings of the International Conference on Automated Planning and Scheduling*, volume 30,
509 pages 38–47, 2020.

510 Brandon Van Over, Bowen Li, Edwin KP Chong, and Ali Pezeshki. A performance bound for
511 the greedy algorithm in a generalized class of string optimization problems. *arXiv preprint*
512 *arXiv:2409.05020*, 2024.

513 Sebastian Tschiatschek, Adish Singla, and Andreas Krause. Selecting sequences of items via
514 submodular maximization. In *Proceedings of the AAAI Conference on Artificial Intelligence*,
515 volume 31, 2017.

516 Henry Gouk, Eibe Frank, Bernhard Pfahringer, and Michael J Cree. Regularisation of neural networks
517 by enforcing lipschitz continuity. *Machine Learning*, 110(2):393–416, 2021.

518 Ronald J Williams and David Zipser. A learning algorithm for continually running fully recurrent
519 neural networks. *Neural computation*, 1(2):270–280, 1989.

520 Alex M Lamb, Anirudh Goyal ALIAS PARTH GOYAL, Ying Zhang, Saizheng Zhang, Aaron C
521 Courville, and Yoshua Bengio. Professor forcing: A new algorithm for training recurrent networks.
522 *Advances in neural information processing systems*, 29, 2016.

523 Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. Scheduled sampling for sequence
524 prediction with recurrent neural networks. *Advances in neural information processing systems*, 28,
525 2015.

526 Anqi Mao, Mehryar Mohri, and Yutao Zhong. Regression with multi-expert deferral. In *International*
527 *Conference on Machine Learning*, pages 34738–34759. PMLR, 2024.

528 Ali Taghibakhshi, Scott MacLachlan, Luke Olson, and Matthew West. Optimization-based algebraic
529 multigrid coarsening using reinforcement learning. *Advances in neural information processing*
530 *systems*, 34:12129–12140, 2021.

531 Matteo Caldana, Paola F Antonietti, et al. A deep learning algorithm to accelerate algebraic multigrid
532 methods in finite element solvers of 3d elliptic pdes. *Computers & Mathematics with Applications*,
533 167:217–231, 2024.

534 Alena Kopaničáková and George Em Karniadakis. Deepnet based preconditioning strategies for
535 solving parametric linear systems of equations. *SIAM Journal on Scientific Computing*, 47(1):
536 C151–C181, 2025.

537 Ru Huang, Ruipeng Li, and Yuanzhe Xi. Learning optimal multigrid smoothers via neural networks.
538 *SIAM Journal on Scientific Computing*, 45(3):S199–S225, 2022.

539 Alexandr Katrutsa, Talgat Daulbaev, and Ivan Oseledets. Black-box learning of multigrid parameters.
540 *Journal of Computational and Applied Mathematics*, 368:112524, 2020.

541 Adar Kahana, Enrui Zhang, Somdatta Goswami, George Karniadakis, Rishikesh Ranade, and Jay
542 Pathak. On the geometry transferability of the hybrid iterative numerical solver for differential
543 equations. *Computational Mechanics*, 72(3):471–484, 2023.

544 Pengzhan Jin, Shuai Meng, and Lu Lu. Mionet: Learning multiple-input operators via tensor product.
545 *SIAM Journal on Scientific Computing*, 44(6):A3490–A3514, 2022.

546 Jun Hu and Pengzhan Jin. A hybrid iterative method based on mionet for pdes: Theory and numerical
547 examples. *Mathematics of Computation*, 2025.

548 Chen Cui, Kai Jiang, Yun Liu, and Shi Shu. Fourier neural solver for large sparse linear algebraic
549 systems. *Mathematics*, 10(21):4014, 2022.

550 Tal Shnitzer, Anthony Ou, Mírian Silva, Kate Soule, Yuekai Sun, Justin Solomon, Neil Thompson,
551 and Mikhail Yurochkin. Large language model routing with benchmark datasets. *arXiv preprint*
552 *arXiv:2309.15789*, 2023.

553 Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt
554 Keutzer, and Shriyash Kaustubh Upadhyay. Routerbench: A benchmark for multi-llm routing
555 system. *arXiv preprint arXiv:2403.12031*, 2024.

556 Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Ruhle, Laks VS
557 Lakshmanan, and Ahmed Hassan Awadallah. Hybrid llm: Cost-efficient and quality-aware query
558 routing. *arXiv preprint arXiv:2404.14618*, 2024.

559 Zhongzhan Huang, Guoming Ling, Yupei Lin, Yandong Chen, Shanshan Zhong, Hefeng Wu, and
560 Liang Lin. Routereval: A comprehensive benchmark for routing llms to explore model-level
561 scaling up in llms. *arXiv preprint arXiv:2503.10657*, 2025.

562 Yu-Neng Chuang, Prathusha Kameswara Sarma, Parikshit Gopalan, John Boccio, Sara Bolouki,
563 Xia Hu, and Helen Zhou. Learning to route llms with confidence tokens. *arXiv preprint*
564 *arXiv:2410.13284*, 2024.

565 Yu-Neng Chuang, Leisheng Yu, Guanchu Wang, Lizhe Zhang, Zirui Liu, Xuanning Cai, Yang Sui,
566 Vladimir Braverman, and Xia Hu. Confident or seek stronger: Exploring uncertainty-based
567 on-device llm routing from benchmarking to generalization. *arXiv preprint arXiv:2502.04428*,
568 2025.

569 Surya Narayanan Hari and Matt Thomson. Tryage: Real-time, intelligent routing of user prompts to
570 large language models. *arXiv preprint arXiv:2308.11601*, 2023.

571 Alireza Mohammadshahi, Arshad Rafiq Shaikh, and Majid Yazdani. Routoo: Learning to route to
572 large language models effectively. *arXiv preprint arXiv:2401.13979*, 2024.

573 Marija Šakota, Maxime Peyrard, and Robert West. Fly-swat or cannon? cost-effective language
574 model choice via meta-modeling. In *Proceedings of the 17th ACM International Conference on*
575 *Web Search and Data Mining*, pages 606–615, 2024.

576 Robert A Jacobs, Michael I Jordan, Steven J Nowlan, and Geoffrey E Hinton. Adaptive mixtures of
577 local experts. *Neural computation*, 3(1):79–87, 1991.

578 Michael I Jordan and Robert A Jacobs. Hierarchical mixtures of experts and the em algorithm. *Neural*
579 *computation*, 6(2):181–214, 1994.

580 Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarz, Andy Davis, Quoc Le, Geoffrey Hinton, and
581 Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv*
582 *preprint arXiv:1701.06538*, 2017.

583 Dmitry Lepikhin, Hyounjoong Lee, Yuanzhong Xu, Dehao Chen, Orhan Firat, Yanping Huang,
584 Maxim Krikun, Noam Shazeer, and Zhifeng Chen. Gshard: Scaling giant models with conditional
585 computation and automatic sharding. *arXiv preprint arXiv:2006.16668*, 2020.

586 William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter
587 models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39,
588 2022.

589 Albert Q Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris
590 Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, et al.
591 Mixtral of experts. *arXiv preprint arXiv:2401.04088*, 2024.

592 Anqi Mao, Mehryar Mohri, and Yutao Zhong. Cross-entropy loss functions: Theoretical analysis and
593 applications. In *International conference on Machine learning*, pages 23803–23828. pmlr, 2023.

594 Michael C Mozer. A focused backpropagation algorithm for temporal pattern recognition. In
595 *Backpropagation*, pages 137–169. Psychology Press, 2013.

596 Anthony J Robinson and Frank Fallside. *The utility driven dynamic error propagation network*,
597 volume 11. University of Cambridge Department of Engineering Cambridge, 1987.

598 Paul J Werbos. Generalization of backpropagation with application to a recurrent gas market model.
599 *Neural networks*, 1(4):339–356, 1988.

600 Lu Lu, Xuhui Meng, Shengze Cai, Zhiping Mao, Somdatta Goswami, Zhongqiang Zhang, and
601 George Em Karniadakis. A comprehensive and fair comparison of two neural operators (with prac-
602 tical extensions) based on FAIR data. *Computer Methods in Applied Mechanics and Engineering*,
603 393:114778, 2022.

604 Stéphane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured
605 prediction to no-regret online learning. In *Proceedings of the Fourteenth International Confer-
606 ence on Artificial Intelligence and Statistics*, pages 627–635. JMLR Workshop and Conference
607 Proceedings, 2011.

608 A Proofs for Section 4

609 A.1 Proof of Proposition A.1

610 **Proposition A.1.** Any prefix monotonically non-increasing sequence supermodular function g is
611 weakly supermodular with respect to all sequences $S \in \Omega^*$ with $\alpha(S) = 1$

612 *Proof.* This proof is adapted from Liberty and Sviridenko [2017].

613 If g is sequence supermodular then,

$$\begin{aligned}
& g(S) - g(S \oplus S') \\
&= \sum_{i=1}^{|S'|} g(S \oplus (S'_1, \dots, S'_{i-1})) - g(S \oplus (S'_1, \dots, S'_i)) \\
&\stackrel{(a)}{\leq} \sum_{i=1}^{|S'|} g(S) - g(S \oplus S'_i) \\
&\leq |S'| \max_{i \in [|S'|]} g(S) - g(S \oplus S'_i)
\end{aligned}$$

614 (a) by supermodularity

615

□

616 A.2 Proof of Theorem 4.1

617 **Theorem 4.1.** Let $g : \Omega^* \rightarrow \mathbb{R}$ be a weakly supermodular function with respect to the optimal
618 solution $O = \arg \min_{S \in \Omega^T} h(S)$ with a supermodularity ratio of $\alpha(O)$ and postfix monotonicity. Let
619 the greedy solution of length T be S^T . If $\phi_T(\alpha) = (1 - \frac{1}{\alpha T})^T$, then

$$g(S^T) \leq (1 - \phi_T(\alpha(O))) g(O) + \phi_T(\alpha(O)) g(\emptyset)$$

620 *Proof.* This proof strategy is inspired by Streeter and Golovin [2008] and Liberty and Sviridenko
 621 [2017]

$$\begin{aligned}
 g(S^t) - g(O) &\stackrel{(a)}{\leq} g(S^t) - g(S^t \oplus O) \\
 &= \sum_{i=1}^{|O|} g(S^t \oplus (o_1, \dots, o_{i-1})) - g(S^t \oplus (o_1, \dots, o_i)) \\
 &\stackrel{(b)}{\leq} \alpha(O) \sum_{i=1}^{|O|} g(S^t) - g(S^t \oplus o_i) \\
 &\leq \alpha(O) |O| \max_{i \in [|O|]} g(S^t) - g(S^t \oplus o_i) \\
 &\leq \alpha(O) T g(S^t) - \alpha(O) T \min_{\omega \in \Omega} g(S^t \oplus \omega) \\
 &= \alpha(O) T g(S^t) - \alpha(O) T g(S^{t+1})
 \end{aligned}$$

622 (a) by μ - postfix monotonicity, (b) by supermodularity.

623 After rearranging the inequality, we get:

$$\begin{aligned}
 g(S^{t+1}) &\leq \frac{1}{\alpha(O)T} (g(O) - (\alpha(O)T - 1) g(S^t)) \\
 &= \frac{1}{\alpha(O)T} g(O) + \left(1 - \frac{1}{\alpha(O)T}\right) g(S^t)
 \end{aligned}$$

624 When recursively applying this inequality, we get:

$$\begin{aligned}
 g(S^T) &\leq \frac{1}{\alpha(O)T} g(O) \sum_{i=0}^{T-1} \left(1 - \frac{1}{\alpha(O)T}\right)^i + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset) \\
 &= \left(1 - \left(1 - \frac{1}{\alpha(O)T}\right)^T\right) g(O) + \left(1 - \frac{1}{\alpha(O)T}\right)^T g(\emptyset)
 \end{aligned}$$

625

□

626 A.3 Proof of Proposition 4.2

627 **Proposition 4.2.** Suppose that for all $j \in [K]$, the error propagation function $I - C_j \circ \mathcal{L}_h^a$ is
 628 ρ_j -Lipschitz continuous with $\rho_j < 1$, and that $(I - C_j \circ \mathcal{L}_h^a)(0_N) = 0_N$. Then, the function h is
 629 weakly supermodular with respect to the optimal solution O , with

$$\alpha(O) = \max \left\{ \frac{4}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\}$$

630 Furthermore, if $I - C_j \circ \mathcal{L}_h^a$ is invertible for all $j \in [K]$, h is also postfix monotonically non-increasing.

631 *Proof.* For brevity, we use the notation $(g_1 \circ \dots \circ g_T)(x) = \circ_{t=1}^T g_t(x)$, where composition is
 632 applied from right to left so that g_T acts first. In this proof, we use a few properties of Lipschitz
 633 continuous functions:

- 634 • **Property 1:** If g is ρ -Lipschitz continuous and $g(0) = 0$, then $\|g(x)\|_2 = \|g(x) - 0\|_2 =$
 635 $\|g(x) - g(0)\|_2 \leq \rho \|x - 0\|_2 = \rho \|x\|_2$
- 636 • **Property 2:** If g_1 and g_2 are Lipschitz continuous functions with Lipschitz constants of ρ_1
 637 and ρ_2 respectively, the Lipschitz constant of $g_1 + g_2$ and $g_1 - g_2$ is $\rho_1 + \rho_2$.

638 • **Property 3:** If g_1 and g_2 are Lipschitz continuous functions with Lipschitz constants of ρ_1
 639 and ρ_2 respectively, the Lipschitz constant of $g_1 \circ g_2$ is $\rho_1 \rho_2$.

640 In order to prove weakly- α -supermodularity, we must first prove prefix monotonicity.

641 **Prefix monotonicity:** Let $S \preceq S'$ where $S' = S \oplus N$.

$$\begin{aligned} h(S') &= \left\| \circ_{t=|S'|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &= \left\| \circ_{t=|S'|}^{|S|+1} (I - C_{S'_t} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\stackrel{(a)}{\leq} \left(\prod_{t=|S|+1}^{|S'|} \rho_{S'_t}^2 \right) \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\leq h(S) \end{aligned}$$

642 (a) by Property 1

643 **Weak supermodularity:** We will upper bound $\alpha(O)$ by providing an upper bound for $h(S) -$

644 $h(S \oplus O)$ and a lower bound for $\sum_{i=1}^T h(S) - h(S \oplus O_i)$.

$$\begin{aligned} h(S) - h(S \oplus O) &= \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 - \left\| \circ_{t=|O|}^1 (I - C_{O_t} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\stackrel{(a)}{\leq} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) - \circ_{t=|O|}^1 (I - C_{O_t} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &= \left\| \left(I - \circ_{t=|O|}^1 (I - C_{O_t} \circ \mathcal{L}_h^a) \right) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\stackrel{(b)}{\leq} \left(1 + \prod_{t=1}^{|O|} \rho_{O_t} \right)^2 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\stackrel{(c)}{\leq} 4 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \end{aligned}$$

645 (a) by reverse triangle property and $h(S) - h(S \oplus S') > 0$ by prefix monotonicity, (b) since the

646 Lipschitz constant of $I - \circ_{t=|S'|}^1 (I - C_{S'_t} \circ \mathcal{L}_h^a)$ is $1 + \prod_{t=1}^{|S'|} \rho_{S'_t}$ by Property 2 and 3, (c) since

647 $\rho_j < 1$

648 To lower bound $\sum_{i=1}^{|O|} h(S) - h(S \oplus O_i)$

$$\begin{aligned} \sum_{i=1}^{|O|} h(S) - h(S \oplus O_i) &= \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 - \left\| (I - C_{O_i} \circ \mathcal{L}_h^a) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &\geq \sum_{i=1}^{|O|} \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 - \rho_{O_i}^2 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \\ &= \left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \left(T - \sum_{i=1}^T \rho_{O_i}^2 \right) \end{aligned}$$

649 Finally,

$$\begin{aligned} \frac{h(S) - h(S \oplus O)}{T \max_i h(S) - h(S \oplus O_i)} &\leq \max \left\{ \frac{4 \left\| \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2}{\left\| \circ_{t=|S|}^1 (I_N - C_{S_t} \circ \mathcal{L}_h^a) \left(e_h^{(0)} \right) \right\|_2^2 \left(T - \sum_{i=1}^T \rho_{O_i}^2 \right)}, 1 \right\} \\ &= \max \left\{ \frac{4}{T - \sum_{i=1}^T \rho_{O_i}^2}, 1 \right\} \end{aligned}$$

650 **Postfix monotonicity:** Let $S' = S \oplus N$.

$$\begin{aligned}
h(S') &= \left\| \circ_{t=|S'|}^1 (I - C_{S'_t} \circ \mathcal{L}_h) e_h^{(0)} \right\|_2^2 \\
&= \left\| \circ_{t=|N|}^1 (I - C_{N_t} \circ \mathcal{L}_h) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h) e_h^{(0)} \right\|_2^2 \\
&\stackrel{(a)}{=} \left\| \circ_{t=|N|}^1 (I - C_{N_t} \circ \mathcal{L}_h) \circ \circ_{t=|S|}^1 (I - C_{S_t} \circ \mathcal{L}_h) \circ \left(\circ_{t=|N|}^1 (I - C_{N_t} \circ \mathcal{L}_h) \right)^{-1} \circ \circ_{t=|N|}^1 (I - C_{N_t} \circ \mathcal{L}_h) e_h^{(0)} \right\|_2^2 \\
&\leq \prod_{t=1}^{|N|} \rho_{N_t}^2 \prod_{t=1}^{|S|} \rho_{S_t}^2 \prod_{t=1}^{|N|} \rho_{N_t}^{-2} \left\| \circ_{t=|N|}^1 (I - C_{N_t} \circ \mathcal{L}_h) e_h^{(0)} \right\|_2^2 \\
&\leq h(N)
\end{aligned}$$

651 (a) due the invertibility of $I_N - C_j \mathcal{L}_h$

652 □

653 A.4 Proof of Theorem A.2

654 **Lemma A.2.** Let $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$ where P is an orthogonal matrix and $\Lambda_j =$
655 $\text{diag}(\lambda_{j1}, \dots, \lambda_{jN})$. If $P^{-1} e_h^{(0)} = z$, then the following equality holds:

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \quad (11)$$

656 where $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$

Proof.

$$\begin{aligned}
h(S) &= \left\| \prod_{t=|S|}^1 (I_N - C_{S_t} \mathcal{L}_h^a) e_h^{(0)} \right\|_2^2 \\
&= \left\| \prod_{t=|S|}^1 (P \Lambda_{S_t} P^{-1}) e_h^{(0)} \right\|_2^2 \\
&= \left\| P \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|_2^2 \\
&= \left\| \prod_{t=|S|}^1 \Lambda_{S_t} P^{-1} e_h^{(0)} \right\|_2^2 \\
&= \sum_{i=1}^N z_i^2 \prod_{t=1}^{|S|} \lambda_{S_t i}^2 \\
&= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)}
\end{aligned}$$

657 □

658 A.5 Proof of Theorem 4.3

659 **Proposition 4.3.** Let $\|I_N - C_j \mathcal{L}_h^a\| \leq 1$ for all $j \in [K]$ and $(I - C_j \mathcal{L}_h^a) = P \Lambda_j P^{-1}$. Then, h is
660 *supermodular*.

661 *Proof.* Let $S \preceq S'$ where $S' = S \oplus B$. By Theorem A.2,

$$h(S) = \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)}$$

662 where $m_j(S) = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=j}$ be the number of times a sequence S calls the solver j . Recall that h is
 663 considered sequence supermodular if $\forall S', S \in \Omega^*$ such that $S \preceq S'$, it holds that

$$h(S) - h(S \oplus \omega) \geq h(S') - h(S' \oplus \omega)$$

664

$$\begin{aligned} h(S) - h(S \oplus \omega) &= \sum_{i=1}^N z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} - \sum_{i=1}^N z_i^2 \lambda_{\omega i}^2 \prod_{k=1}^K \lambda_{ki}^{2m_j(S)} \\ &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \end{aligned}$$

665 Similarly,

$$\begin{aligned} h(S') - h(S \oplus \omega) &= \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S')} \\ &\stackrel{(a)}{=} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2(m_j(S) + m_j(B))} \\ &\stackrel{(b)}{\leq} \sum_{i=1}^N (1 - \lambda_{\omega i}^2) z_i^2 \prod_{j=1}^K \lambda_{ji}^{2m_j(S)} \\ &= h(S) - h(S \oplus \omega) \end{aligned}$$

666 (a) Since $m_j(S') = \sum_{t=1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S'_t=j} + \sum_{t=|S|+1}^{|S'|} \mathbf{1}_{S'_t=j} = \sum_{t=1}^{|S|} \mathbf{1}_{S_t=k} + \sum_{t=1}^{|B|} \mathbf{1}_{B_t=j} =$
 667 $m_j(S) + m_j(B)$, (b) since $\rho(I_N - C_j \mathcal{L}_h^a) < 1$ and $m_j(B) \geq 0$ for all $j \in [K]$

668 □

669 B Proofs for Section 5

670 B.1 Proof of Theorem B.1

671 **Lemma B.1.** For any set of preconditioning functions $\mathcal{C}$, any discrete operator $\mathcal{L}_h^a$, any router r , any
 672 $a_h, f_h, u_h^{(t)}, u_h \in \mathcal{A} \times \mathcal{F} \times \mathcal{U} \times \mathcal{U}$, the following equality holds true:

$$\begin{aligned} l_{route}(r, a_h, f_h, u_h^{(t)}, u_h) &= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\ &\quad - (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) (u_h - u_h^{(t)}) \right\|_2^2 \end{aligned}$$

673 *Proof.* Note that $\sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} = K - 1$

$$\begin{aligned}
l_{\text{route}} \left(r, a_h, f_h, u_h^{(t)}, u_h \right) &= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) = j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 - (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&= \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 - \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad + \sum_{j=1}^K \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&\quad - (K - 1) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2
\end{aligned}$$

674

$$\begin{aligned}
&= \sum_{j=1}^K \left(\sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 - \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \right) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K - 2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&= \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - (K - 2) \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2
\end{aligned}$$

675

□

676 **B.2 Proof of Theorem B.2**

677 **Proposition B.2.** For any router r defined by $r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$, any $a_h \in \mathcal{A}$,
678 $f_h \in \mathcal{F}$, and $u_h^{(t)}, u_h \in \mathcal{U}$, the routing loss l_{route} satisfies:

$$\log(2) l_{\text{route}} \left(r, a_h, f_h, u_h^{(t)}, u_h \right) \leq \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)$$

679 *Proof.* By Theorem B.1, we know that

$$\begin{aligned}
\log(2)l_{\text{route}}\left(r, a_h, f_h, u_h^{(t)}, u_h\right) &= \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\quad - \log(2) (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&\leq \log(2) \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\
&\stackrel{(a)}{\leq} - \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \log \left(\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \\
&= \Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, u_h)
\end{aligned}$$

680 (a) if $r(a_h, f_h, u_h^{(t)}) \neq j$, $\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} < 0.5$ which implies that

$$681 -\log \left(\frac{\exp(g_j(a, f, u^{(t)}))}{\sum_{k=1}^K \exp(g_k(a, f, u^{(t)}))} \right) \geq \log(2) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j}$$

682 $\square$

683 B.3 Proof of Theorem 5.1

684 **Theorem 5.1.** Let $\tilde{c}_j(a_h, u_h^{(t)}, u_h) < \bar{E} < \infty$ for all $j \in [K]$. If there exists $j \in [K]$ such that
685 $\tilde{c}_j(a_h, u_h^{(t)}, u_h) > E_{\min} > 0$, then, for any collection of solvers $\{C_j\}_{j=1}^K$ and linear discrete
686 operator $\mathcal{L}_h^a$, Ψ is Bayes consistent surrogate for l_{route} .

687 *Proof.* For a given a_h, f_h , let u_h be $\mathcal{G}_h(a_h, f_h)$ where $\mathcal{G}_h$ denotes the solution operator acting on the
688 grid G_h . Furthermore, let's consider routers of the form

$$r(a, f, u^{(t)}) = \operatorname{argmax}_{j \in [K]} g_j(a, f, u^{(t)})$$

689 For a given $a_h, f_h, u_h^{(t)} \in \mathcal{A} \times \mathcal{F} \times \mathcal{U}$, let the optimal loss under l_{route} be
690 $l_{\text{route}}^*(a_h, f_h, u_h^{(t)}) = \inf_{\tilde{r}} l_{\text{route}}(\tilde{r}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h))$. Similarly, let the optimal loss un-
691 der Ψ be $\Psi^*(a_h, f_h, u_h^{(t)}) = \inf_{\tilde{\mathbf{g}}} \Psi(\tilde{\mathbf{g}}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h))$. Let $B_j(a_h, f_h, u_h^{(t)}) =$
692 $\sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(\mathcal{G}_h(a_h, f_h) - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j}$.

$$\begin{aligned}
&l_{\text{route}}\left(r, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)\right) - l_{\text{route}}^*\left(a_h, f_h, u_h^{(t)}\right) \\
&\stackrel{(a)}{=} \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&\quad - \inf_{\tilde{r}} \sum_{j=1}^K \sum_{k=1}^K \left\| (I - C_k \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \mathbf{1}_{k \neq j} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} + (K-2) \sum_{j=1}^K \left\| (I - C_j \circ \mathcal{L}_h^a) \left(u_h - u_h^{(t)} \right) \right\|_2^2 \\
&= \sum_{j=1}^K B_j(a_h, f_h, u_h^{(t)}) \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K B_j(a_h, f_h, u_h^{(t)}) \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \\
&= \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \left(\sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \right)
\end{aligned}$$

693 (a) by Theorem B.1

694 Let $\mathcal{X} = \mathcal{A} \times \mathcal{F} \times \mathcal{U}$ and $\mathcal{Y} = [K]$. Let $\mathcal{P}_{\mathcal{X}}$ denote the degenerate distribution supported at the
695 point $(a_h, f_h, u_h^{(t)})$. We define the conditional distribution - $P(Y = j \mid X = (a_h, f_h, u_h^{(t)})) =$
696 $\frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})}$ for $j \in [K]$. The risk and optimal risk of 0 - 1 loss under this distribution can
697 be written as:

$$\begin{aligned}\mathcal{R}_{0-1}(r) &= \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} \\ \mathcal{R}_{0-1}^* &= \inf_r \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j}\end{aligned}$$

698 If $r(a_h, f_h, u_h^{(t)}) = \arg \max_{j \in [K]} g_j(a_h, f_h, u_h^{(t)})$ for all $x \in \mathcal{X}$, then the he risk and optimal risk of
699 cross entropy loss ($l_{ce}(\mathbf{g}, x, y) - \log \left(\frac{\exp(g_y(x))}{\sum_{k=1}^K \exp(g_k(x))} \right)$) under this distribution can be written as:

$$\begin{aligned}\mathcal{R}_{ce}(\mathbf{g}) &= - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \\ \mathcal{R}_{ce}^* &= \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right)\end{aligned}$$

700 From Theorem 3.1 of [Mao et al., 2023], $\mathcal{R}_{0-1}(r) - \mathcal{R}_{0-1}^* \leq \Gamma^{-1}(\mathcal{R}_{ce}(\mathbf{g}) - \mathcal{R}_{ce}^*)$ if
701 $r(a_h, f_h, u_h^{(t)}) = \arg \max_{j \in [K]} g_j(a_h, f_h, u_h^{(t)})$ where $\Gamma(z) = \frac{1+z}{2} \log(1+z) + \frac{1-z}{2} \log(1-z)$.
702 Then,

$$\begin{aligned}
& l_{\text{route}} \left(r, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h) \right) - l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \\
&= \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \left(\sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{r(a_h, f_h, u_h^{(t)}) \neq j} - \inf_{\tilde{r}} \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \mathbf{1}_{\tilde{r}(a_h, f_h, u_h^{(t)}) \neq j} \right) \\
&\leq \sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)}) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&\stackrel{(a)}{\leq} \bar{E} K (K-1) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{\sum_{k=1}^K B_k(a_h, f_h, u_h^{(t)})} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&\stackrel{(b)}{\leq} \bar{E} K (K-1) \Gamma^{-1} \left(- \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right. \\
&\quad \left. - \inf_{\mathbf{g}} - \sum_{j=1}^K \frac{B_j(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \log \left(\frac{\exp(g_j(a_h, f_h, u_h^{(t)}))}{\sum_{k=1}^K \exp(g_k(a_h, f_h, u_h^{(t)}))} \right) \right) \\
&= \bar{E} K (K-1) \Gamma^{-1} \left(\frac{\Psi(\mathbf{g}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \right)
\end{aligned}$$

703 (a) since $\left\| (I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)}) \right\|_2^2 < \bar{E}$ for all $j \in [K]$, (b) since Γ^{-1} is non-decreasing and

704 $\exists j \in [K]$ such that $\left\| (I - C_j \circ \mathcal{L}_h^a)(e_h^{(t)}) \right\|_2^2 > E_{\min}$

705 Finally,

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \mathcal{R}_{\text{route}}(r_n) - \mathcal{R}_{\text{route}}^* \\
&\stackrel{(a)}{=} \lim_{n \rightarrow \infty} \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[l_{\text{route}} \left(r_n, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h) \right) - l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \right] \\
&\leq \lim_{n \rightarrow \infty} \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[\bar{E} K (K-1) \Gamma^{-1} \left(\frac{\Psi(\tilde{\mathbf{g}}, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)})}{(K-1)E_{\min}} \right) \right] \\
&\stackrel{(b)}{\leq} \lim_{n \rightarrow \infty} \bar{E} K (K-1) \Gamma^{-1} \left(\frac{\mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[\Psi(\mathbf{g}_n, a_h, f_h, u_h^{(t)}, \mathcal{G}_h(a_h, f_h)) - \Psi^*(a_h, f_h, u_h^{(t)}) \right]}{(K-1)E_{\min}} \right)
\end{aligned}$$

$$\begin{aligned}
&= \lim_{n \rightarrow \infty} \bar{E}K(K-1)\Gamma^{-1} \left(\frac{\mathcal{R}_\Psi(\mathbf{g}_n) - \mathcal{R}_\Psi^*}{(K-1)E_{min}} \right) \\
&\stackrel{(c)}{=} \bar{E}K(K-1)\Gamma^{-1} \left(\frac{\lim_{n \rightarrow \infty} \mathcal{R}_\Psi(\mathbf{g}_n) - \mathcal{R}_\Psi^*}{(K-1)E_{min}} \right) \\
&= \bar{E}K(K-1)\Gamma^{-1}(0) \\
&\stackrel{(d)}{=} 0
\end{aligned}$$

707 (a) $\mathcal{R}_{\text{route}}^* = \mathbb{E}_{a_h, f_h \sim \mathcal{P}_{\mathcal{A} \times \mathcal{F}}} \left[l_{\text{route}}^* \left(a_h, f_h, u_h^{(t)} \right) \right]$ since the infimum is taken over all measurable
708 functions, (b) by Jensen's inequality since Γ^{-1} is concave, (c) by continuity of Γ^{-1} at 0, (d) $\Gamma^{-1}(0) =$
709 0

710

□

711 C Training Details

712 Data for both DeepONet and the routers is sampled from a zero-mean Hierarchical Gaussian Random
713 Field on the periodic domain with covariance operator $\alpha(-\Delta + \beta I)^{-\gamma}$, where α, β, γ are sampled
714 as:

$$\begin{aligned}
\alpha &\sim \text{Log-Uniform}(0.01, 100) \\
\beta &\sim \text{Log-Uniform}(0.1, 1000) \\
\gamma &\sim \text{Uniform}(\{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0\})
\end{aligned}$$

715 Given (α, β, γ) , samples are generated in the Fourier space. For each non-zero frequency mode
716 k , we draw an independent complex coefficient from a Gaussian distribution with mean 0 and
717 variance $\alpha(4\pi^2\|k\|_2^2 + \beta)^{-\gamma}$. We enforce a Hermitian symmetry to obtain a real-valued field, set the
718 zero-frequency (DC) mode to 0 to obtain a zero mean field, and apply the inverse Discrete Fourier
719 Transform to obtain the field in physical space.

720 For each sample, we compute reference solutions with a least squares solver and treat them as ground
721 truth.

722 This data is used to trained our DeepONet models and LSTM routers. All the models were imple-
723 mented using PyTorch and all the models were trained on one Nvidia A40 GPU.

724 Table 4 contains all hyperparameter details for the DeepONet. DeepONet took 30 minutes to train.
725 We then use the model with the best validation loss.

Table 4: Hyperparameter settings for DeepONet

Hyperparameter	Value
Learning rate	1e-3
Branch Dimension	128
Hidden dimension for branch net	256
No. of hidden layers in branch net	4
Hidden dimension for trunk net	256
No. of hidden layers in trunk net	4
Gradient Clipping Norm	1.0
Weight Decay	0.005
Batch size	256
Training samples	10000
Validation samples	2000
Epochs	1000

726 The routers are LSTM models. Along with the inputs specified in Section 5, namely $a_h, f_h, u_h^{(t)}$,
727 we also supply the routers with the iteration index t , current residual $f_h - \mathcal{L}_h^a u_h^{(t)}$, and the routing

728 decisions from the previous iteration. While the iteration index and residual are deterministic
729 functions of $(a_h, f_h, u_h^{(t)})$ and align with the theory presented in Section 5, incorporating previous
730 routing decisions is a mild deviation from the idealized setting. Nevertheless, this modification is
731 motivated by the same considerations that justify the use of scheduled sampling: under deployment,
732 routing decisions are predicted autoregressively and influence future inputs, whereas training under
733 teacher forcing assumes independence from past predictions. Including previous routing decisions
734 bridges the gap induced by exposure bias by providing the model with trajectory-level information.

735 All the routers are trained with scheduled sampling. We use a warm-up of e_w epochs with teacher-
736 forcing probability $p_{tf}(e) = ss_{start}$. After the warm-up, the p_{tf} decays geometrically by a factor of
737 $\gamma_{tf} < 1$ per epoch and is floored by ss_{end} :

$$p_{tf}(e) = \begin{cases} ss_{start} & e \leq e_w \\ \max(ss_{start}\gamma_{tf}^{e-e_w}, ss_{end}) & e > e_w \end{cases}$$

738 At each time step, with probability $p_{tf}(e)$, we feed the teacher-forced greedy iterate; otherwise, we
739 feed the router’s own predicted iterate.

740 Since LSTMs on long rollouts can suffer from exploding/vanishing gradients, we use truncated
741 backpropagation through time (TBPTT) [Mozer, 2013, Robinson and Fallside, 1987, Werbos, 1988]:
742 the forward pass unrolls the entire trajectory, but gradients are propagated only through the most
743 recent $w_{bptt}(e)$ steps at epoch e . Hidden states are passed forward between segments, while earlier
744 segments are treated as stop-gradient.

745 We employ a curriculum learning approach analogous to scheduled sampling. Let T_{max} be the
746 horizon (300 iterations). With a warm-up of e_w epochs,

$$w_{bptt}(e) = \begin{cases} w_{start} & e \leq e_w \\ \min\left(T_{max}, w_{start}\gamma_{bptt}^{\lfloor \frac{e-e_w}{f_{bptt}} \rfloor}\right) & e > e_w \end{cases} \quad (12)$$

747 so the window grows geometrically by a factor of $\gamma_{bptt} > 1$ every f_{bptt} epochs and is capped at the
748 full trajectory length.

749 Table 5 contains all hyperparameter details for the LSTM routers. The routers for the Paired Solver
750 experiment took a maximum of 4 hours and 30 minutes to train while the routers for the Solver
751 Ensemble experiment took a maximum of 6 hours to train. We then use the model with the best
752 validation loss for testing. The architecture and dataset size is larger for the Solver ensemble
753 experiments to encourage the model to learn some of the nuanced differences between the classes.

Table 5: Hyperparameter settings for routers in Paired solver and Solver Ensemble Experiments

Hyperparameter	Paired Solver	Solver Ensemble
Learning rate	1e-3	1e-3
Hidden dimension	256	512
No. of hidden layers	4	3
Gradient Clipping Norm	1.0	1.0
Weight Decay	0.005	0.005
Batch size	32	32
Training samples	256	1024
Validation samples	32	128
Epochs	200	200
ss_{start}	1.0	1.0
γ_{tf}	0.95	0.95
ss_{end}	0.0	0.0
w_{start}	50	50
γ_{bptt}	1.25	1.25
e_w	10	10
f_{bptt}	1	1

D Additional Experimental Results

We present additional visualizations and tables that further highlight the strengths of our method. In the paired solver experiments, we include Damped Jacobi with $\omega = 0.67$ and Successive Over-Relaxation (SOR) with $\omega = 1.5$. We also compare against a True Greedy oracle, which has access to true error at each step (See Section 4). This comparison demonstrates that the learned router closely approximates the behavior of the greedy policy.

D.1 Error Comparisons with Paired t -Test Results

We supplement the paired-solver experiments in Table 1 with additional solver families (Jacobi (0.67) and SOR (1.5)) and statistical significance analysis. For each solver family, we compare the learned greedy router against baseline methods (single-solver only and HINTS) using paired t -tests with a one-sided alternative hypothesis that the learned router achieves lower error or AUC. Results are reported in Table 6.

Across all five solver families, the resulting p -values indicate statistically significant improvements of the learned greedy router over both single-solver and HINTS baselines.

We support the Large Solver ensemble experiments in Table 3 with a statistical significance analysis. We use paired t -tests across the 128 test instances. For each baseline solver j (e.g., Jacobi, GS, SymGS, etc.), we compare the performance of the solver ensemble $\mathcal{W}$ against the corresponding pairwise learned router $\text{Router}(\text{NO} \cup \{j\})$ if $j \in \mathcal{W}$. The test is performed on per-instance differences in final error (and AUC), using a one-sided alternative hypothesis that the solver ensemble achieves lower error than the baseline.

Across both Poisson and convection–diffusion problems, the solver ensembles yield statistically significant improvements over all pairwise baselines in nearly every setting, with p -values typically well below 0.01. The gains are enhanced as the ensemble size increases, while the performance of the learned router remains close to that of the true greedy oracle. These results support the claim that enlarging the solver set provides measurable benefits and that our learned policy effectively captures the resulting improvements. Results are reported in Table 7.

Table 6: Final error and AUC of squared L^2 error (lower is better). Values are mean ($\pm$ standard error (s.e.)) over 128 test instances; both mean and s.e. of error are reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Statistical significance is assessed via paired t -tests comparing each baseline (single-solver only and HINTS) against the learned greedy router, using a one-sided alternative that the learned router achieves lower error/AUC. Reported p -values correspond to these tests.

Equation	Poisson				ConvDiff			
Methods	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ $ p -value	AUC	AUC p -value	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ $ p -value	AUC	AUC p -value
Jacobi-related solvers								
Jacobi Only	0.383 (1.029)	$< 10^{-3}$	0.821 (2.154)	$< 10^{-3}$	0.136 (0.364)	$< 10^{-3}$	0.312 (0.788)	$< 10^{-3}$
HINTS-Jacobi	0.759 (0.016)	$< 10^{-3}$	0.393 (0.590)	$< 10^{-3}$	0.447 (0.013)	$< 10^{-3}$	0.202 (0.256)	$< 10^{-3}$
Learned Greedy-Jacobi	0.054 (0.142)	-	0.165 (0.368)	-	0.033 (0.097)	-	0.098 (0.239)	-
True-Greedy-Jacobi	0.021 (0.018)	-	0.094 (0.170)	-	0.012 (0.012)	-	0.049 (0.091)	-
GS-related solvers								
GS only	0.019 (0.051)	$< 10^{-3}$	0.435 (1.141)	$< 10^{-3}$	$< 10^{-3}$	0.006	0.088 (0.219)	$< 10^{-3}$
HINTS-GS	0.724 (0.000)	$< 10^{-3}$	0.325 (0.488)	$< 10^{-3}$	0.456 (0.000)	$< 10^{-3}$	0.130 (0.154)	0.0
Learned Greedy-GS	0.002 (0.004)	-	0.083 (0.152)	-	$< 10^{-3}$	-	0.031 (0.078)	-
True-Greedy-GS	0.001 (0.001)	-	0.057 (0.107)	-	$< 10^{-3}$	-	0.019 (0.038)	-
SymGS-related solvers								
SymGS only	$< 10^{-3}$	$< 10^{-3}$	0.227 (0.593)	$< 10^{-3}$	$< 10^{-3}$	0.028	0.071 (0.178)	$< 10^{-3}$
HINTS-SymGS	0.709 (0.000)	$< 10^{-3}$	0.252 (0.380)	$< 10^{-3}$	0.463 (0.000)	$< 10^{-3}$	0.116 (0.131)	$< 10^{-3}$
Learned Greedy-SymGS	$< 10^{-3}$	-	0.052 (0.102)	-	$< 10^{-3}$	-	0.022 (0.039)	-
True-Greedy-SymGS	$< 10^{-3}$	-	0.034 (0.066)	-	$< 10^{-3}$	-	0.016 (0.032)	-
Jacobi (0.67)-related solvers								
Jacobi (0.67) only	1.066 (2.861)	$< 10^{-3}$	1.128 (2.954)	$< 10^{-3}$	0.349 (0.931)	$< 10^{-3}$	0.428 (1.077)	$< 10^{-3}$
HINTS-Jacobi (0.67)	0.789 (0.000)	$< 10^{-3}$	0.429 (0.642)	0.005	0.473 (0.000)	$< 10^{-3}$	0.223 (0.284)	$< 10^{-3}$
Learned Greedy-Jacobi (0.67)	0.232 (0.654)	-	0.309 (0.791)	-	0.069 (0.102)	-	0.117 (0.210)	-
True-Greedy-Jacobi (0.67)	0.057 (0.041)	-	0.131 (0.233)	-	0.026 (0.021)	-	0.065 (0.117)	-
SOR (1.5)-related solvers								
SOR (1.5) only	$< 10^{-3}$	0.057	0.156 (0.409)	$< 10^{-3}$	$< 10^{-3}$	0.086	0.008 (0.020)	0.988
HINTS-SOR (1.5)	0.700 (0.000)	$< 10^{-3}$	0.207 (0.316)	$< 10^{-3}$	0.462 (0.000)	$< 10^{-3}$	0.021 (0.020)	$< 10^{-3}$
Learned Greedy-SOR (1.5)	$< 10^{-3}$	-	0.052 (0.143)	-	$< 10^{-3}$	-	0.011 (0.030)	-
True-Greedy-SOR (1.5)	$< 10^{-3}$	-	0.032 (0.071)	-	$< 10^{-3}$	-	0.007 (0.019)	-

Table 7: Final error and AUC of squared L^2 error (lower is better). Values are mean ($\pm$ standard error (s.e.)) over 128 test instances; both mean and s.e. of error are reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Statistical significance is assessed via paired t -tests comparing each solver ensemble against the corresponding pairwise learned router (e.g., NO + Jacobi, NO + GS), using a one-sided alternative that the ensemble achieves lower error/AUC. Reported p -values correspond to these tests.

$\mathcal{W}$	$\ e_h^{(T)}\ \times 10^3$	AUC	Jacobi p -value	GS p -value	SymGS p -value	Jacobi (0.67) p -value	SOR (1.5) p -value
Poisson							
Learned Greedy {Jacobi, GS}	0.003 (0.002)	0.068 (0.131)	$< 10^{-3}$	$< 10^{-3}$	-	-	-
True Greedy {Jacobi, GS}	0.001 (0.001)	0.057 (0.107)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS}	$< 10^{-3}$	0.038 (0.071)	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	-	-
True Greedy {Jacobi, GS, SymGS}	$< 10^{-3}$	0.034 (0.066)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	$< 10^{-3}$	0.042 (0.074)	$< 10^{-3}$	$< 10^{-3}$	0.004	$< 10^{-3}$	-
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	$< 10^{-3}$	0.034 (0.066)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.039 (0.076)	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	0.04
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.031 (0.069)	-	-	-	-	-
ConvDiff							
Learned Greedy {Jacobi, GS}	$< 10^{-3}$	0.021 (0.040)	$< 10^{-3}$	0.005	-	-	-
True Greedy {Jacobi, GS}	$< 10^{-3}$	0.019 (0.038)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS}	$< 10^{-3}$	0.020 (0.036)	$< 10^{-3}$	0.004	$< 10^{-3}$	-	-
True Greedy {Jacobi, GS, SymGS}	$< 10^{-3}$	0.016 (0.032)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	$< 10^{-3}$	0.020 (0.039)	$< 10^{-3}$	0.003	0.003	$< 10^{-3}$	-
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	$< 10^{-3}$	0.016 (0.032)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.008 (0.019)	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	0.004
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.007 (0.018)	-	-	-	-	-

780 D.2 Convergence Histories

781 See Figure 4 for convergence histories for the Jacobi (0.67)-related solvers and Successive Over-
782 relaxation (1.5)-related solvers.

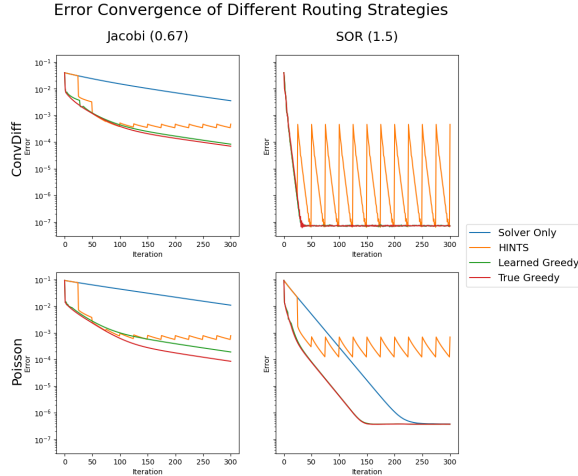


Figure 4: Convergence histories for representative test instances. Rows: ConvDiff (top) and Poisson (bottom). Columns: Jacobi (0.67) and Successive Over-relaxation (1.5). Greedy yields near-monotone decay and the lowest errors, whereas HINTS shows sawtooth behaviors.

783 D.3 Prediction and Error Visualizations

784 Figures 5 and 6 provide qualitative visualizations of sample predictions across routing strategies and
785 numerical solvers for both equations. While these prediction may appear identical to the ground
786 truth, Figures 7 and 8 reveals distinct error patterns with varying magnitudes across all predictions.
787 In particular, our learned method consistently achieves the lower-magnitude errors compared to its
788 respective baselines.

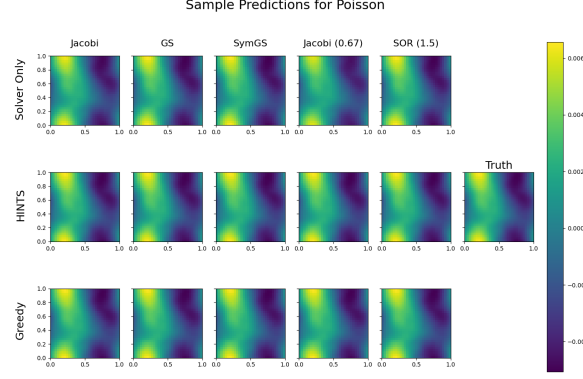


Figure 5: Sample predictions for the Poisson equation across different solver pairings. Each column corresponds to a numerical solver (Jacobi, Gauss–Seidel (GS), Symmetric GS, Damped Jacobi, and Successive Over-relaxation), while rows show predictions from the corresponding solver-only baseline (top), HINTS (middle), and the learned greedy router (bottom). The ground truth solution is shown on the right.

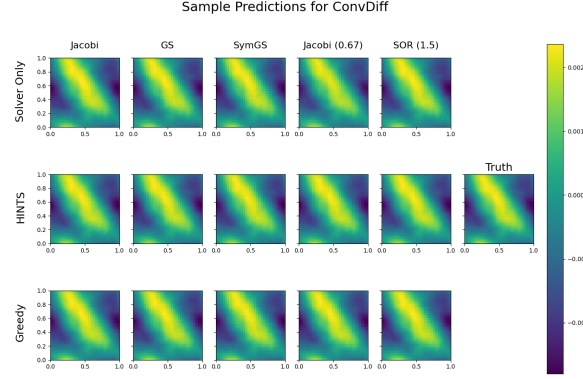


Figure 6: Sample predictions for the convection diffusion equation across different solver pairings. Each column corresponds to a numerical solver (Jacobi, Gauss–Seidel (GS), Symmetric GS, Damped Jacobi, and Successive Over-relaxation), while rows show predictions from the corresponding solver-only baseline (top), HINTS (middle), and the learned greedy router (bottom). The ground truth solution is shown on the right.

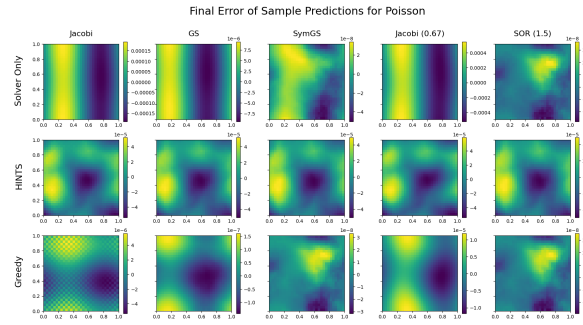


Figure 7: Pointwise error maps for Poisson sample predictions across solver families. Columns correspond to different numerical solvers, while rows show solver-only (top), HINTS (middle), and learned greedy router (bottom). Errors are computed as $u_h - u_h^{(T)} = e_h^{(T)}$. The learned greedy router consistently yields lower-magnitude errors compared to both solver-only and HINTS baselines.

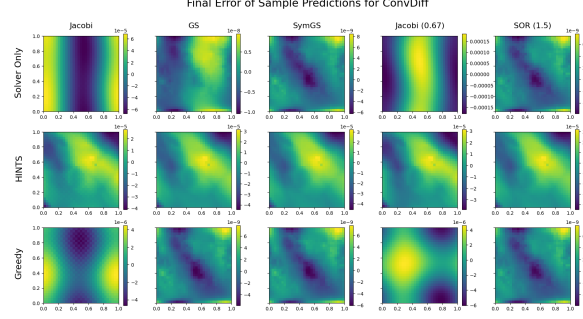


Figure 8: Pointwise error maps for convection diffusion sample predictions across solver families. Columns correspond to different numerical solvers, while rows show solver-only (top), HINTS (middle), and learned greedy router (bottom). Errors are computed as $u_h - u_h^{(T)} = e_h^{(T)}$. The learned greedy router consistently yields lower-magnitude errors compared to both solver-only and HINTS baselines.

789 D.4 Residual Comparison

Table 8: Final residual $\|\mathcal{L}_h^a e_h^{(T)}\|$ and AUC of residuals over iterations (lower is better). Values are reported as mean ($\pm$ standard error (s.e.)) over 128 test instances; residuals are scaled by $\times 10^{-3}$. If the residual norm is reported as $< 10^{-3}$, the corresponding raw value is $< 10^{-6}$. Statistical significance is evaluated via paired t -tests comparing each baseline (single-solver only and HINTS) to the learned greedy router under a one-sided alternative hypothesis that the learned router achieves lower residual and AUC. Reported p -values correspond to these tests.

Equation	Poisson				ConvDiff			
Methods	$\ \mathcal{L}_h^a e_h^{(T)}\ \times 10^3$	$\ \mathcal{L}_h^a e_h^{(T)}\ $	AUC	AUC p-value	$\ \mathcal{L}_h^a e_h^{(T)}\ \times 10^3$	$\ \mathcal{L}_h^a e_h^{(T)}\ $ p-value	AUC	AUC p-value
Jacobi-related solvers								
Jacobi only	23.939 (66.755)	0.021	43.475 (109.764)	$< 10^{-3}$	27.208 (74.555)	0.012	45.523 (114.774)	$< 10^{-3}$
HINTS-Jacobi	193.327 (37.362)	$< 10^{-3}$	39.763 (70.671)	$< 10^{-3}$	266.421 (34.699)	$< 10^{-3}$	42.851 (70.748)	$< 10^{-3}$
Learned Greedy-Jacobi	20.870 (62.246)	-	32.041 (83.647)	-	20.786 (66.511)	-	34.844 (84.666)	-
GS-related solvers								
GS only	0.761 (2.043)	$< 10^{-3}$	20.776 (52.270)	$< 10^{-3}$	0.002 (0.005)	$< 10^{-3}$	10.720 (26.892)	$< 10^{-3}$
HINTS-GS	183.359 (0.000)	$< 10^{-3}$	22.581 (27.984)	$< 10^{-3}$	246.877 (0.000)	$< 10^{-3}$	18.654 (19.201)	$< 10^{-3}$
Learned Greedy-GS	0.092 (0.145)	-	11.960 (27.863)	-	0.001 (0.003)	-	7.322 (16.968)	-
SymGS-related solvers								
SymGS only	0.004 (0.009)	$< 10^{-3}$	10.343 (25.957)	$< 10^{-3}$	0.002 (0.004)	0.004	8.604 (21.776)	$< 10^{-3}$
HINTS-SymGS	185.601 (0.001)	$< 10^{-3}$	15.489 (18.255)	$< 10^{-3}$	245.629 (0.000)	$< 10^{-3}$	16.515 (15.990)	$< 10^{-3}$
Learned Greedy-SymGS	0.002 (0.005)	-	7.149 (18.364)	-	0.001 (0.002)	-	6.481 (13.190)	-
Jacobi (0.67)-related solvers								
Jacobi (0.67) only	42.246 (112.968)	$< 10^{-3}$	55.250 (138.258)	$< 10^{-3}$	45.451 (121.431)	$< 10^{-3}$	56.747 (141.901)	$< 10^{-3}$
HINTS-Jacobi (0.67)	188.719 (0.008)	$< 10^{-3}$	38.970 (53.035)	$< 10^{-3}$	256.407 (0.005)	$< 10^{-3}$	39.604 (52.578)	$< 10^{-3}$
Learned Greedy-Jacobi (0.67)	9.290 (26.109)	-	35.905 (86.881)	-	9.010 (13.426)	-	32.516 (83.488)	-
SOR (1.5)-related solvers								
SOR (1.5) only	0.003 (0.008)	0.5	9.044 (22.962)	$< 10^{-3}$	0.002 (0.004)	0.189	2.817 (7.499)	$< 10^{-3}$
HINTS-SOR (1.5)	188.187 (0.001)	$< 10^{-3}$	15.894 (19.565)	$< 10^{-3}$	248.226 (0.001)	$< 10^{-3}$	11.420 (7.498)	$< 10^{-3}$
Learned Greedy-SOR (1.5)	0.003 (0.008)	-	7.784 (19.939)	-	0.002 (0.004)	-	5.530 (19.679)	-

Table 9: Final residual and AUC of squared L^2 residual for varying numbers of solvers. Values are mean ($\pm$ standard error (s.e.)) over 128 test instances; both mean and s.e. are reported in $\times 10^{-3}$.

Equation	Poisson		ConvDiff	
$\mathcal{W}$	$\ \mathcal{L}_h^a e_h^{(T)}\ $	AUC	$\ \mathcal{L}_h^a e_h^{(T)}\ $	AUC
{Jacobi, GS}	0.135 (0.096)	11.815 (33.768)	0.023 (0.044)	5.765 (14.581)
{Jacobi, GS, SymGS}	0.002 (0.004)	5.118 (12.920)	0.002 (0.002)	4.810 (11.264)
{Jacobi, GS, SymGS, Jacobi (0.67)}	0.013 (0.011)	6.120 (14.651)	0.010 (0.013)	4.859 (11.473)
{Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	0.004 (0.006)	8.196 (22.516)	0.001 (0.002)	3.209 (8.676)

790 Table 8 summarizes the performance of single-solver schedules, HINTS, and greedy with respect
791 to the final residuals $r_h^{(T)} = \|f_h - \mathcal{L}_h^a u_h^{(T)}\|$ or $\|\mathcal{L}_h^a e_h^{(T)}\|$ and its AUC $AUC_T = \sum_{t=1}^T \|r_h^{(t)}\|_2^2$.
792 Greedy outperforms its HINTS and single-solver counterparts in most equations. We must note that
793 our greedy router is trained to reduce error, not residual. The same error can induce very different
794 residuals depending on the spectrum $\mathcal{L}_h^a$. Table 9 exhibits how residuals are affected by the number of
795 solvers in the solver ensemble. Similar to error, we observe both the final residual and AUC decrease
796 as the number of solvers increase.

797 D.5 Fourier mode-wise error comparison

798 We assess frequency-resolved performance by projecting the error onto the discrete Fourier basis.
 799 Table 10 report, for modes 1, 5, and 10, the mode-wise final error and mode-wise AUC, comparing
 800 single-solver baselines, HINTS, and the greedy router. As a result of including a deep learning
 801 model, Greedy consistently achieves the smallest mode-1 error/AUC across equations and solver
 802 families. For modes 5 and 10, single-solver schedules sometimes have an edge, reflecting the
 803 tendency of classical smoothers to damp high-frequency components more aggressively than ML
 804 surrogates (spectral bias). Overall, greedy delivers more uniform convergence across the spectrum:
 805 it routes to whichever solver most decreases the full L^2 error, and by Parseval’s identity $|e_h^{(t)}|_2^2 =$
 806 $\sum_m |\hat{u}_m^{(t)} - \hat{u}_m|^2$, reductions in the objective correspond to reducing energy across all modes rather
 807 than giving preferential treatment to a subset. Additionally, in Table 11, we observe that all mode-wise
 808 errors/AUCs reduce with the inclusion of more solvers.

Table 10: Final error and AUC of squared L^2 error for Mode 1, 5, and 10 (lower is better) for Poisson and ConvDiff. Values are mean ($\pm$ standard error (s.e.)) over 128 test instances; both mean and s.e. of final errors are reported in $\times 10^{-3}$. If the residual norm is reported as $< 10^{-3}$, the corresponding raw value is $< 10^{-6}$.

Method	Mode 1		Poisson Mode 5		Mode 10		Mode 1		ConvDiff Mode 5		Mode 10	
	Final Error	AUC	Final Error	AUC	Final Error	AUC	Final Error	AUC	Final Error	AUC	Final Error	AUC
Jacobi Only	0.135 (0.295)	3.231 (7.056)	$< 10^{-3}$	0.003 (0.010)	$< 10^{-3}$	0.000 (0.001)	0.079 (0.171)	1.084 (2.369)	$< 10^{-3}$	0.003 (0.011)	$< 10^{-3}$	0.000 (0.001)
HINTS-Jacobi	5.730 (0.240)	2.734 (3.371)	0.028 (0.002)	0.004 (0.012)	0.006 (0.000)	0.000 (0.001)	0.994 (0.220)	0.697 (1.091)	0.061 (0.001)	0.005 (0.012)	0.022 (0.001)	0.001 (0.001)
Learned Greedy-Jacobi	0.032 (0.085)	0.841 (2.081)	$< 10^{-3}$	0.009 (0.028)	$< 10^{-3}$	0.001 (0.003)	0.021 (0.055)	0.324 (0.780)	$< 10^{-3}$	0.009 (0.025)	$< 10^{-3}$	0.001 (0.002)
GS-related solvers												
GS Only	0.002 (0.004)	1.695 (3.702)	$< 10^{-3}$	0.003 (0.011)	$< 10^{-3}$	0.000 (0.001)	$< 10^{-3}$	0.205 (0.452)	$< 10^{-3}$	0.002 (0.006)	$< 10^{-3}$	$< 10^{-3}$
HINTS-GS	5.377 (0.000)	2.113 (2.498)	0.027 (0.000)	0.005 (0.012)	0.007 (0.000)	0.000 (0.001)	0.661 (0.000)	0.271 (0.432)	0.065 (0.000)	0.003 (0.006)	0.019 (0.000)	$< 10^{-3}$
Learned Greedy-GS	0.000 (0.001)	0.465 (1.125)	$< 10^{-3}$	0.009 (0.029)	$< 10^{-3}$	0.001 (0.002)	$< 10^{-3}$	0.075 (0.151)	$< 10^{-3}$	0.005 (0.017)	$< 10^{-3}$	0.001 (0.001)
SymGS-related solvers												
SymGS Only	$< 10^{-3}$	0.883 (1.919)	$< 10^{-3}$	0.002 (0.006)	$< 10^{-3}$	0.000 (0.001)	$< 10^{-3}$	0.180 (0.400)	$< 10^{-3}$	0.001 (0.005)	$< 10^{-3}$	0.000 (0.001)
HINTS-SymGS	5.085 (0.000)	1.432 (1.700)	0.025 (0.000)	0.002 (0.007)	0.007 (0.000)	0.000 (0.001)	0.672 (0.000)	0.240 (0.381)	0.070 (0.000)	0.002 (0.005)	0.019 (0.000)	0.000 (0.001)
Learned Greedy-SymGS	$< 10^{-3}$	0.334 (0.943)	$< 10^{-3}$	0.007 (0.025)	$< 10^{-3}$	0.001 (0.002)	$< 10^{-3}$	0.079 (0.193)	$< 10^{-3}$	0.004 (0.012)	$< 10^{-3}$	0.001 (0.001)
Jacobi (0.67)-related solvers												
Jacobi (0.67) Only	1.063 (2.321)	4.771 (10.421)	$< 10^{-3}$	0.005 (0.017)	$< 10^{-3}$	$< 10^{-3}$	0.428 (0.934)	1.535 (3.352)	$< 10^{-3}$	0.004 (0.016)	$< 10^{-3}$	$< 10^{-3}$
HINTS-Jacobi (0.67)	5.804 (0.001)	2.983 (3.666)	0.029 (0.000)	0.007 (0.019)	0.006 (0.000)	$< 10^{-3}$	1.080 (0.000)	0.776 (1.158)	0.065 (0.000)	0.008 (0.017)	0.021 (0.000)	$< 10^{-3}$
Learned Greedy-Jacobi (0.67)	0.390 (1.195)	1.804 (5.363)	$< 10^{-3}$	0.017 (0.058)	$< 10^{-3}$	0.001 (0.002)	0.096 (0.208)	0.435 (0.910)	$< 10^{-3}$	0.013 (0.044)	$< 10^{-3}$	0.001 (0.004)
SOR (1.5)-related solvers												
SOR (1.5) Only	$< 10^{-3}$	0.727 (1.594)	$< 10^{-3}$	0.005 (0.019)	$< 10^{-3}$	0.001 (0.002)	$< 10^{-3}$	0.037 (0.082)	$< 10^{-3}$	0.002 (0.007)	$< 10^{-3}$	0.000 (0.001)
HINTS-SOR (1.5)	4.826 (0.000)	1.239 (1.484)	0.027 (0.000)	0.007 (0.020)	0.008 (0.000)	0.001 (0.002)	0.645 (0.000)	0.080 (0.082)	0.068 (0.000)	0.005 (0.007)	0.019 (0.000)	0.001 (0.001)
Learned Greedy-SOR (1.5)	$< 10^{-3}$	0.302 (0.757)	$< 10^{-3}$	0.010 (0.032)	$< 10^{-3}$	0.001 (0.003)	$< 10^{-3}$	0.049 (0.119)	$< 10^{-3}$	0.007 (0.030)	$< 10^{-3}$	0.001 (0.002)

Table 11: Final error and AUC of squared L^2 error of Mode 1, 5, and 10 for varying numbers of solvers. Values are mean ($\pm$ standard error (s.e.)) over 128 test instances; both mean and s.e. are reported in $\times 10^{-3}$.

Equation	Poisson				ConvDiff			
	Mode 1 Error	Mode 1 AUC	Mode 5 Error	Mode 5 AUC	Mode 1 Error	Mode 1 AUC	Mode 5 Error	Mode 5 AUC
JW Mode								
[Jacobi, GS]	0.000 (0.001)	0.267 (0.571)	$< 10^{-3}$	0.006 (0.025)	$< 10^{-3}$	0.001 (0.005)	$< 10^{-3}$	0.055 (0.112)
[Jacobi, GS, SymGS]	0.000 (0.001)	0.185 (0.415)	$< 10^{-3}$	0.003 (0.009)	$< 10^{-3}$	0.000 (0.002)	$< 10^{-3}$	0.054 (0.114)
[Jacobi, GS, SymGS, Jacobi (0.67)]	0.000 (0.001)	0.172 (0.333)	$< 10^{-3}$	0.003 (0.010)	$< 10^{-3}$	0.000 (0.002)	$< 10^{-3}$	0.056 (0.119)
[Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)]	0.000 (0.001)	0.210 (0.399)	$< 10^{-3}$	0.006 (0.023)	$< 10^{-3}$	0.001 (0.004)	$< 10^{-3}$	0.026 (0.061)

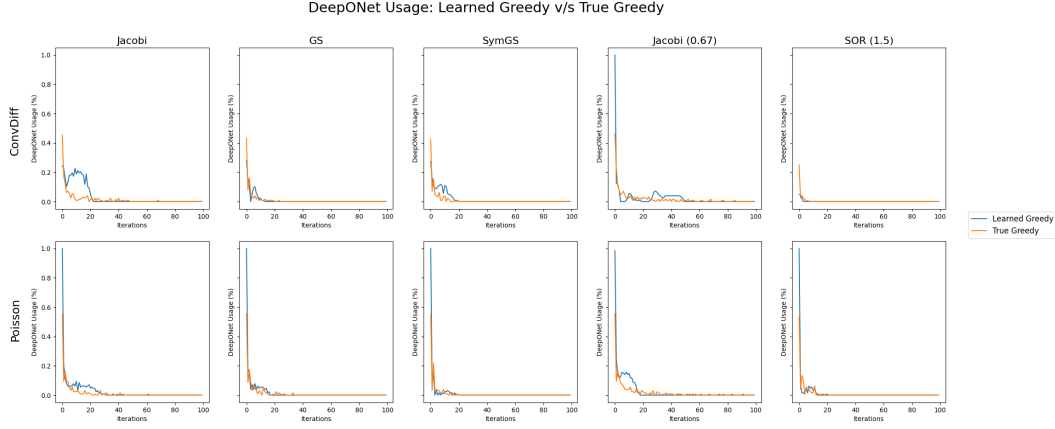


Figure 9: DeepONet usage over iterations for learned and true greedy routing across solver families. Columns correspond to different numerical solvers, and rows correspond to the two equations. The learned router closely follows the true greedy policy, capturing its nonuniform usage of DeepONet across iterations. For readability, we display the first 100 iterations (out of 300 total)

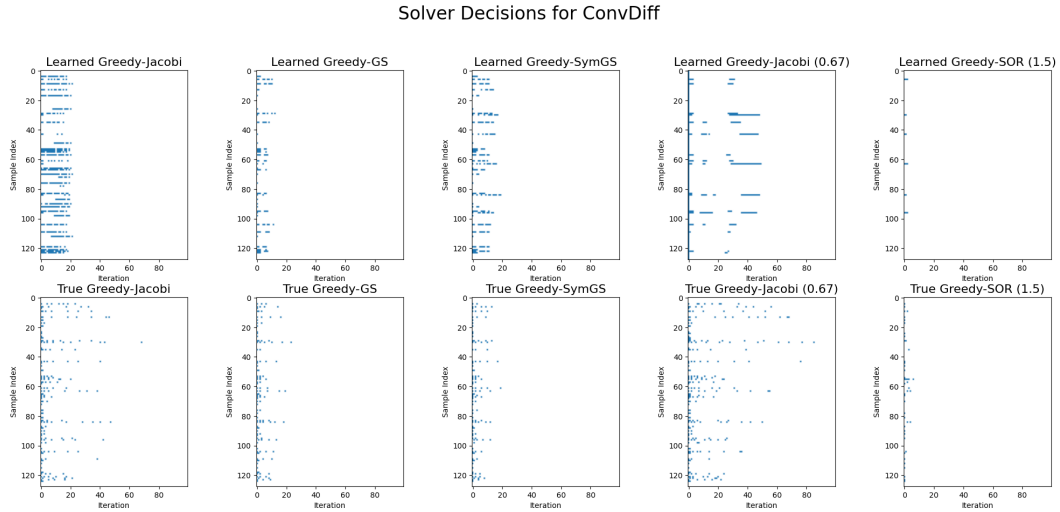


Figure 10: Solver selection patterns for the convection–diffusion equation across different solver pairings and test samples. Each column corresponds to a solver family, while rows show decisions from the learned greedy router (top) and the true greedy policy (bottom). Colored entries indicate iterations where the neural operator is selected, with blank regions corresponding to numerical solver updates. The learned router exhibits instance-dependent routing behavior.

Solver Decisions for Poisson

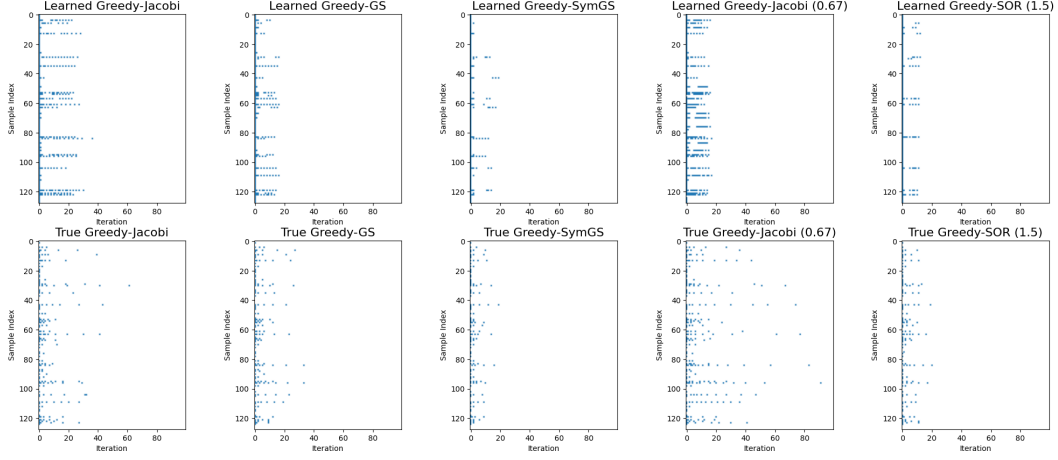


Figure 11: Solver selection patterns for the Poisson equation across different solver pairings and test samples. Each column corresponds to a solver family, while rows show decisions from the learned greedy router (top) and the true greedy policy (bottom). Colored entries indicate iterations where the neural operator is selected, with blank regions corresponding to numerical solver updates. The learned router exhibits instance-dependent routing behavior.

Figure 9 highlights the proportion of DeepONet calls across iterations, while Figures 10 and 11 visualize the routing decisions of the learned and true greedy policies across all test samples and solver families.

Two key observations emerge. First, the routing strategy is highly sample-dependent, indicating that a fixed routing schedule across all instances would be suboptimal and can slow convergence. Second, the learned routing policy closely tracks the behavior of the true greedy policy, providing empirical support for the approximation guarantees discussed in Section 5. In particular, the learned policy can be viewed as a smoothed approximation of the true greedy strategy, capturing its overall structure while exhibiting minor deviations due to learning and finite-sample effects.

Table 12: Average component selection frequencies for the Poisson equation. Rows indicate the router access set. Entries are mean ($\pm$ s.e.) over 128 test instances. Here NO denotes the neural operator; “-” indicates that the component is not available to the router.

$\mathcal{W}$	Jacobi	GS	SSOR	Jacobi (0.67)	SOR (1.5)	DeepONet
{Jacobi, GS}	0.405 (0.143)	0.588 (0.134)	-	-	-	0.007 (0.011)
{Jacobi, GS, SSOR}	0.000 (0.000)	0.000 (0.000)	0.996 (0.005)	-	-	0.004 (0.005)
{Jacobi, GS, SSOR, Jacobi (0.67)}	0.227 (0.112)	0.212 (0.118)	0.434 (0.212)	0.121 (0.102)	-	0.005 (0.007)
{Jacobi, GS, SSOR, Jacobi (0.67), SOR (1.5)}	0.009 (0.033)	0.570 (0.057)	0.260 (0.040)	0.000 (0.000)	0.156 (0.013)	0.005 (0.005)

Table 13: Average component selection frequencies for the Convection-Diffusion equation. Rows indicate the router access set. Entries are mean ($\pm$ s.e.) over 128 test instances. Here NO denotes the neural operator; “-” indicates that the component is not available to the router.

$\mathcal{W}$	Jacobi	GS	SSOR	Jacobi (0.67)	SOR (1.5)	DeepONet
{Jacobi, GS}	0.417 (0.307)	0.579 (0.307)	-	-	-	0.004 (0.006)
{Jacobi, GS, SSOR}	0.109 (0.109)	0.396 (0.110)	0.493 (0.175)	-	-	0.003 (0.005)
{Jacobi, GS, SSOR, Jacobi (0.67)}	0.168 (0.096)	0.190 (0.166)	0.410 (0.227)	0.229 (0.146)	-	0.003 (0.004)
{Jacobi, GS, SSOR, Jacobi (0.67), SOR (1.5)}	0.307 (0.150)	0.061 (0.131)	0.088 (0.170)	0.202 (0.089)	0.342 (0.066)	0.001 (0.001)

Tables 12 and 13 report solver selection frequencies across different solver ensembles. Several interesting observations emerge from these results.

First, the router typically relies on a small subset of solvers, with larger ensembles often leading to the dominance of a single solver (e.g., SOR(1.5) in convection–diffusion). This explains the diminishing returns observed in the multi-solver setting, as the solver ensemble grows.

Additionally, the neural operator is selected only rarely across all configurations. This reflects the strength of the numerical solvers rather than a limitation of our method. Our framework is designed to exploit this structure by relying primarily on numerical updates and invoking the learned component only when beneficial, demonstrating that selective use of learned corrections can yield performance gains without frequent deployment like in HINTS or similar fixed schedules.

Finally, the non-zero standard deviations indicate that solver usage varies across test instances, suggesting that no single global routing strategy emerges.

D.7 Results on a finer grid

We have replicated the Paired Solver and Solver Ensemble experiments for a grid of 63×63 .

In Table 14, we notice that, similar to the case with 31×31 grid, the learned greedy router closely matches the performance of the true greedy oracle and outperforms its single solver and HINTS baselines with great statistical significance.

Similar to Table 3, we notice, in Table 15, the learned greedy router seems to outperform the various pairwise configurations mostly with great statistical significance.

Table 14: Final error and AUC of squared L^2 error (lower is better). Values are mean ($\pm$ standard error (s.e.)) over 128 test instances for the grid 63×63 ; both mean and s.e. of error are reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Statistical significance is assessed via paired t -tests comparing each baseline (single-solver only and HINTS) against the learned greedy router, using a one-sided alternative that the learned router achieves lower error/AUC. Reported p -values correspond to these tests.

Equation	Poisson				ConvDiff			
Methods	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ $ p -value	AUC	AUC p -value	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ $ p -value	AUC	AUC p -value
Jacobi-related solvers								
Jacobi Only	4.483 (12.776)	$< 10^{-3}$	2.092 (5.889)	$< 10^{-3}$	1.470 (4.164)	0.001	0.757 (2.116)	0.001
HINTS-Jacobi	0.823 (0.163)	0.043	0.571 (1.205)	0.021	1.512 (0.433)	$< 10^{-3}$	0.565 (0.723)	$< 10^{-3}$
Learned Greedy-Jacobi	0.506 (2.201)	-	0.429 (1.293)	-	0.654 (1.603)	-	0.393 (0.969)	-
True-Greedy-Jacobi	0.199 (0.277)	-	0.289 (0.661)	-	0.317 (0.593)	-	0.277 (0.635)	-
GS-related solvers								
GS only	2.094 (6.009)	$< 10^{-3}$	1.519 (4.296)	$< 10^{-3}$	0.357 (1.025)	0.004	0.430 (1.211)	0.003
HINTS-GS	0.631 (0.003)	$< 10^{-3}$	0.444 (0.939)	$< 10^{-3}$	0.996 (0.004)	$< 10^{-3}$	0.369 (0.397)	0.03
Learned Greedy-GS	0.138 (0.581)	-	0.228 (0.623)	-	0.219 (0.697)	-	0.282 (0.820)	-
True-Greedy-GS	0.072 (0.064)	-	0.180 (0.403)	-	0.053 (0.065)	-	0.132 (0.279)	-
SymGS-related solvers								
SymGS only	1.318 (3.541)	$< 10^{-3}$	1.106 (2.978)	$< 10^{-3}$	0.267 (0.725)	0.012	0.362 (0.974)	0.014
HINTS-SymGS	0.566 (0.002)	$< 10^{-3}$	0.375 (0.782)	$< 10^{-3}$	1.007 (0.000)	$< 10^{-3}$	0.367 (0.329)	$< 10^{-3}$
Learned Greedy-SymGS	0.162 (0.155)	-	0.169 (0.310)	-	0.141 (0.355)	-	0.239 (0.599)	-
True-Greedy-SymGS	0.084 (0.094)	-	0.130 (0.266)	-	0.146 (0.285)	-	0.130 (0.241)	-
Jacobi (0.67)-related solvers								
Jacobi (0.67) only	5.833 (16.544)	$< 10^{-3}$	2.365 (6.639)	$< 10^{-3}$	1.985 (5.595)	0.005	0.876 (2.443)	0.004
HINTS-Jacobi (0.67)	1.177 (0.445)	$< 10^{-3}$	0.687 (1.414)	$< 10^{-3}$	1.970 (0.989)	$< 10^{-3}$	0.652 (0.883)	0.001
Learned Greedy-Jacobi (0.67)	0.693 (1.959)	-	0.489 (1.217)	-	1.060 (2.422)	-	0.517 (1.211)	-
True-Greedy-Jacobi (0.67)	0.396 (0.741)	-	0.406 (0.979)	-	0.528 (1.133)	-	0.352 (0.831)	-
SOR (1.5)-related solvers								
SOR (1.5) only	0.115 (0.331)	$< 10^{-3}$	0.652 (1.850)	$< 10^{-3}$	$< 10^{-3}$	0.056	0.078 (0.220)	0.002
HINTS-SOR (1.5)	0.480 (0.000)	$< 10^{-3}$	0.304 (0.680)	$< 10^{-3}$	0.608 (0.000)	$< 10^{-3}$	0.147 (0.165)	$< 10^{-3}$
Learned Greedy-SOR (1.5)	0.017 (0.084)	-	0.148 (0.516)	-	$< 10^{-3}$	-	0.039 (0.090)	-
True-Greedy-SOR (1.5)	0.004 (0.003)	-	0.081 (0.188)	-	$< 10^{-3}$	-	0.030 (0.067)	-

Table 15: Final error and AUC of squared L^2 error (lower is better). Values are mean ($\pm$ standard error (s.e.)) over 128 test instances for the grid 63×63 ; both mean and s.e. of error are reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Statistical significance is assessed via paired t -tests comparing each solver ensemble against the corresponding pairwise learned router (e.g., NO + Jacobi, NO + GS), using a one-sided alternative that the ensemble achieves lower error/AUC. Reported p -values correspond to these tests.

$\mathcal{W}$	$\ e_h^{(T)}\ \times 10^3$	AUC	Jacobi p -value	GS p -value	SymGS p -value	Jacobi (0.67) p -value	SOR (1.5) p -value
Poisson							
Learned Greedy {Jacobi, GS}	0.077 (0.064)	0.188 (0.410)	0.002	0.08	-	-	-
True Greedy {Jacobi, GS}	0.072 (0.064)	0.180 (0.403)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS}	0.112 (0.086)	0.151 (0.275)	0.002	0.015	$< 10^{-3}$	-	-
True Greedy {Jacobi, GS, SymGS}	0.054 (0.059)	0.126 (0.264)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	0.109 (0.103)	0.152 (0.282)	0.002	0.013	$< 10^{-3}$	$< 10^{-3}$	-
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	0.054 (0.059)	0.126 (0.264)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	0.014 (0.006)	0.092 (0.205)	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	0.054
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	0.004 (0.003)	0.082 (0.189)	-	-	-	-	-
ConvDiff							
Learned Greedy {Jacobi, GS}	0.136 (0.447)	0.170 (0.394)	$< 10^{-3}$	0.015	-	-	-
True Greedy {Jacobi, GS}	0.053 (0.065)	0.132 (0.279)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS}	0.063 (0.098)	0.142 (0.335)	$< 10^{-3}$	0.006	0.001	-	-
True Greedy {Jacobi, GS, SymGS}	0.025 (0.037)	0.105 (0.220)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	0.114 (0.216)	0.181 (0.413)	$< 10^{-3}$	0.024	0.011	$< 10^{-3}$	-
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67)}	0.025 (0.037)	0.105 (0.220)	-	-	-	-	-
Learned Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.037 (0.087)	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	$< 10^{-3}$	0.069
True Greedy {Jacobi, GS, SymGS, Jacobi (0.67), SOR (1.5)}	$< 10^{-3}$	0.030 (0.067)	-	-	-	-	-

838 D.8 Results with Dirichlet Boundary Conditions

839 We have replicated the Paired Solver and Solver Ensemble experiments for solutions with zero
840 dirichlet boundary conditions.

841 In Table 16, we notice that, similar to PDEs with periodic boundary conditions, the learned greedy
842 router closely matches the performance of the true greedy oracle and outperforms its single solver
843 and HINTS baselines with great statistical significance. Despite the learned router matching the
844 performance of the true greedy oracle, we observe that the learned router doesn't seem to provide
845 much improvement over single-solver baseline in the case of SOR (1.5)-related solvers. This may be
846 a result of the deepnet correction always being weaker relative to the SOR (1.5) correction in terms
847 of error reduction.

848 Similar to Table 3, we notice, in Table 15, the learned greedy router seems to outperform the various
849 pairwise configurations mostly with great statistical significance.

Table 16: Final error and AUC of squared L^2 error (lower is better). Values are mean ($\pm$ standard error (s.e.)) over 128 test instances with dirichlet boundaries; both mean and s.e. of error are reported in $\times 10^{-3}$. If a standard error is not shown, it is $< 10^{-3}$ in the reported units (raw $< 10^{-6}$). Statistical significance is assessed via paired t -tests comparing each baseline (single-solver only and HINTS) against the learned greedy router, using a one-sided alternative that the learned router achieves lower error/AUC. Reported p -values correspond to these tests.

Equation	Poisson				ConvDiff			
Methods	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ p$ -value	AUC	AUC p -value	$\ e_h^{(T)}\ \times 10^3$	$\ e_h^{(T)}\ p$ -value	AUC	AUC p -value
Jacobi-related solvers								
Jacobi Only	0.704 (2.151)	0.001	0.703 (1.929)	$< 10^{-3}$	0.003 (0.006)	0.057	0.169 (0.430)	$< 10^{-3}$
HINTS-Jacobi	2.674 (7.315)	$< 10^{-3}$	0.869 (2.396)	$< 10^{-3}$	0.474 (0.001)	$< 10^{-3}$	0.199 (0.254)	$< 10^{-3}$
Learned Greedy-Jacobi	0.243 (0.851)	-	0.236 (0.719)	-	0.003 (0.006)	-	0.076 (0.162)	-
True-Greedy-Jacobi	0.042 (0.047)	-	0.099 (0.252)	-	0.003 (0.007)	-	0.042 (0.089)	-
GS-related solvers								
GS only	0.138 (0.431)	0.001	0.408 (1.139)	$< 10^{-3}$	0.003 (0.006)	0.028	0.057 (0.146)	$< 10^{-3}$
HINTS-GS	2.668 (7.316)	$< 10^{-3}$	0.841 (2.360)	$< 10^{-3}$	0.470 (0.001)	$< 10^{-3}$	0.115 (0.137)	$< 10^{-3}$
Learned Greedy-GS	0.032 (0.158)	-	0.115 (0.433)	-	0.003 (0.006)	-	0.024 (0.048)	-
True-Greedy-GS	0.010 (0.015)	-	0.063 (0.166)	-	0.003 (0.006)	-	0.017 (0.036)	-
SymGS-related solvers								
SymGS only	2.924 (8.865)	0.01	0.926 (2.740)	0.006	1.331 (3.687)	0.004	0.413 (1.133)	0.004
HINTS-SymGS	2.748 (7.375)	0.008	0.856 (2.427)	0.008	0.917 (1.015)	0.006	0.332 (0.697)	0.003
Learned Greedy-SymGS	1.333 (5.715)	-	0.416 (1.722)	-	0.628 (1.468)	-	0.196 (0.449)	-
True-Greedy-SymGS	0.342 (0.906)	-	0.147 (0.375)	-	0.184 (0.323)	-	0.067 (0.118)	-
Jacobi (0.67)-related solvers								
Jacobi (0.67) only	1.265 (3.768)	$< 10^{-3}$	0.910 (2.468)	$< 10^{-3}$	0.008 (0.021)	0.003	0.253 (0.642)	$< 10^{-3}$
HINTS-Jacobi (0.67)	2.686 (7.311)	$< 10^{-3}$	0.886 (2.412)	$< 10^{-3}$	0.487 (0.002)	$< 10^{-3}$	0.220 (0.279)	$< 10^{-3}$
Learned Greedy-Jacobi (0.67)	0.134 (0.275)	-	0.215 (0.660)	-	0.004 (0.010)	-	0.127 (0.393)	-
True-Greedy-Jacobi (0.67)	0.096 (0.220)	-	0.145 (0.408)	-	0.004 (0.013)	-	0.061 (0.131)	-
SOR (1.5)-related solvers								
SOR (1.5) only	0.004 (0.011)	0.012	0.145 (0.412)	$< 10^{-3}$	0.003 (0.006)	1.0	0.006 (0.014)	1.0
HINTS-SOR (1.5)	2.650 (7.321)	$< 10^{-3}$	0.791 (2.295)	$< 10^{-3}$	0.468 (0.001)	$< 10^{-3}$	0.014 (0.014)	$< 10^{-3}$
Learned Greedy-SOR (1.5)	0.004 (0.011)	-	0.039 (0.091)	-	0.003 (0.006)	-	0.006 (0.014)	-
True-Greedy-SOR (1.5)	0.004 (0.011)	-	0.028 (0.070)	-	0.003 (0.006)	-	0.006 (0.014)	-

E Cost-Aware Routing and Wall-Clock Evaluation

The experiments of Section 7 compare methods per iteration, which favors methods whose iterations are expensive. This appendix evaluates the end-to-end *wall-clock time to a target accuracy* of the learned router against HINTS and the classical solver alone, for all five classical solvers and three PDEs, on a 128×128 grid ($N^2 = 128^2$ unknowns), using the cost-aware instantiation of the greedy rule introduced at the end of Section 4. Besides the Poisson and convection–diffusion equations of Section 7 we add *anisotropic diffusion*, $-\epsilon \partial_{x_1}^2 u - \partial_{x_2}^2 u = f$ with $\epsilon = 10^{-2}$ (diffusion tensor $\text{diag}(\epsilon, 1)$), on the same periodic domain with the same forcing distribution. It is the standard model problem on which point-relaxation methods fail to smooth: an error component oscillatory in x_1 and smooth in x_2 is damped by only $(1 - \epsilon)/(1 + \epsilon) \approx 0.98$ per Jacobi sweep, and geometric multigrid with point smoothing loses its grid-independent convergence (line relaxation or semi-coarsening would be needed). The same pipeline is then applied to solver ensembles, to larger grids and against stronger classical baselines.

E.1 Protocol

Fast, matched implementations. A wall-clock comparison is meaningful only if the classical baseline is not handicapped by implementation. All solvers are therefore re-implemented for the periodic constant-coefficient setting as $O(N^2)$ operations: Jacobi and damped Jacobi as 5-point stencil sweeps, Gauss–Seidel, SOR and symmetric GS via cached sparse triangular factorizations. Their one-step iterates match the dense implementations used in the rest of the paper to 10^{-12} in double precision. Reference solutions are computed by FFT diagonalization of the same discrete operator, i.e., they are exact solutions of the discrete system, so true errors are available for evaluation at no charged cost. All timings are single-threaded CPU measurements (Apple M4 Pro, numpy/scipy/PyTorch, one thread) on an otherwise idle machine.

Corrector. The neural corrector is a DeepONet applied to the residual $r = f_h - \mathcal{L}_h^a u_h^{(t)}$ and evaluated on a 64×64 sensor grid, i.e., on the $4 \times$ coarser grid obtained by band-limited (FFT) restriction to the modes $|k_1|, |k_2| \leq 31$ (for the anisotropic equation the sensor grid is 128×32 , i.e., full resolution along x_1 , so that the band contains every mode the point smoothers cannot damp; the sensor grid is the only PDE-specific hyperparameter of the pipeline); its output is prolonged back to the fine grid by exact trigonometric interpolation, so the correction is band-limited by construction and never injects error into the high-frequency band that the classical smoother owns. The trunk net is the fixed orthonormal Fourier basis of that band ($p = 3969$ basis functions, as in POD-DeepONet [Lu et al., 2022]), and the branch net is a linear layer on the 4096 sensor values; it is fit by least squares on 64,000 exact (residual, error) pairs of the discrete operator drawn from a mixture of spectral shapes (forcing-like, residual-like, white and randomly tilted spectra), and reaches a validation relative error of 1.7×10^{-6} on every mode of the band. The corrector is applied scale-equivariantly, $C_{\text{NO}}(r) = \|Rr\| \mathcal{G}_\theta(Rr/\|Rr\|)$, which preserves $C_{\text{NO}}(0) = 0$ (Proposition 4.2). One corrector call costs ≈ 0.6 – 0.7 ms at 128^2 (two FFTs and a 4096×4096 matrix–vector product), i.e., about 12 Jacobi sweeps, 3 GS/SOR sweeps, or 2 symmetric-GS sweeps (Table 20). The same corrector is used by HINTS and by all routing policies.

Cost-equalised macro-actions. Per-iteration costs c_j (residual evaluation plus the update) are measured once per pairing on the benchmark machine (minimum over seven repeated blocks of the block median, cached), the unit of cost u is one corrector call, and every operation is applied $m_j = \max\{1, \text{round}(u/c_j)\}$ times per decision (Table 20); for the five stationary solvers this equalises costs to within 8%. An operation dearer than the unit (a multigrid cycle) is applied once and compared per unit of cost, $\|e_j\|^{u/(m_j c_j)}$; the surrogate weights $\tilde{c}_j$ use the same exponent. The cost-aware oracle runs Algorithm 1 on these macro-actions with the true error; the *greedy oracle* row runs the cost-agnostic Algorithm 1 on single operations, as in the main text.

Lightweight router. The learned router selects macro-actions from $6 + K$ scalar features, all $O(1)$ given the residual norm that any stopping test computes: the log relative residual, its per-iteration change over the last macro-action and an exponential moving average thereof, the log epoch index, a one-hot encoding of the previous macro-action, and the log counts of corrector calls so far and of iterations since the last call. It is a two-hidden-layer MLP (64 units) evaluated in numpy, costing

902 $\approx 5 \mu\text{s}$ per decision independently of N . It is trained with the surrogate of Section 5.1, with the
 903 per-state weights $\tilde{c}_k$ normalised by $\sum_k \tilde{c}_k$ (which does not change the Bayes decision), on states
 904 visited by oracle rollouts with ε -exploration (probability 0.1), followed by two DAgger rounds
 905 [Ross et al., 2011] in which the current router’s own rollouts are relabelled by the oracle—the batch
 906 analogue of the scheduled sampling used in Section C. Training uses 128 instances per round drawn
 907 from the same GRF distribution as the test set but with a different seed; the router of every pairing
 908 and ensemble is trained in under two minutes.

909 **Baselines and metric.** HINTS applies the corrector every τ -th iteration; we report $\tau = 25$ (the
 910 schedule used in the main text and in Zhang et al. [2024]) and the best of $\tau \in \{5, 10, 25, 50\}$ selected
 911 *per cell* on the test instances themselves (an optimistic baseline). For each of 64 test instances we
 912 record, after every iteration, the true relative error $\|e_h^{(t)}\|/\|u_h\|$ (untimed, benchmark-only) and the
 913 cumulative charged wall-clock time of a replay that executes only the chosen operations (residual,
 914 update, and—for the learned router—every feature and decision computation; oracle decisions are
 915 not charged). We report the median first time at which the error drops below a tolerance ε , and paired
 916 per-instance median speedups. We highlight $\varepsilon = h^2 = 6.1 \times 10^{-5}$: the discretization error of the
 917 second-order scheme scales as $O(h^2)$, so driving the algebraic error far below h^2 does not improve
 918 the computed solution. We also report the iteration-based metrics of the main text (final relative error
 919 and $\text{AUC} = \sum_{t=1}^T \|e_h^{(t)}\|/\|u_h\|$ over $T = 300$ iterations).

920 E.2 Pairwise results

921 Table 2 (main text) reports the time to truncation-level accuracy for all 15 pairings; the tables below
 922 give the comparison against HINTS at three tolerances (Table 17), the iteration-based metrics of the
 923 main text (Table 18), corrector-call and iteration counts (Table 19), the measured costs (Table 20),
 924 and the full tolerance sweeps (Tables 21 and 22).

925 **Uniform gains.** In every one of the 15 solver–equation pairings the learned router is the fastest
 926 deployable method at $\varepsilon = h^2$: $238\times$ – $1983\times$ faster than the solver alone, $2.68\times$ – $11.8\times$ faster than
 927 HINTS with $\tau=25$, and $1.21\times$ – $3.22\times$ faster than the best fixed period chosen per pairing on the test
 928 set (all paired Wilcoxon $p < 0.01$). The gains are not confined to the truncation level: at $\varepsilon = 10^{-8}$
 929 the router is still $1.33\times$ – $5.26\times$ faster than HINTS ($\tau=25$) and $1.07\times$ – $2.74\times$ faster than the per-
 930 tolerance best period (Table 17), and the best period itself varies across pairings and tolerances ($\tau=5$
 931 at truncation level, $\tau \in \{5, 10, 50\}$ at 10^{-8}), so no single fixed schedule is competitive everywhere.
 932 The learned router reproduces the cost-aware oracle’s decisions almost exactly (the same median
 933 iteration and call counts in Table 19; median times within $0.98\times$ – $1.06\times$ of the oracle, the difference
 934 being its own decision cost and timer noise).

935 **Where the time goes.** Table 19 explains the numbers: with a corrector that removes the smooth
 936 band to 10^{-6} in one call, the cost-aware oracle and the router call it first and reach h^2 after one or
 937 two further sweeps, i.e., in about two iterations, whereas HINTS reaches the same accuracy only
 938 at its first scheduled call (iteration τ), having spent $\tau-1$ sweeps on an error that is almost entirely
 939 smooth. Below h^2 the remaining error is the high-frequency band that the corrector cannot see; the
 940 router therefore stops calling it (one to three calls in total down to 10^{-8} ; Figure 13), whereas a fixed
 941 schedule keeps paying for a call every τ iterations—HINTS ($\tau=5$) executes hundreds of useless calls
 942 on the Jacobi pairings (Table 21). The cost-agnostic greedy oracle (Algorithm 1 on single operations)
 943 makes the same decisions as its cost-aware counterpart here, because after the first call the corrector
 944 is never better than a single sweep on the remaining high-frequency error; the two rules differ only
 945 through the granularity of their decisions, and the learned router pays for neither.

946 **Anisotropic diffusion.** The anisotropic equation is where the two components of the hybrid are
 947 most complementary and where the classical baselines are weakest: the point smoothers cannot
 948 damp the x_1 -oscillatory modes at all (Table 23: none of the five solvers alone reaches h^2 within
 949 60,000 iterations), whereas the corrector, whose sensor grid keeps full resolution along x_1 , removes
 950 exactly those modes together with the smooth ones in a single call and leaves the smoother only the
 951 x_2 -oscillatory band, which it damps at the isotropic rate. Consequently the learned router reaches
 952 h^2 in about 1 ms for every pairing—the same absolute time as on the isotropic equations—while
 953 HINTS ($\tau=25$) needs 3–10 ms and the best fixed period 1.2–2.4 ms (Tables 2 and 17); the router

again matches the cost-aware oracle. The multigrid and Krylov baselines are not run for this equation: with point smoothing, the V-cycle’s contraction factor degrades from ≈ 0.03 to ≈ 0.85 per cycle (line relaxation or semi-coarsening would be required), so the comparison of Section E.4 would not be informative.

Iteration-based metrics. Table 18 confirms that the iteration-count conclusions of Section 7 hold on the finer grid: the router’s AUC over $T = 300$ iterations is three to four orders of magnitude below HINTS’ and four to five below the solver-only baseline (all $p < 10^{-3}$), and its error after T iterations is at the stopping threshold of the runs (10^{-9}) except for undamped Jacobi, where all hybrids stall at the near-Nyquist band.

Table 17: Paired median speedup of the learned router over HINTS at three tolerances (64 instances; bold: $\geq 1.1 \times$ with one-sided Wilcoxon $p < 0.01$). The right block compares against the fixed period that is best for that tolerance and pairing, chosen on the test set.

Equation	Solver	vs. HINTS ($\tau=25$)			vs. best fixed τ		
		$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$
Poisson	Jacobi	2.89 $\times$	2.68 $\times$	1.55 $\times$	1.19 $\times$ ($\tau=5$)	1.21 $\times$ ($\tau=5$)	1.22 $\times$ ($\tau=50$)
	Jacobi (0.67)	3.20 $\times$	2.90 $\times$	1.78 $\times$	1.31 $\times$ ($\tau=5$)	1.28 $\times$ ($\tau=5$)	1.09 $\times$ ($\tau=10$)
	GS	7.46 $\times$	6.75 $\times$	3.25 $\times$	1.99 $\times$ ($\tau=5$)	1.94 $\times$ ($\tau=5$)	1.25 $\times$ ($\tau=5$)
	SymGS	11.8 $\times$	9.21 $\times$	5.26 $\times$	2.71 $\times$ ($\tau=5$)	2.34 $\times$ ($\tau=5$)	1.81 $\times$ ($\tau=5$)
	SOR (1.5)	7.63 $\times$	10.3 $\times$	2.65 $\times$	1.97 $\times$ ($\tau=5$)	3.22 $\times$ ($\tau=5$)	1.25 $\times$ ($\tau=10$)
ConvDiff	Jacobi	3.34 $\times$	2.95 $\times$	1.33 $\times$	1.27 $\times$ ($\tau=5$)	1.34 $\times$ ($\tau=5$)	1.17 $\times$ ($\tau=50$)
	Jacobi (0.67)	3.64 $\times$	3.14 $\times$	1.77 $\times$	1.39 $\times$ ($\tau=5$)	1.32 $\times$ ($\tau=5$)	1.07 $\times$ ($\tau=10$)
	GS	7.71 $\times$	6.76 $\times$	3.78 $\times$	2.08 $\times$ ($\tau=5$)	2.60 $\times$ ($\tau=5$)	1.17 $\times$ ($\tau=5$)
	SymGS	11.6 $\times$	8.70 $\times$	4.75 $\times$	2.69 $\times$ ($\tau=5$)	2.40 $\times$ ($\tau=5$)	1.62 $\times$ ($\tau=5$)
	SOR (1.5)	7.36 $\times$	8.89 $\times$	2.58 $\times$	2.03 $\times$ ($\tau=5$)	2.40 $\times$ ($\tau=5$)	1.31 $\times$ ($\tau=5$)
AnisoDiff	Jacobi	3.19 $\times$	3.16 $\times$	1.44 $\times$	1.27 $\times$ ($\tau=5$)	1.25 $\times$ ($\tau=5$)	1.21 $\times$ ($\tau=50$)
	Jacobi (0.67)	3.43 $\times$	3.31 $\times$	1.43 $\times$	1.40 $\times$ ($\tau=5$)	1.34 $\times$ ($\tau=5$)	1.17 $\times$ ($\tau=10$)
	GS	7.80 $\times$	7.61 $\times$	4.05 $\times$	2.08 $\times$ ($\tau=5$)	1.99 $\times$ ($\tau=5$)	2.74 $\times$ ($\tau=5$)
	SymGS	12.1 $\times$	11.8 $\times$	3.71 $\times$	2.81 $\times$ ($\tau=5$)	2.72 $\times$ ($\tau=5$)	1.36 $\times$ ($\tau=5$)
	SOR (1.5)	7.85 $\times$	7.36 $\times$	2.40 $\times$	2.13 $\times$ ($\tau=5$)	2.04 $\times$ ($\tau=5$)	1.82 $\times$ ($\tau=10$)

Table 18: Iteration-based metrics of the main text on the 128×128 grid: relative error after $T = 300$ iterations and AUC of the relative error over T iterations, mean (standard error) over 64 instances; p -values are one-sided paired t -tests of the baseline’s AUC against the learned router’s.

Method	Poisson			ConvDiff			AnisoDiff		
	$\ e_h^{(T)}\ /\ u_h\ $	AUC	p	$\ e_h^{(T)}\ /\ u_h\ $	AUC	p	$\ e_h^{(T)}\ /\ u_h\ $	AUC	p
Jacobi-related solvers									
Solver only	7.99e-01 (4.5e-03)	2.68e+02 (8.9e-01)	$< 10^{-3}$	7.43e-01 (9.1e-03)	2.57e+02 (1.9e+00)	$< 10^{-3}$	9.96e-01 (1.5e-04)	2.99e+02 (2.3e-02)	$< 10^{-3}$
HINTS ($\tau=25$)	7.10e-06 (1.9e-06)	2.37e+01 (1.1e-02)	$< 10^{-3}$	1.55e-05 (4.2e-06)	2.36e+01 (2.7e-02)	$< 10^{-3}$	7.65e-07 (2.1e-07)	2.40e+01 (1.7e-04)	$< 10^{-3}$
Learned router (ours)	6.86e-06 (1.8e-06)	7.15e-03 (1.9e-03)	-	1.50e-05 (4.1e-06)	1.55e-02 (4.1e-03)	-	7.45e-07 (2.0e-07)	6.80e-04 (1.8e-04)	-
Cost-aware oracle	6.86e-06 (1.8e-06)	7.15e-03 (1.9e-03)	-	1.50e-05 (4.1e-06)	1.55e-02 (4.1e-03)	-	7.45e-07 (2.0e-07)	6.80e-04 (1.8e-04)	-
Jacobi (0.67)-related solvers									
Solver only	8.56e-01 (3.8e-03)	2.77e+02 (7.1e-01)	$< 10^{-3}$	8.11e-01 (7.9e-03)	2.69e+02 (1.5e+00)	$< 10^{-3}$	9.97e-01 (1.0e-04)	3.00e+02 (1.6e-02)	$< 10^{-3}$
HINTS ($\tau=25$)	1.15e-12 (1.0e-13)	2.38e+01 (8.3e-03)	$< 10^{-3}$	1.46e-12 (2.7e-13)	2.37e+01 (2.1e-02)	$< 10^{-3}$	3.34e-10 (4.6e-11)	2.40e+01 (1.2e-04)	$< 10^{-3}$
Learned router (ours)	7.57e-10 (2.9e-11)	2.49e-03 (6.3e-04)	-	7.96e-10 (1.6e-11)	5.39e-03 (1.4e-03)	-	8.97e-10 (7.4e-12)	2.82e-04 (6.6e-05)	-
Cost-aware oracle	8.19e-10 (1.2e-11)	2.49e-03 (6.3e-04)	-	8.18e-10 (1.3e-11)	5.39e-03 (1.4e-03)	-	8.96e-10 (7.4e-12)	2.82e-04 (6.6e-05)	-
GS-related solvers									
Solver only	6.53e-01 (5.2e-03)	2.42e+02 (1.2e+00)	$< 10^{-3}$	5.46e-01 (9.9e-03)	2.20e+02 (2.4e+00)	$< 10^{-3}$	9.92e-01 (2.7e-04)	2.99e+02 (4.3e-02)	$< 10^{-3}$
HINTS ($\tau=25$)	7.14e-10 (5.4e-11)	2.30e+01 (3.5e-02)	$< 10^{-3}$	3.64e-12 (2.3e-13)	2.32e+01 (5.4e-02)	$< 10^{-3}$	9.79e-10 (1.4e-12)	2.40e+01 (3.4e-04)	$< 10^{-3}$
Learned router (ours)	7.66e-10 (1.8e-11)	1.94e-03 (4.9e-04)	-	6.84e-10 (2.8e-11)	4.03e-03 (1.0e-03)	-	8.15e-10 (1.3e-11)	1.95e-04 (4.6e-05)	-
Cost-aware oracle	7.08e-10 (2.5e-11)	1.94e-03 (4.9e-04)	-	6.73e-10 (3.2e-11)	4.03e-03 (1.0e-03)	-	8.02e-10 (1.5e-11)	1.95e-04 (4.6e-05)	-
SymGS-related solvers									
Solver only	4.49e-01 (4.6e-03)	2.03e+02 (1.3e+00)	$< 10^{-3}$	3.77e-01 (9.1e-03)	1.84e+02 (2.7e+00)	$< 10^{-3}$	9.84e-01 (4.8e-04)	2.98e+02 (7.7e-02)	$< 10^{-3}$
HINTS ($\tau=25$)	7.14e-10 (5.4e-11)	2.30e+01 (3.5e-02)	$< 10^{-3}$	3.64e-12 (2.3e-13)	2.26e+01 (8.1e-02)	$< 10^{-3}$	7.48e-10 (3.2e-11)	2.40e+01 (6.5e-04)	$< 10^{-3}$
Learned router (ours)	6.50e-10 (2.4e-11)	1.46e-03 (3.7e-04)	-	5.19e-10 (3.4e-11)	3.16e-03 (8.0e-04)	-	7.53e-10 (2.2e-11)	1.23e-04 (2.9e-05)	-
Cost-aware oracle	6.32e-10 (2.5e-11)	1.46e-03 (3.7e-04)	-	5.00e-10 (3.2e-11)	3.16e-03 (8.0e-04)	-	7.54e-10 (1.8e-11)	1.23e-04 (2.9e-05)	-
SOR (1.5)-related solvers									
Solver only	3.09e-01 (3.5e-03)	1.72e+02 (1.3e+00)	$< 10^{-3}$	1.23e-01 (3.4e-03)	1.17e+02 (1.9e+00)	$< 10^{-3}$	9.76e-01 (6.8e-04)	2.96e+02 (1.1e-01)	$< 10^{-3}$
HINTS ($\tau=25$)	2.16e-10 (1.3e-11)	2.26e+01 (4.6e-02)	$< 10^{-3}$	8.06e-10 (2.3e-11)	2.12e+01 (1.2e-01)	$< 10^{-3}$	9.09e-10 (1.6e-11)	2.40e+01 (9.5e-04)	$< 10^{-3}$
Learned router (ours)	8.74e-10 (9.7e-12)	3.30e-03 (8.4e-04)	-	8.08e-10 (1.3e-11)	6.49e-03 (1.7e-03)	-	8.67e-10 (9.3e-12)	2.87e-04 (6.9e-05)	-
Cost-aware oracle	8.36e-10 (1.5e-11)	3.29e-03 (8.4e-04)	-	7.77e-10 (1.8e-11)	6.49e-03 (1.7e-03)	-	8.63e-10 (9.4e-12)	2.87e-04 (6.9e-05)	-

Table 19: Median number of iterations and corrector calls needed to reach $\varepsilon = h^2$ on the 128×128 grid.

Equation	Solver	HINTS ($\tau=25$)		Cost-aware oracle		Learned router	
		iters	NO calls	iters	NO calls	iters	NO calls
Poisson	Jacobi	25	1	2	1	2	1
	Jacobi (0.67)	25	1	2	1	2	1
	GS	26	1	2	1	2	1
	SymGS	26	1	2	1	2	1
	SOR (1.5)	50	2	2	1	2	1
ConvDiff	Jacobi	25	1	2	1	2	1
	Jacobi (0.67)	25	1	4	1	4	1
	GS	30	1	2	1	2	1
	SymGS	26	1	2	1	2	1
	SOR (1.5)	50	2	4	1	4	1
AnisoDiff	Jacobi	25	1	1	1	1	1
	Jacobi (0.67)	25	1	1	1	1	1
	GS	25	1	1	1	1	1
	SymGS	25	1	1	1	1	1
	SOR (1.5)	25	1	1	1	1	1

Table 20: Measured single-thread per-iteration costs (residual + update) on the 128×128 grid and the resulting macro-action sizes $m_j = \max\{1, \text{round}(u/c_j)\}$ (u : corrector call).

Equation	Solver	classical iteration	corrector iteration	m_{solver}	m_{NO}
Poisson	Jacobi	52 μs	635 μs	12	1
	Jacobi (0.67)	52 μs	639 μs	12	1
	GS	200 μs	643 μs	3	1
	SymGS	366 μs	635 μs	2	1
	SOR (1.5)	197 μs	635 μs	3	1
ConvDiff	Jacobi	72 μs	658 μs	9	1
	Jacobi (0.67)	71 μs	659 μs	9	1
	GS	234 μs	673 μs	3	1
	SymGS	378 μs	665 μs	2	1
	SOR (1.5)	223 μs	660 μs	3	1
AnisoDiff	Jacobi	63 μs	695 μs	11	1
	Jacobi (0.67)	64 μs	692 μs	11	1
	GS	215 μs	711 μs	3	1
	SymGS	382 μs	711 μs	2	1
	SOR (1.5)	219 μs	730 μs	3	1

Table 21: Poisson on the 128×128 grid: median wall-clock time to relative error ε for every policy and pairing (64 instances; paired median speedup over solver-only in parentheses; † n : n instances did not reach the tolerance within the iteration cap).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	536.0 ms	806.1 ms	1.15 s	1.36 s	1.62 s	2.14 s
	HINTS ($\tau=25$)	2.61 ms (193 $\times$)	2.61 ms (294 $\times$)	2.65 ms (412 $\times$)	2.89 ms (478 $\times$)	3.56 ms (447 $\times$)	226.1 ms (10.0 $\times$)
	HINTS ($\tau=10$)	1.42 ms (355 $\times$)	1.42 ms (543 $\times$)	1.46 ms (738 $\times$)	1.50 ms (864 $\times$)	3.89 ms (518 $\times$)	329.2 ms (6.94 $\times$)
	HINTS ($\tau=5$)	1.06 ms (463 $\times$)	1.06 ms (719 $\times$)	1.10 ms (984 $\times$)	1.15 ms (1153 $\times$)	6.30 ms (432 $\times$)	545.2 ms (4.19 $\times$)
	HINTS ($\tau=50$)	3.80 ms (122 $\times$)	3.80 ms (187 $\times$)	4.16 ms (253 $\times$)	4.22 ms (300 $\times$)	5.29 ms (322 $\times$)	182.8 ms (12.5 $\times$)
	Greedy oracle (Alg. 1)	773 μ s (668 $\times$)	803 μ s (996 $\times$)	854 μ s (1283 $\times$)	964 μ s (1350 $\times$)	2.94 ms (680 $\times$)	155.1 ms (14.4 $\times$)
	Cost-aware oracle	862 μ s (605 $\times$)	880 μ s (901 $\times$)	942 μ s (1199 $\times$)	1.02 ms (1268 $\times$)	2.94 ms (623 $\times$)	154.1 ms (13.9 $\times$)
	Learned router (ours)	884 μ s (597 $\times$)	918 μ s (868 $\times$)	987 μ s (1154 $\times$)	1.09 ms (1202 $\times$)	2.98 ms (655 $\times$)	151.3 ms (14.4 $\times$)
Jacobi (0.67)	Solver only	777.3 ms	1.16 s	1.62 s	1.93 s	2.33 s	3.16 s
	HINTS ($\tau=25$)	2.71 ms (290 $\times$)	2.71 ms (437 $\times$)	2.71 ms (621 $\times$)	2.71 ms (749 $\times$)	2.74 ms (874 $\times$)	5.16 ms (631 $\times$)
	HINTS ($\tau=10$)	1.46 ms (540 $\times$)	1.46 ms (819 $\times$)	1.46 ms (1161 $\times$)	1.50 ms (1335 $\times$)	1.61 ms (1467 $\times$)	3.03 ms (1058 $\times$)
	HINTS ($\tau=5$)	1.11 ms (713 $\times$)	1.11 ms (1074 $\times$)	1.13 ms (1464 $\times$)	1.17 ms (1681 $\times$)	1.91 ms (1332 $\times$)	4.09 ms (747 $\times$)
	HINTS ($\tau=50$)	4.42 ms (183 $\times$)	4.42 ms (278 $\times$)	4.42 ms (391 $\times$)	4.42 ms (468 $\times$)	4.43 ms (557 $\times$)	8.66 ms (377 $\times$)
	Greedy oracle (Alg. 1)	796 μ s (988 $\times$)	818 μ s (1455 $\times$)	915 μ s (1904 $\times$)	1.09 ms (1970 $\times$)	1.50 ms (1575 $\times$)	2.83 ms (1112 $\times$)
	Cost-aware oracle	805 μ s (981 $\times$)	820 μ s (1424 $\times$)	922 μ s (1836 $\times$)	1.09 ms (1892 $\times$)	1.53 ms (1684 $\times$)	2.93 ms (1106 $\times$)
	Learned router (ours)	822 μ s (958 $\times$)	839 μ s (1402 $\times$)	912 μ s (1863 $\times$)	1.10 ms (1869 $\times$)	1.51 ms (1650 $\times$)	2.93 ms (1093 $\times$)
GS	Solver only	854.7 ms	1.29 s	1.83 s	2.17 s	2.60 s	3.45 s
	HINTS ($\tau=25$)	6.43 ms (134 $\times$)	6.43 ms (203 $\times$)	6.65 ms (277 $\times$)	12.4 ms (177 $\times$)	12.6 ms (208 $\times$)	13.9 ms (250 $\times$)
	HINTS ($\tau=10$)	2.88 ms (296 $\times$)	2.88 ms (450 $\times$)	3.14 ms (574 $\times$)	5.65 ms (383 $\times$)	5.65 ms (458 $\times$)	8.38 ms (412 $\times$)
	HINTS ($\tau=5$)	1.72 ms (502 $\times$)	1.72 ms (759 $\times$)	1.99 ms (906 $\times$)	3.36 ms (645 $\times$)	3.63 ms (711 $\times$)	5.55 ms (621 $\times$)
	HINTS ($\tau=50$)	11.8 ms (73.1 $\times$)	11.8 ms (110 $\times$)	12.0 ms (152 $\times$)	20.2 ms (108 $\times$)	23.4 ms (111 $\times$)	23.7 ms (146 $\times$)
	Greedy oracle (Alg. 1)	814 μ s (1066 $\times$)	835 μ s (1563 $\times$)	960 μ s (1872 $\times$)	1.29 ms (1622 $\times$)	2.53 ms (999 $\times$)	4.34 ms (778 $\times$)
	Cost-aware oracle	818 μ s (1061 $\times$)	827 μ s (1571 $\times$)	953 μ s (1892 $\times$)	1.31 ms (1603 $\times$)	2.55 ms (990 $\times$)	4.37 ms (773 $\times$)
	Learned router (ours)	832 μ s (1041 $\times$)	850 μ s (1544 $\times$)	1.01 ms (1809 $\times$)	1.34 ms (1563 $\times$)	2.61 ms (969 $\times$)	4.45 ms (758 $\times$)
SymGS	Solver only	720.5 ms	1.08 s	1.52 s	1.81 s	2.17 s	2.91 s
	HINTS ($\tau=25$)	10.2 ms (71.3 $\times$)	10.2 ms (108 $\times$)	10.3 ms (149 $\times$)	11.8 ms (155 $\times$)	20.2 ms (109 $\times$)	20.2 ms (146 $\times$)
	HINTS ($\tau=10$)	4.33 ms (167 $\times$)	4.33 ms (252 $\times$)	4.53 ms (341 $\times$)	6.89 ms (266 $\times$)	8.48 ms (259 $\times$)	9.10 ms (325 $\times$)
	HINTS ($\tau=5$)	2.35 ms (308 $\times$)	2.35 ms (465 $\times$)	2.60 ms (585 $\times$)	4.38 ms (412 $\times$)	4.63 ms (477 $\times$)	6.94 ms (425 $\times$)
	HINTS ($\tau=50$)	19.6 ms (36.9 $\times$)	19.6 ms (55.9 $\times$)	19.6 ms (78.4 $\times$)	20.8 ms (88.2 $\times$)	39.1 ms (56.4 $\times$)	39.1 ms (75.4 $\times$)
	Greedy oracle (Alg. 1)	814 μ s (882 $\times$)	829 μ s (1312 $\times$)	1.09 ms (1395 $\times$)	1.44 ms (1285 $\times$)	2.20 ms (1029 $\times$)	3.87 ms (772 $\times$)
	Cost-aware oracle	817 μ s (883 $\times$)	829 μ s (1319 $\times$)	1.13 ms (1330 $\times$)	1.49 ms (1205 $\times$)	2.22 ms (1004 $\times$)	3.86 ms (766 $\times$)
	Learned router (ours)	843 μ s (856 $\times$)	853 μ s (1289 $\times$)	1.13 ms (1331 $\times$)	1.49 ms (1198 $\times$)	2.28 ms (981 $\times$)	3.88 ms (754 $\times$)
SOR (1.5)	Solver only	281.1 ms	424.4 ms	600.9 ms	713.1 ms	855.7 ms	1.15 s
	HINTS ($\tau=25$)	6.37 ms (44.3 $\times$)	6.37 ms (67.6 $\times$)	12.2 ms (49.4 $\times$)	12.5 ms (57.5 $\times$)	12.5 ms (69.3 $\times$)	18.5 ms (62.4 $\times$)
	HINTS ($\tau=10$)	2.85 ms (98.0 $\times$)	2.85 ms (148 $\times$)	5.59 ms (107 $\times$)	5.59 ms (127 $\times$)	5.83 ms (149 $\times$)	8.56 ms (135 $\times$)
	HINTS ($\tau=5$)	1.70 ms (167 $\times$)	1.72 ms (251 $\times$)	3.37 ms (180 $\times$)	3.41 ms (208 $\times$)	5.08 ms (168 $\times$)	9.38 ms (122 $\times$)
	HINTS ($\tau=50$)	11.9 ms (23.3 $\times$)	11.9 ms (35.6 $\times$)	17.0 ms (35.3 $\times$)	23.8 ms (30.1 $\times$)	23.8 ms (36.2 $\times$)	35.0 ms (32.7 $\times$)
	Greedy oracle (Alg. 1)	783 μ s (354 $\times$)	814 μ s (528 $\times$)	1.03 ms (583 $\times$)	1.74 ms (419 $\times$)	3.77 ms (236 $\times$)	7.48 ms (158 $\times$)
	Cost-aware oracle	784 μ s (357 $\times$)	800 μ s (527 $\times$)	1.02 ms (582 $\times$)	1.87 ms (380 $\times$)	3.94 ms (219 $\times$)	6.81 ms (168 $\times$)
	Learned router (ours)	804 μ s (343 $\times$)	823 μ s (511 $\times$)	1.02 ms (576 $\times$)	1.82 ms (390 $\times$)	3.88 ms (223 $\times$)	7.05 ms (160 $\times$)

Table 22: Convection–diffusion on the 128×128 grid: median wall-clock time to relative error ε for every policy and pairing (64 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	692.2 ms	1.07 s	1.51 s	1.80 s	2.16 s	2.90 s
	HINTS ($\tau=25$)	3.41 ms (205 $\times$)	3.41 ms (312 $\times$)	3.44 ms (436 $\times$)	3.46 ms (516 $\times$)	9.31 ms (306 $\times$)	384.4 ms (7.40 $\times$)
	HINTS ($\tau=10$)	1.76 ms (392 $\times$)	1.76 ms (598 $\times$)	1.80 ms (829 $\times$)	1.88 ms (972 $\times$)	12.8 ms (229 $\times$)	536.2 ms (5.30 $\times$)
	HINTS ($\tau=5$)	1.26 ms (539 $\times$)	1.26 ms (821 $\times$)	1.28 ms (1145 $\times$)	1.47 ms (1236 $\times$)	20.4 ms (140 $\times$)	838.4 ms (3.39 $\times$)
	HINTS ($\tau=50$)	5.76 ms (121 $\times$)	5.76 ms (184 $\times$)	5.76 ms (261 $\times$)	5.85 ms (307 $\times$)	9.40 ms (250 $\times$)	321.5 ms (8.89 $\times$)
	Greedy oracle (Alg. 1)	851 μ s (792 $\times$)	874 μ s (1198 $\times$)	992 μ s (1532 $\times$)	1.31 ms (1317 $\times$)	7.99 ms (332 $\times$)	281.0 ms (10.2 $\times$)
	Cost-aware oracle	961 μ s (706 $\times$)	979 μ s (1073 $\times$)	1.15 ms (1319 $\times$)	1.43 ms (1206 $\times$)	8.26 ms (337 $\times$)	281.8 ms (10.1 $\times$)
	Learned router (ours)	1.00 ms (694 $\times$)	1.01 ms (1062 $\times$)	1.14 ms (1317 $\times$)	1.52 ms (1146 $\times$)	8.15 ms (313 $\times$)	292.7 ms (9.72 $\times$)
Jacobi (0.67)	Solver only	1.01 s	1.55 s	2.22 s	2.65 s	3.21 s	4.32 s
	HINTS ($\tau=25$)	3.40 ms (296 $\times$)	3.40 ms (461 $\times$)	3.40 ms (656 $\times$)	3.40 ms (784 $\times$)	3.47 ms (926 $\times$)	6.53 ms (663 $\times$)
	HINTS ($\tau=10$)	1.77 ms (572 $\times$)	1.77 ms (880 $\times$)	1.77 ms (1234 $\times$)	1.79 ms (1445 $\times$)	2.06 ms (1575 $\times$)	3.79 ms (1139 $\times$)
	HINTS ($\tau=5$)	1.28 ms (782 $\times$)	1.28 ms (1208 $\times$)	1.31 ms (1670 $\times$)	1.49 ms (1760 $\times$)	2.62 ms (1221 $\times$)	5.62 ms (764 $\times$)
	HINTS ($\tau=50$)	5.76 ms (174 $\times$)	5.76 ms (269 $\times$)	5.76 ms (386 $\times$)	5.76 ms (461 $\times$)	5.81 ms (551 $\times$)	11.3 ms (381 $\times$)
	Greedy oracle (Alg. 1)	849 μ s (1185 $\times$)	885 μ s (1789 $\times$)	1.10 ms (2036 $\times$)	1.38 ms (1877 $\times$)	1.91 ms (1668 $\times$)	3.56 ms (1177 $\times$)
	Cost-aware oracle	856 μ s (1182 $\times$)	878 μ s (1766 $\times$)	1.08 ms (1974 $\times$)	1.37 ms (1885 $\times$)	1.88 ms (1624 $\times$)	3.61 ms (1164 $\times$)
	Learned router (ours)	893 μ s (1145 $\times$)	912 μ s (1741 $\times$)	1.10 ms (1983 $\times$)	1.43 ms (1817 $\times$)	1.94 ms (1620 $\times$)	3.70 ms (1139 $\times$)
GS	Solver only	749.7 ms	1.15 s	1.63 s	1.94 s	2.34 s	3.14 s
	HINTS ($\tau=25$)	7.03 ms (107 $\times$)	7.03 ms (164 $\times$)	8.27 ms (198 $\times$)	13.7 ms (141 $\times$)	13.7 ms (170 $\times$)	20.5 ms (152 $\times$)
	HINTS ($\tau=10$)	3.21 ms (233 $\times$)	3.21 ms (356 $\times$)	5.02 ms (330 $\times$)	6.34 ms (312 $\times$)	6.35 ms (370 $\times$)	9.41 ms (337 $\times$)
	HINTS ($\tau=5$)	1.90 ms (391 $\times$)	1.90 ms (604 $\times$)	3.60 ms (453 $\times$)	3.66 ms (529 $\times$)	4.28 ms (547 $\times$)	6.24 ms (502 $\times$)
	HINTS ($\tau=50$)	13.3 ms (56.5 $\times$)	13.3 ms (86.7 $\times$)	14.2 ms (115 $\times$)	26.4 ms (74.0 $\times$)	26.4 ms (89.1 $\times$)	28.2 ms (112 $\times$)
	Greedy oracle (Alg. 1)	863 μ s (855 $\times$)	887 μ s (1283 $\times$)	1.23 ms (1341 $\times$)	1.75 ms (1108 $\times$)	3.28 ms (712 $\times$)	5.32 ms (587 $\times$)
	Cost-aware oracle	869 μ s (838 $\times$)	909 μ s (1262 $\times$)	1.32 ms (1222 $\times$)	1.84 ms (1048 $\times$)	3.26 ms (717 $\times$)	5.46 ms (582 $\times$)
	Learned router (ours)	892 μ s (819 $\times$)	906 μ s (1239 $\times$)	1.35 ms (1243 $\times$)	1.89 ms (1060 $\times$)	3.29 ms (704 $\times$)	5.59 ms (570 $\times$)
SymGS	Solver only	716.5 ms	1.10 s	1.56 s	1.85 s	2.23 s	2.99 s
	HINTS ($\tau=25$)	10.7 ms (67.4 $\times$)	10.7 ms (103 $\times$)	11.1 ms (141 $\times$)	21.1 ms (88.1 $\times$)	21.2 ms (106 $\times$)	21.6 ms (139 $\times$)
	HINTS ($\tau=10$)	4.55 ms (157 $\times$)	4.55 ms (240 $\times$)	5.02 ms (311 $\times$)	8.94 ms (207 $\times$)	8.94 ms (249 $\times$)	13.2 ms (226 $\times$)
	HINTS ($\tau=5$)	2.52 ms (287 $\times$)	2.52 ms (436 $\times$)	2.96 ms (529 $\times$)	4.92 ms (378 $\times$)	4.93 ms (455 $\times$)	7.35 ms (408 $\times$)
	HINTS ($\tau=50$)	20.7 ms (34.5 $\times$)	20.7 ms (53.0 $\times$)	21.1 ms (73.6 $\times$)	40.9 ms (45.5 $\times$)	41.4 ms (54.1 $\times$)	41.4 ms (72.4 $\times$)
	Greedy oracle (Alg. 1)	895 μ s (785 $\times$)	911 μ s (1198 $\times$)	1.25 ms (1254 $\times$)	1.66 ms (1130 $\times$)	2.97 ms (752 $\times$)	4.91 ms (618 $\times$)
	Cost-aware oracle	886 μ s (795 $\times$)	905 μ s (1212 $\times$)	1.27 ms (1236 $\times$)	1.73 ms (1084 $\times$)	3.14 ms (693 $\times$)	4.48 ms (657 $\times$)
	Learned router (ours)	908 μ s (772 $\times$)	924 μ s (1182 $\times$)	1.28 ms (1211 $\times$)	1.72 ms (1086 $\times$)	3.20 ms (698 $\times$)	4.54 ms (647 $\times$)
SOR (1.5)	Solver only	177.4 ms	270.0 ms	383.6 ms	457.3 ms	550.1 ms	737.6 ms
	HINTS ($\tau=25$)	6.91 ms (25.3 $\times$)	6.92 ms (38.5 $\times$)	13.6 ms (28.0 $\times$)	13.6 ms (33.4 $\times$)	13.6 ms (40.1 $\times$)	20.2 ms (36.1 $\times$)
	HINTS ($\tau=10$)	3.16 ms (54.9 $\times$)	3.27 ms (82.9 $\times$)	6.21 ms (61.4 $\times$)	6.21 ms (73.1 $\times$)	9.20 ms (59.6 $\times$)	10.7 ms (68.5 $\times$)
	HINTS ($\tau=5$)	1.85 ms (92.3 $\times$)	1.98 ms (137 $\times$)	3.64 ms (104 $\times$)	4.36 ms (107 $\times$)	5.88 ms (93.1 $\times$)	9.62 ms (76.0 $\times$)
	HINTS ($\tau=50$)	13.4 ms (13.2 $\times$)	13.4 ms (20.0 $\times$)	26.6 ms (14.5 $\times$)	26.6 ms (17.2 $\times$)	26.6 ms (20.7 $\times$)	39.8 ms (18.7 $\times$)
	Greedy oracle (Alg. 1)	876 μ s (199 $\times$)	919 μ s (300 $\times$)	1.39 ms (273 $\times$)	2.43 ms (188 $\times$)	4.34 ms (130 $\times$)	7.66 ms (98.8 $\times$)
	Cost-aware oracle	900 μ s (198 $\times$)	922 μ s (294 $\times$)	1.63 ms (240 $\times$)	2.72 ms (167 $\times$)	4.55 ms (121 $\times$)	7.97 ms (93.9 $\times$)
	Learned router (ours)	910 μ s (195 $\times$)	934 μ s (294 $\times$)	1.59 ms (238 $\times$)	2.67 ms (168 $\times$)	4.50 ms (120 $\times$)	8.18 ms (91.7 $\times$)

Table 23: Anisotropic diffusion on the 128×128 grid: median wall-clock time to relative error ε for every policy and pairing (64 instances). The point smoothers alone do not reach h^2 within 60,000 iterations (their times are lower bounds).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	$>4.62 \text{ s}^{164}$	$>4.62 \text{ s}^{164}$	$>4.62 \text{ s}^{164}$	$>4.62 \text{ s}^{164}$	$>4.62 \text{ s}^{164}$	$>4.62 \text{ s}^{164}$
	HINTS ($\tau=25$)	2.90 ms ($> 10^3 \times$)	2.90 ms ($> 10^3 \times$)	2.90 ms ($> 10^3 \times$)	2.92 ms ($> 10^3 \times$)	2.98 ms ($> 10^3 \times$)	67.6 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	1.55 ms ($> 10^3 \times$)	1.55 ms ($> 10^3 \times$)	1.55 ms ($> 10^3 \times$)	1.56 ms ($> 10^3 \times$)	1.64 ms ($> 10^3 \times$)	101.0 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	1.15 ms ($> 10^3 \times$)	1.15 ms ($> 10^3 \times$)	1.15 ms ($> 10^3 \times$)	1.16 ms ($> 10^3 \times$)	1.34 ms ($> 10^3 \times$)	165.7 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	4.73 ms ($> 10^3 \times$)	4.73 ms ($> 10^3 \times$)	4.73 ms ($> 10^3 \times$)	4.79 ms ($> 10^3 \times$)	4.86 ms ($> 10^3 \times$)	58.8 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	801 μs ($> 10^3 \times$)	801 μs ($> 10^3 \times$)	826 μs ($> 10^3 \times$)	872 μs ($> 10^3 \times$)	1.66 ms ($> 10^3 \times$)	49.6 ms ($> 10^3 \times$)
	Cost-aware oracle	876 μs ($> 10^3 \times$)	876 μs ($> 10^3 \times$)	892 μs ($> 10^3 \times$)	925 μs ($> 10^3 \times$)	1.45 ms ($> 10^3 \times$)	49.9 ms ($> 10^3 \times$)
	Learned router (ours)	903 μs ($> 10^3 \times$)	903 μs ($> 10^3 \times$)	911 μs ($> 10^3 \times$)	975 μs ($> 10^3 \times$)	1.55 ms ($> 10^3 \times$)	55.6 ms ($> 10^3 \times$)
Jacobi (0.67)	Solver only	$>4.52 \text{ s}^{164}$	$>4.52 \text{ s}^{164}$	$>4.52 \text{ s}^{164}$	$>4.52 \text{ s}^{164}$	$>4.52 \text{ s}^{164}$	$>4.52 \text{ s}^{164}$
	HINTS ($\tau=25$)	2.85 ms ($> 10^3 \times$)	2.85 ms ($> 10^3 \times$)	2.85 ms ($> 10^3 \times$)	2.85 ms ($> 10^3 \times$)	2.89 ms ($> 10^3 \times$)	5.46 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	1.52 ms ($> 10^3 \times$)	1.52 ms ($> 10^3 \times$)	1.52 ms ($> 10^3 \times$)	1.54 ms ($> 10^3 \times$)	1.67 ms ($> 10^3 \times$)	4.21 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	1.16 ms ($> 10^3 \times$)	1.16 ms ($> 10^3 \times$)	1.18 ms ($> 10^3 \times$)	1.22 ms ($> 10^3 \times$)	2.10 ms ($> 10^3 \times$)	6.89 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	4.72 ms ($> 10^3 \times$)	4.72 ms ($> 10^3 \times$)	4.72 ms ($> 10^3 \times$)	4.72 ms ($> 10^3 \times$)	4.73 ms ($> 10^3 \times$)	9.22 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	787 μs ($> 10^3 \times$)	787 μs ($> 10^3 \times$)	810 μs ($> 10^3 \times$)	875 μs ($> 10^3 \times$)	1.83 ms ($> 10^3 \times$)	3.70 ms ($> 10^3 \times$)
	Cost-aware oracle	802 μs ($> 10^3 \times$)	802 μs ($> 10^3 \times$)	823 μs ($> 10^3 \times$)	895 μs ($> 10^3 \times$)	1.61 ms ($> 10^3 \times$)	3.81 ms ($> 10^3 \times$)
	Learned router (ours)	831 μs ($> 10^3 \times$)	831 μs ($> 10^3 \times$)	850 μs ($> 10^3 \times$)	881 μs ($> 10^3 \times$)	1.62 ms ($> 10^3 \times$)	3.86 ms ($> 10^3 \times$)
GS	Solver only	$>13.63 \text{ s}^{164}$	$>13.63 \text{ s}^{164}$	$>13.63 \text{ s}^{164}$	$>13.63 \text{ s}^{164}$	$>13.63 \text{ s}^{164}$	$>13.63 \text{ s}^{164}$
	HINTS ($\tau=25$)	6.48 ms ($> 10^3 \times$)	6.48 ms ($> 10^3 \times$)	6.48 ms ($> 10^3 \times$)	6.50 ms ($> 10^3 \times$)	12.1 ms ($> 10^3 \times$)	23.7 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	2.93 ms ($> 10^3 \times$)	2.93 ms ($> 10^3 \times$)	2.93 ms ($> 10^3 \times$)	2.95 ms ($> 10^3 \times$)	5.60 ms ($> 10^3 \times$)	18.4 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	1.73 ms ($> 10^3 \times$)	1.73 ms ($> 10^3 \times$)	1.74 ms ($> 10^3 \times$)	1.78 ms ($> 10^3 \times$)	3.42 ms ($> 10^3 \times$)	16.5 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	12.1 ms ($> 10^3 \times$)	12.1 ms ($> 10^3 \times$)	12.1 ms ($> 10^3 \times$)	12.1 ms ($> 10^3 \times$)	16.5 ms ($> 10^3 \times$)	29.8 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	812 μs ($> 10^3 \times$)	812 μs ($> 10^3 \times$)	831 μs ($> 10^3 \times$)	938 μs ($> 10^3 \times$)	2.17 ms ($> 10^3 \times$)	5.87 ms ($> 10^3 \times$)
	Cost-aware oracle	802 μs ($> 10^3 \times$)	802 μs ($> 10^3 \times$)	833 μs ($> 10^3 \times$)	961 μs ($> 10^3 \times$)	2.10 ms ($> 10^3 \times$)	5.33 ms ($> 10^3 \times$)
	Learned router (ours)	827 μs ($> 10^3 \times$)	827 μs ($> 10^3 \times$)	852 μs ($> 10^3 \times$)	987 μs ($> 10^3 \times$)	2.06 ms ($> 10^3 \times$)	5.40 ms ($> 10^3 \times$)
SymGS	Solver only	$>23.15 \text{ s}^{164}$	$>23.15 \text{ s}^{164}$	$>23.15 \text{ s}^{164}$	$>23.15 \text{ s}^{164}$	$>23.15 \text{ s}^{164}$	$>23.15 \text{ s}^{164}$
	HINTS ($\tau=25$)	10.3 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$)	10.4 ms ($> 10^3 \times$)	19.9 ms ($> 10^3 \times$)	20.5 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	4.41 ms ($> 10^3 \times$)	4.41 ms ($> 10^3 \times$)	4.41 ms ($> 10^3 \times$)	4.44 ms ($> 10^3 \times$)	8.39 ms ($> 10^3 \times$)	9.56 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	2.40 ms ($> 10^3 \times$)	2.40 ms ($> 10^3 \times$)	2.40 ms ($> 10^3 \times$)	2.46 ms ($> 10^3 \times$)	4.69 ms ($> 10^3 \times$)	7.45 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	19.8 ms ($> 10^3 \times$)	19.8 ms ($> 10^3 \times$)	19.8 ms ($> 10^3 \times$)	19.9 ms ($> 10^3 \times$)	37.2 ms ($> 10^3 \times$)	39.4 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	811 μs ($> 10^3 \times$)	811 μs ($> 10^3 \times$)	832 μs ($> 10^3 \times$)	947 μs ($> 10^3 \times$)	1.95 ms ($> 10^3 \times$)	5.63 ms ($> 10^3 \times$)
	Cost-aware oracle	826 μs ($> 10^3 \times$)	826 μs ($> 10^3 \times$)	844 μs ($> 10^3 \times$)	961 μs ($> 10^3 \times$)	2.03 ms ($> 10^3 \times$)	5.59 ms ($> 10^3 \times$)
	Learned router (ours)	856 μs ($> 10^3 \times$)	856 μs ($> 10^3 \times$)	871 μs ($> 10^3 \times$)	929 μs ($> 10^3 \times$)	2.02 ms ($> 10^3 \times$)	5.47 ms ($> 10^3 \times$)
SOR (1.5)	Solver only	$>13.80 \text{ s}^{163}$	$>13.80 \text{ s}^{164}$	$>13.80 \text{ s}^{164}$	$>13.80 \text{ s}^{164}$	$>13.80 \text{ s}^{164}$	$>13.80 \text{ s}^{164}$
	HINTS ($\tau=25$)	6.62 ms ($> 10^3 \times$)	6.62 ms ($> 10^3 \times$)	6.62 ms ($> 10^3 \times$)	7.57 ms ($> 10^3 \times$)	13.0 ms ($> 10^3 \times$)	17.9 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	3.01 ms ($> 10^3 \times$)	3.01 ms ($> 10^3 \times$)	3.01 ms ($> 10^3 \times$)	4.38 ms ($> 10^3 \times$)	5.92 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	1.78 ms ($> 10^3 \times$)	1.78 ms ($> 10^3 \times$)	1.80 ms ($> 10^3 \times$)	3.41 ms ($> 10^3 \times$)	4.09 ms ($> 10^3 \times$)	13.6 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	12.2 ms ($> 10^3 \times$)	12.2 ms ($> 10^3 \times$)	12.2 ms ($> 10^3 \times$)	12.5 ms ($> 10^3 \times$)	24.3 ms ($> 10^3 \times$)	24.3 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	813 μs ($> 10^3 \times$)	813 μs ($> 10^3 \times$)	843 μs ($> 10^3 \times$)	994 μs ($> 10^3 \times$)	2.64 ms ($> 10^3 \times$)	7.60 ms ($> 10^3 \times$)
	Cost-aware oracle	823 μs ($> 10^3 \times$)	823 μs ($> 10^3 \times$)	862 μs ($> 10^3 \times$)	974 μs ($> 10^3 \times$)	2.61 ms ($> 10^3 \times$)	6.71 ms ($> 10^3 \times$)
	Learned router (ours)	845 μs ($> 10^3 \times$)	845 μs ($> 10^3 \times$)	869 μs ($> 10^3 \times$)	966 μs ($> 10^3 \times$)	2.64 ms ($> 10^3 \times$)	7.68 ms ($> 10^3 \times$)

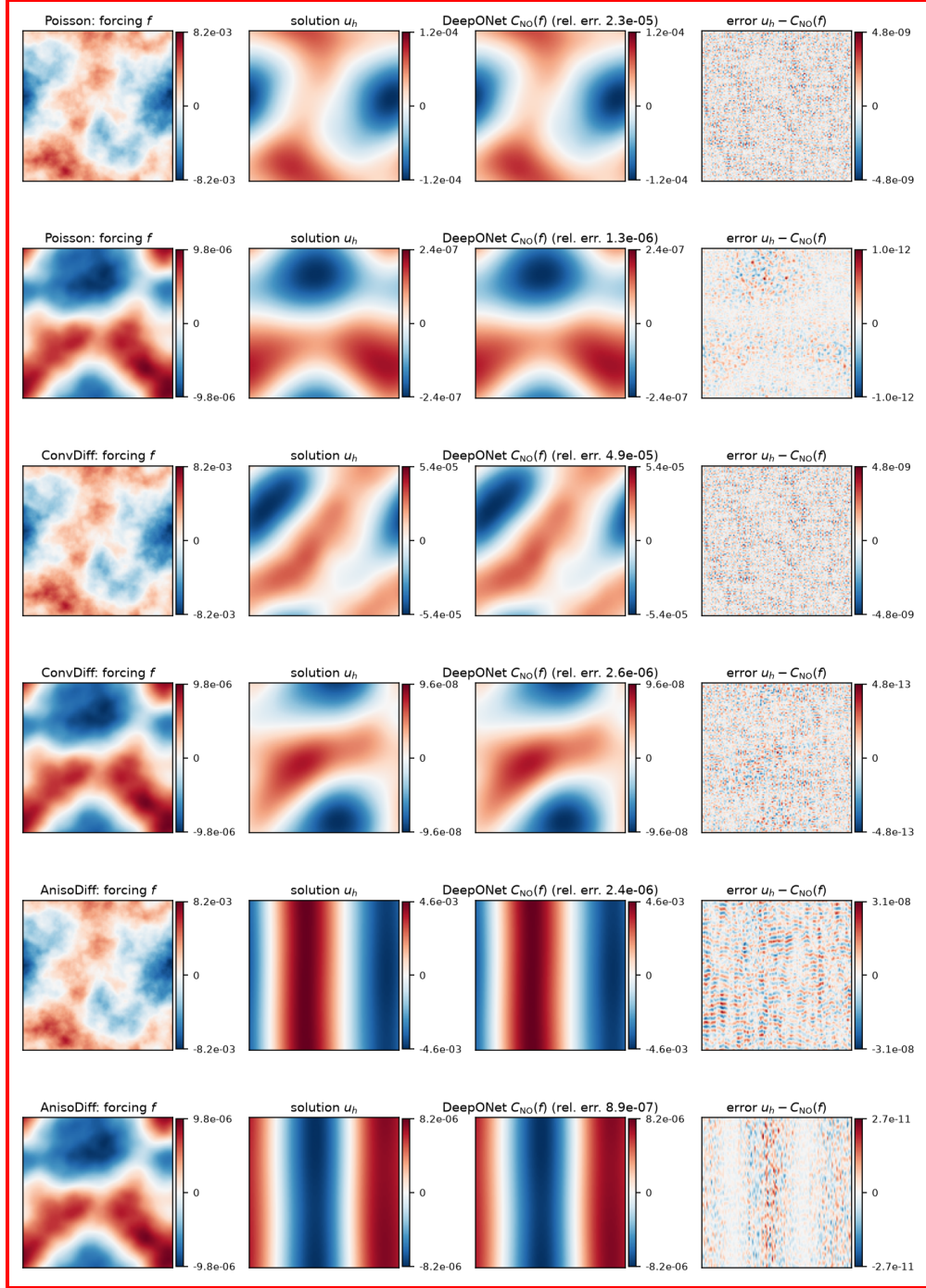


Figure 12: One-shot predictions of the corrector on two test forcings per equation (rows): forcing, exact discrete solution, DeepONet output $C_{NO}(f)$ and its error. The corrector solves the band $|k| \leq 31$ to $\approx 10^{-6}$; the remaining error is the high-frequency content of the solution outside the band, which the classical smoother removes in a few sweeps.

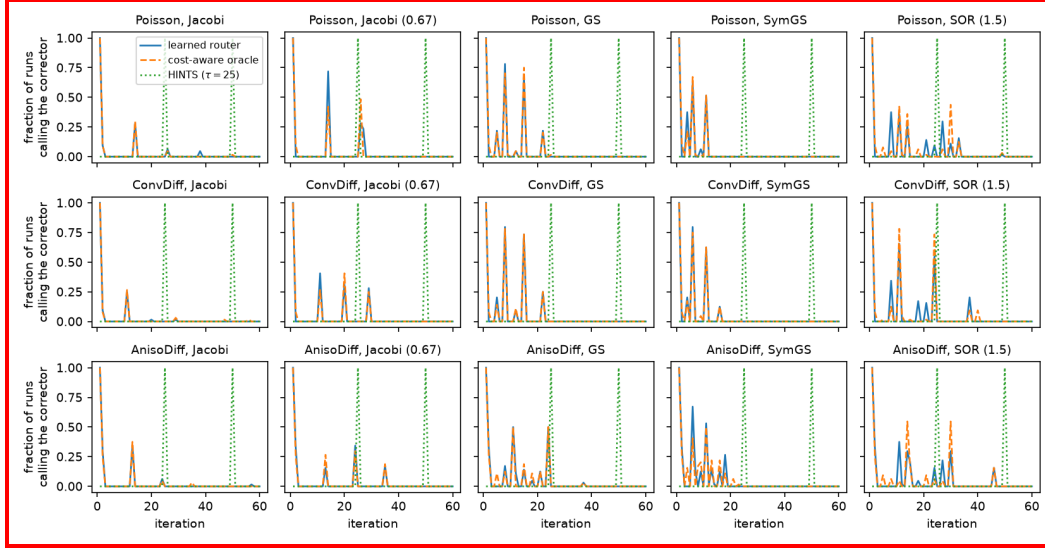


Figure 13: Corrector usage: fraction of test runs that execute a corrector call at each iteration, for the learned router, the cost-aware oracle and HINTS ($\tau = 25$). The router calls the corrector once at the start, spends a solver-dependent number of sweeps on the high-frequency band, and calls it again only when the smooth error has re-emerged; HINTS waits 24 sweeps before its first call.

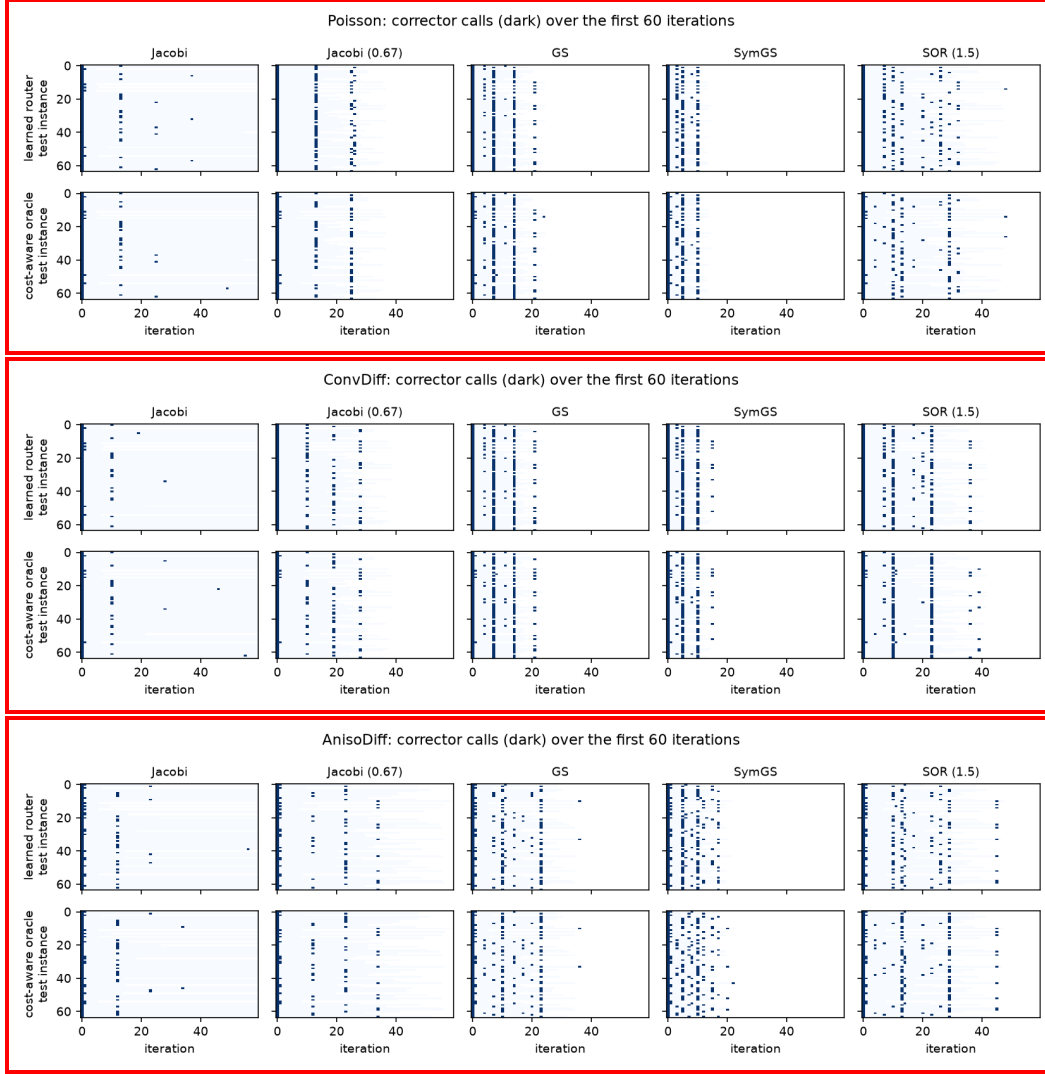


Figure 14: Routing decisions (dark: corrector call) of the learned router (top rows) and the cost-aware oracle (bottom rows) over the first 60 iterations of the stored test instances, for every pairing on the 128^2 grid (Poisson, convection–diffusion and anisotropic diffusion).

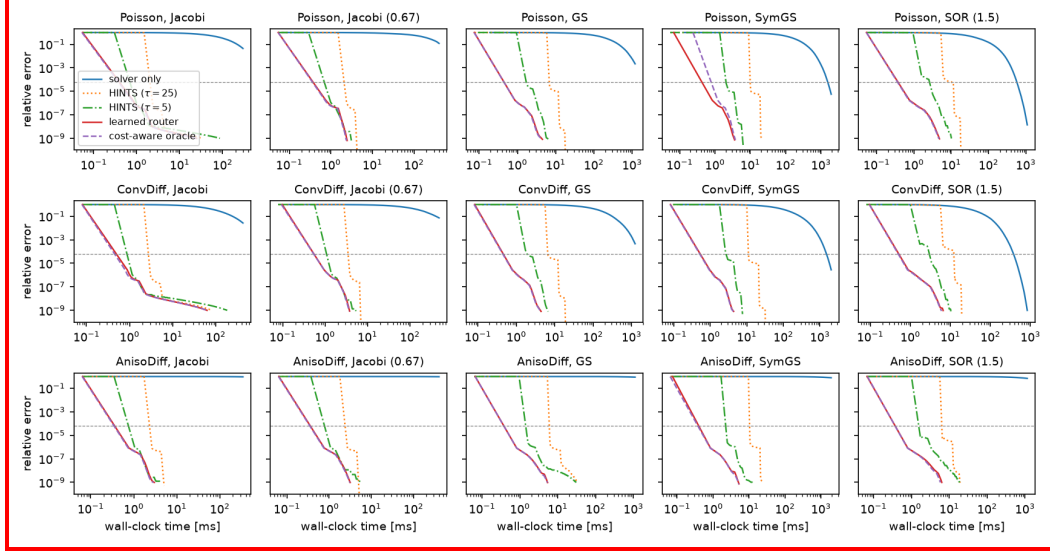


Figure 15: Relative error against charged wall-clock time for one test instance per pairing. The dashed horizontal line is $\varepsilon = h^2$.

963 E.3 Solver ensembles

964 The ensemble experiments of Section 7 are repeated in wall-clock terms with the cost-aware router
 965 over $\text{NO} \cup \mathcal{W}$ for the same four ensembles $\mathcal{W}$ (macro-action sizes are set from the measured costs
 966 of all members, $m_j = \text{round}(c_{\max}/c_j)$). HINTS does not apply to ensembles, so the comparison
 967 is against every member solver alone and against the best pairwise router $\text{Router}(\text{NO} \cup \{\text{solver}\})$,
 968 $\text{solver} \in \mathcal{W}$, from Table 2.

969 **Results.** Table 24 shows that the ensemble router is competitive with, but does not systematically
 970 improve on, the best pairwise router of its members: across the 12 ensemble–equation cells the ratio
 971 of median times to h^2 (best pairwise router over ensemble router) ranges from $0.82\times$ to $1.14\times$, i.e.,
 972 within about $\pm 20\%$, while the ensemble router is $380\times \rightarrow 10^3\times$ faster than the fastest member
 973 solver alone (on anisotropic diffusion no member reaches h^2 within the cap), and it does not need
 974 to know in advance which member is fastest (the best pairwise member differs between Jacobi,
 975 damped Jacobi and Gauss–Seidel across cells). On anisotropic diffusion the ensemble router is
 976 within 3% of the best pairwise router in all four cells, and the gap to its oracle is at most 4%. The
 977 cost-aware *oracle* over the full ensemble is faster than every pairwise router in every cell (Table 24),
 978 so the ensembles do contain exploitable diversity; the gap between the ensemble router and its oracle
 979 (significant in the larger ensembles, Table 45) is imitation error. It has a clear cause: with several
 980 cost-equalised smoothers in the ensemble the oracle’s choice among them depends on the spectral
 981 content of the current error, which the router’s residual-norm history features only reflect indirectly,
 982 and the surrogate’s weights are small for such near-ties, so the router agrees with the oracle on only
 983 55–83% of its own decisions (against 88–99% for the pairwise routers) and occasionally picks a
 984 smoother that is a few percent worse per unit cost. Deciding *when* to call the corrector—the decision
 985 that dominates the wall-clock gains—is learned equally well in the ensembles. Table 25 lists how
 986 the iterations up to the h^2 crossing are distributed: the corrector is called once (or twice) and the
 987 remaining iterations go to whichever smoother the router selects for the high-frequency band, with an
 988 instance-dependent mix, as in Section D.6.

Table 24: Ensembles: median wall-clock time to $\varepsilon = h^2$ on the 128×128 grid (64 instances). The ensemble router is compared with the fastest member solver alone, with the fastest pairwise router of its members (paired median speedup in parentheses), and with the cost-aware oracle over the same ensemble.

Equation	$\mathcal{W}$	Best solver only	Best pairwise router	Router($\text{NO} \cup \mathcal{W}$)	Oracle($\text{NO} \cup \mathcal{W}$)
Poisson	$\{Jacobi, GS\}$	1.15 s (Jacobi)	987 μs (Jacobi)	964 μs (1.09 $\times$ vs pairwise)	834 μs
	$\{Jacobi, GS, SymGS\}$	1.15 s (Jacobi)	987 μs (Jacobi)	951 μs (1.07 $\times$ vs pairwise)	854 μs
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	1.15 s (Jacobi)	912 μs (Jacobi (0.67))	868 μs (0.98 $\times$ vs pairwise)	842 μs
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	600.9 ms (SOR (1.5))	912 μs (Jacobi (0.67))	1.12 ms (0.86 $\times$ vs pairwise)	900 μs
ConvDiff	$\{Jacobi, GS\}$	1.51 s (Jacobi)	1.14 ms (Jacobi)	1.25 ms (1.09 $\times$ vs pairwise)	974 μs
	$\{Jacobi, GS, SymGS\}$	1.51 s (Jacobi)	1.14 ms (Jacobi)	1.01 ms (1.16 $\times$ vs pairwise)	996 μs
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	1.51 s (Jacobi)	1.10 ms (Jacobi (0.67))	996 μs (1.00 $\times$ vs pairwise)	1.07 ms
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	383.6 ms (SOR (1.5))	1.10 ms (Jacobi (0.67))	1.01 ms (1.06 $\times$ vs pairwise)	1.08 ms
AnisoDiff	$\{Jacobi, GS\}$	>4.62 s [†] (none)	852 μs (GS)	898 μs (1.01 $\times$ vs pairwise)	870 μs
	$\{Jacobi, GS, SymGS\}$	>4.62 s [†] (none)	852 μs (GS)	880 μs (1.03 $\times$ vs pairwise)	857 μs
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	>4.52 s [†] (none)	850 μs (Jacobi (0.67))	880 μs (1.00 $\times$ vs pairwise)	858 μs
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	>4.52 s [†] (none)	850 μs (Jacobi (0.67))	890 μs (1.00 $\times$ vs pairwise)	863 μs

Table 25: Ensembles: average fraction of iterations (up to the h^2 crossing) spent in each component by the learned router, mean (standard deviation) over 64 instances; “-” marks components not in the ensemble.

Equation	$\mathcal{W}$	Jacobi	GS	SymGS	Jacobi (0.67)	SOR (1.5)	DeepONet
Poisson	$\{Jacobi, GS\}$	0.228 (0.365)	0.144 (0.216)	-	-	-	0.628 (0.386)
	$\{Jacobi, GS, SymGS\}$	0.237 (0.337)	0.043 (0.134)	0.086 (0.165)	-	-	0.634 (0.378)
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	0.235 (0.297)	0.008 (0.062)	0.101 (0.207)	0.000 (0.000)	-	0.656 (0.352)
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	0.094 (0.195)	0.020 (0.051)	0.182 (0.265)	0.046 (0.125)	0.007 (0.023)	0.650 (0.366)
ConvDiff	$\{Jacobi, GS\}$	0.089 (0.255)	0.310 (0.337)	-	-	-	0.600 (0.373)
	$\{Jacobi, GS, SymGS\}$	0.279 (0.354)	0.067 (0.156)	0.055 (0.103)	-	-	0.599 (0.374)
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	0.305 (0.340)	0.000 (0.000)	0.022 (0.038)	0.110 (0.214)	-	0.563 (0.410)
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	0.166 (0.302)	0.028 (0.127)	0.000 (0.000)	0.233 (0.375)	0.000 (0.000)	0.573 (0.399)
AnisoDiff	$\{Jacobi, GS\}$	0.173 (0.320)	0.000 (0.000)	-	-	-	0.827 (0.320)
	$\{Jacobi, GS, SymGS\}$	0.052 (0.153)	0.000 (0.000)	0.097 (0.229)	-	-	0.851 (0.273)
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.185 (0.339)	-	0.815 (0.339)
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.185 (0.339)	0.000 (0.000)	0.815 (0.339)

989 E.4 Stronger classical baselines, larger grids and overheads

990 The comparisons above are against the stationary solvers of Section 7. This section adds the classical
991 methods a practitioner would actually reach for on these problems, scales the grid up to 512×512 ,
992 and accounts explicitly for every overhead.

993 **Baselines.** All baselines run on the same $O(N^2)$ stencil with the same accounting (first iteration at
994 which the true error is below ε ; timed replay of exactly that many iterations). (i) *FFT direct solve*: for
995 the constant-coefficient periodic problems used here the discrete operator is diagonalised by the FFT,
996 so the system can be solved directly with two FFTs; this is the reference solver we use to compute
997 ground truth and it is included as the lower bound on the time any iterative method can achieve on this
998 problem class. (ii) *Geometric multigrid*: V(2,2)-cycles with lexicographic Gauss–Seidel smoothing,
999 full-weighting restriction, bilinear prolongation, rediscrretised coarse operators and a direct solve on
1000 the 32×32 coarsest grid (error contraction ≈ 0.03 per cycle). A V-cycle is a stationary iteration, so
1001 multigrid is also used as an ensemble member below. (iii) *Krylov methods* (SciPy implementations
1002 driven by the stencil matvec): conjugate gradients, CG preconditioned by symmetric GS and by a
1003 symmetric V-cycle for Poisson; BiCGSTAB, multigrid-preconditioned BiCGSTAB and restarted
1004 GMRES(20) for convection–diffusion (GMRES is accounted at the granularity of its restart cycles).

Table 26: Strong classical baselines on the 128×128 grid: median time to relative error ε over 64 instances (paired median speedup of the learned router with its best stationary pairing over each method in parentheses; > 1 means the router is faster) and median iterations to h^2 . The FFT solve is a direct method for this problem class and bounds every iterative solver from below.

Equation	Method	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$	iters to h^2
Poisson	FFT direct solve	317 μ s (0.36 $\times$)	317 μ s (0.36 $\times$)	317 μ s (0.36 $\times$)	317 μ s (0.36 $\times$)	1
	Multigrid V(2,2) alone	2.75 ms (2.96 $\times$)	3.99 ms (3.62 $\times$)	5.47 ms (5.42 $\times$)	7.97 ms (8.11 $\times$)	3
	CG	9.37 ms (9.44 $\times$)	11.5 ms (11.8 $\times$)	14.2 ms (14.9 $\times$)	16.3 ms (17.6 $\times$)	162
	PCG (SymGS)	20.3 ms (20.5 $\times$)	30.8 ms (32.4 $\times$)	40.3 ms (44.0 $\times$)	52.5 ms (56.4 $\times$)	75
	PCG (multigrid)	2.97 ms (2.99 $\times$)	4.35 ms (3.85 $\times$)	5.84 ms (5.48 $\times$)	7.36 ms (7.44 $\times$)	3
	HINTS ($\tau=25$), Jacobi (0.67)	2.71 ms (2.90 $\times$)	2.71 ms (2.90 $\times$)	2.74 ms (2.94 $\times$)	5.16 ms (5.57 $\times$)	25
	Learned router, {NO, Jacobi (0.67)}	839 μ s	912 μ s	1.51 ms	2.93 ms	2
	Learned router, {NO, Multigrid}	783 μ s	2.05 ms	3.33 ms	4.72 ms	2
	Cost-aware oracle, {NO, Multigrid}	773 μ s	2.02 ms	3.29 ms	4.66 ms	2
ConvDiff	FFT direct solve	376 μ s (0.37 $\times$)	376 μ s (0.37 $\times$)	376 μ s (0.37 $\times$)	376 μ s (0.37 $\times$)	1
	Multigrid V(2,2) alone	3.06 ms (2.92 $\times$)	4.52 ms (3.71 $\times$)	6.15 ms (5.56 $\times$)	8.92 ms (7.78 $\times$)	3
	BiCGSTAB	56.0 ms (49.3 $\times$)	67.6 ms (61.2 $\times$)	82.2 ms (74.9 $\times$)	93.1 ms (86.3 $\times$)	338
	BiCGSTAB (multigrid)	3.12 ms (3.01 $\times$)	6.17 ms (5.31 $\times$)	6.17 ms (6.11 $\times$)	9.21 ms (8.87 $\times$)	2
	GMRES(20)	301.7 ms (272 $\times$)	434.1 ms (430 $\times$)	615.9 ms (574 $\times$)	882.8 ms (817 $\times$)	2100
	HINTS ($\tau=25$), Jacobi (0.67)	3.40 ms (3.14 $\times$)	3.40 ms (3.14 $\times$)	3.47 ms (3.22 $\times$)	6.53 ms (6.05 $\times$)	25
	Learned router, {NO, Jacobi (0.67)}	912 μ s	1.10 ms	1.94 ms	3.70 ms	4
	Learned router, {NO, Multigrid}	873 μ s	2.32 ms	3.83 ms	5.50 ms	2
	Cost-aware oracle, {NO, Multigrid}	851 μ s	2.33 ms	3.83 ms	5.49 ms	2

Results at 128^2 . With the same implementation technology (numpy stencils and scipy sparse triangular solves for every classical operation; a BLAS matrix–vector product for the corrector), the learned router with its best stationary pairing reaches truncation-level accuracy $3.62 \times - 3.71 \times$ faster than multigrid alone (three V-cycles), $3.85 \times - 5.31 \times$ faster than the best multigrid-preconditioned Krylov method and an order of magnitude faster than the unpreconditioned or SymGS-preconditioned Krylov solvers; the only faster method is the FFT direct solve, by $2.87 \times - 2.91 \times$, which exists only because the test problems have constant coefficients and periodic boundaries (it is not an iterative method and is the reference we solve against). The reason the hybrid beats multigrid here is the corrector’s accuracy on the smooth band: one call removes the modes $|k| \leq 31$ to 10^{-6} at the price of a single 4096×4096 matrix–vector product, whereas a V(2,2)-cycle contracts the whole spectrum by ≈ 0.03 at the price of four Gauss–Seidel sweeps on the fine grid plus the coarse-grid work. Table 26 also reports the router over {NO, multigrid}: the cost-aware oracle and the learned router call the corrector once and then let the V-cycle remove the rest, and are $1.93 \times - 2.03 \times$ faster than multigrid alone, i.e., adding a strong classical member to the ensemble is handled by the same construction without any change (the multigrid cycle is dearer than the unit, so it is the case in which the per-unit-cost comparison of Section 4 matters).

Scaling with the grid. Table 27 repeats the Jacobi and Gauss–Seidel pairings on 256×256 and 512×512 grids (32 and 16 instances; correctors retrained per grid with the same recipe, keeping the sensor grid at 64×64 , i.e., the corrector always handles the modes $|k| \leq 31$, and the router retrained per pairing). The corrector’s cost is dominated by a 4096×4096 matrix–vector product and is therefore nearly grid-independent, while every classical iteration grows as N^2 , so the corrector becomes relatively cheaper as N grows; the classical member, however, has to cover a growing fraction of the spectrum. The solver-only baselines at 512^2 are capped at 8000 iterations and reported as lower bounds; restarted GMRES, which was two orders of magnitude slower than the router at 128^2 and 256^2 , is not run at 512^2 .

What the scaling shows. At 256^2 the router remains the fastest iterative method for every pairing and all three equations: $1.58 \times - 10.9 \times$ faster than HINTS ($\tau=25$) and $1.11 \times - 2.33 \times$ faster than the per-cell best period over the 13 cells (Table 30; the smallest margins are the anisotropic Jacobi pairing, where $\tau=5$ is already close to the oracle’s schedule), and on the isotropic equations $4.6 - 6.7 \times$ faster than multigrid alone and $5 - 6 \times$ faster than multigrid-preconditioned CG/BiCGSTAB (Table 37); the FFT direct solve is $1.7 - 2.6 \times$ faster than the router. At 512^2 the picture splits by pairing (Table 38). With SOR and multigrid the router is still $3.1 \times$ and $2.1 \times$ faster than the best fixed schedule and $1.3 - 2.4 \times$ faster than multigrid alone; the {NO, multigrid} router (39 ms on Poisson) is the fastest iterative method we measured at this size. With Jacobi and Gauss–Seidel, however, the corrector’s fixed band $|k| \leq 31$ leaves the smoother all modes $k \geq 32$, whose slowest member contracts by only 0.98 per Jacobi sweep at $N = 512$ (against 0.71 at $N = 128$): every policy then needs the same ≈ 80 sweeps after a single corrector call, the wall-clock time is dominated by sweeps, and the

1042 schedule barely matters—the router is $1.11\times$ faster than HINTS on Poisson (GS and Jacobi) but only
1043 on par with a fixed $\tau=50$ schedule on convection–diffusion with Jacobi, where its imitation of the
1044 oracle also degrades (4 of 16 instances do not reach h^2 within the cap; the oracle reaches all 16 in the
1045 same 84 iterations as HINTS). This is the expected limit of a corrector whose band does not grow
1046 with the grid: the hybrid’s advantage comes from removing the modes the smoother cannot damp,
1047 and at 512^2 a 64×64 sensor grid no longer covers them. A corrector with a band that scales with
1048 N (at the price of a dense matrix–vector product that grows like the fourth power of the band) or a
1049 multigrid member in the ensemble restores the gains, as the multigrid pairing shows.

Table 27: Scaling: median time to $\varepsilon = h^2$ for the Jacobi and GS pairings against the strongest classical alternatives on each grid (paired median speedup of the learned router over each method in parentheses; † n : n instances did not reach the tolerance within the iteration cap).

Equation	N	Solver only	HINTS ($\tau=25$)	HINTS ($\tau=5$)	Learned router	Cost-aware oracle	Multigrid	PCG/BiCGSTAB (MG)	FFT direct
Poisson, Jacobi	128^2	1.15 s	2.65 ms (2.68 $\times$)	1.10 ms (1.21 $\times$)	987 μ s	942 μ s	3.99 ms (3.43 $\times$)	4.35 ms (3.57 $\times$)	317 μ s (0.33 $\times$)
	256^2	>7.61 s ^{†32}	5.95 ms (2.52 $\times$)	2.47 ms (1.56 $\times$)	2.38 ms	2.32 ms	14.1 ms (6.70 $\times$)	15.2 ms (6.30 $\times$)	1.32 ms (0.58 $\times$)
	512^2	>9.85 s ^{†16}	87.0 ms (1.11 $\times$)	126.3 ms (1.56 $\times$)	78.8 ms ^{†2}	74.2 ms	81.9 ms (1.14 $\times$)	85.6 ms (1.26 $\times$)	7.65 ms (0.10 $\times$)
Poisson, GS	128^2	1.83 s	6.65 ms (6.75 $\times$)	1.99 ms (1.94 $\times$)	1.01 ms	953 μ s	3.99 ms (3.39 $\times$)	4.35 ms (3.52 $\times$)	317 μ s (0.33 $\times$)
	256^2	28.83 s	22.0 ms (5.77 $\times$)	6.07 ms (1.70 $\times$)	3.92 ms	3.78 ms	14.1 ms (4.15 $\times$)	15.2 ms (3.85 $\times$)	1.32 ms (0.36 $\times$)
	512^2	>29.96 s ^{†16}	142.8 ms (1.11 $\times$)	162.5 ms (1.21 $\times$)	131.5 ms	130.0 ms	81.9 ms (0.69 $\times$)	85.6 ms (0.76 $\times$)	7.65 ms (0.06 $\times$)
ConvDiff, Jacobi	128^2	1.51 s	3.44 ms (2.95 $\times$)	1.28 ms (1.34 $\times$)	1.14 ms	1.15 ms	4.52 ms (3.54 $\times$)	6.17 ms (5.18 $\times$)	376 μ s (0.36 $\times$)
	256^2	>12.73 s ^{†32}	9.37 ms (1.92 $\times$)	4.97 ms (1.57 $\times$)	4.76 ms	4.81 ms	18.8 ms (4.60 $\times$)	24.0 ms (5.06 $\times$)	1.56 ms (0.39 $\times$)
	512^2	>13.71 s ^{†16}	153.8 ms (1.04 $\times$)	250.2 ms (1.35 $\times$)	148.5 ms ^{†4}	165.2 ms	99.7 ms (0.68 $\times$)	93.5 ms (0.65 $\times$)	8.06 ms (0.05 $\times$)
ConvDiff, GS	128^2	1.63 s	8.27 ms (6.76 $\times$)	3.60 ms (2.60 $\times$)	1.35 ms	1.32 ms	4.52 ms (3.29 $\times$)	6.17 ms (4.42 $\times$)	376 μ s (0.34 $\times$)
	256^2	30.05 s	31.0 ms (5.52 $\times$)	10.5 ms (1.53 $\times$)	7.05 ms	7.15 ms	18.8 ms (3.08 $\times$)	24.0 ms (3.47 $\times$)	1.56 ms (0.26 $\times$)
	512^2	>32.99 s ^{†16}	214.1 ms (1.05 $\times$)	212.5 ms (1.20 $\times$)	179.9 ms	185.9 ms	99.7 ms (0.56 $\times$)	93.5 ms (0.54 $\times$)	8.06 ms (0.05 $\times$)
AnisoDiff, Jacobi	128^2	>4.62 s ^{†64}	2.90 ms (3.16 $\times$)	1.15 ms (1.25 $\times$)	911 μ s	892 μ s	—	—	—
	256^2	>9.29 s ^{†16}	14.9 ms (1.58 $\times$)	10.3 ms (1.11 $\times$)	9.55 ms	9.42 ms	—	—	—
AnisoDiff, GS	128^2	>13.63 s ^{†64}	6.48 ms (7.61 $\times$)	1.74 ms (1.99 $\times$)	852 μ s	833 μ s	—	—	—
	256^2	>34.18 s ^{†16}	29.7 ms (3.18 $\times$)	12.8 ms (1.37 $\times$)	9.30 ms	9.32 ms	—	—	—

1050 **Overheads and amortisation.** Table 28 lists the measured cost of every operation as a function
1051 of N , including one decision of the LSTM router of Section 7 (full-state input of size $4N^2$, hidden
1052 size 256, 4 layers): at 128^2 one LSTM decision costs 2.4 ms, about 46 Jacobi sweeps, which is why
1053 the wall-clock study uses the scalar-feature router ($\approx 5 \mu$ s, independent of N). The learned router’s
1054 decisions therefore account for well under 1% of its solve time, and its corrector calls for 50–70% of
1055 the time to h^2 (one or two calls). Table 29 accounts for training: the corrector (shared with HINTS)
1056 is fit in minutes, the router in seconds to a few minutes per pairing, and the break-even number of
1057 solves after which the router’s own training cost is recovered relative to HINTS ($\tau=25$) with the
1058 same corrector.

Table 28: Measured single-thread cost of one operation as a function of the grid size (Poisson; median of repeated calls). Multigrid uses a direct solve below 64^2 and is not listed there.

Operation	$N = 32$	$N = 64$	$N = 128$	$N = 256$	$N = 512$
Jacobi iteration	14 μ s	25 μ s	53 μ s	191 μ s	1.07 ms
GS iteration	25 μ s	64 μ s	209 μ s	820 μ s	3.74 ms
SymGS iteration	38 μ s	109 μ s	378 μ s	1.48 ms	6.85 ms
Multigrid V(2,2) cycle	—	390 μ s	1.35 ms	5.27 ms	25.35 ms
FFT direct solve	41 μ s	77 μ s	268 μ s	1.21 ms	6.47 ms
DeepONet corrector call	—	—	663 μ s	1.13 ms	3.62 ms
Router decision (ours)	5 μ s	5 μ s	5 μ s	5 μ s	5 μ s
LSTM router decision (Section 7)	236 μ s	696 μ s	2.43 ms	9.55 ms	37.46 ms

Table 29: Training-cost amortisation for the GS pairing: corrector training (data generation and least-squares fit, CPU), router training (oracle rollouts and two DAGger rounds), median wall-clock saved per solve at $\varepsilon = h^2$ relative to HINTS ($\tau=25$) and to the solver alone, and the number of solves after which the router’s training cost is recovered relative to HINTS.

Equation	N	Corrector data + fit	Router training (GS pairing)	Saving per solve vs HINTS ($\tau=25$)	Break-even solves (router)	Saving per solve vs solver only
Poisson	128 ²	82 s	8 s	5.64 ms	1419	1.83 s
	256 ²	98 s	34 s	18.1 ms	1883	28.82 s
	512 ²	504 s	229 s	11.4 ms	20139	–
ConvDiff	128 ²	79 s	8 s	6.92 ms	1157	1.63 s
	256 ²	106 s	33 s	24.0 ms	1376	30.05 s
	512 ²	493 s	254 s	34.2 ms	7423	–

1059 E.5 Complete results on the 256×256 and 512×512 grids

1060 The tables of Section E.4 summarise the larger grids at $\varepsilon = h^2$ for two pairings; the following tables
1061 repeat the full per-tolerance results of the 128² study (Tables 2, 17 and 19 to 23) on the 256×256
1062 grid (32 test instances and all six pairings on the isotropic equations; 16 instances and the Jacobi, GS
1063 and SOR pairings on anisotropic diffusion) and on the 512×512 grid (16 instances; Jacobi, GS, SOR
1064 and multigrid pairings; solver-only runs capped at 8000 iterations and reported as lower bounds).
1065 Correctors and routers are retrained per grid with the recipe of Section E; the corrector’s sensor grid
1066 stays at 64×64 (band $|k| \leq 31$).

Table 30: 256×256 : median wall-clock time to $\varepsilon = h^2 = 1.5 \times 10^{-5}$ (32 instances; paired median speedup over solver-only in parentheses; † n : n instances did not reach the tolerance within the 40,000-iteration cap, in which case the median time at the cap is given as a lower bound).

Equation	Solver	Solver only	HINTS ($\tau=25$)	HINTS (best τ)	Greedy oracle	Cost-aware oracle	Learned router (ours)
Poisson	Jacobi	>7.61 s ^{†32}	5.95 ms ($> 10^3 \times$)	2.47 ms ($> 10^3 \times$; $\tau=5$)	2.30 ms ($> 10^3 \times$)	2.32 ms ($> 10^3 \times$)	2.38 ms ($> 10^3 \times$)
	Jacobi (0.67)	>7.74 s ^{†32}	6.41 ms ($> 10^3 \times$)	3.43 ms ($> 10^3 \times$; $\tau=5$)	2.93 ms ($> 10^3 \times$)	2.77 ms ($> 10^3 \times$)	2.83 ms ($> 10^3 \times$)
	GS	28.83 s	22.0 ms (1316 $\times$)	6.07 ms (4681 $\times$; $\tau=5$)	3.85 ms (7702 $\times$)	3.78 ms (7797 $\times$)	3.92 ms (7591 $\times$)
	SymGS	26.53 s	37.3 ms (712 $\times$)	8.71 ms (3052 $\times$; $\tau=5$)	4.18 ms (7002 $\times$)	4.19 ms (7038 $\times$)	4.22 ms (7027 $\times$)
	SOR (1.5)	9.60 s	40.7 ms (235 $\times$)	9.06 ms (1065 $\times$; $\tau=5$)	3.69 ms (2650 $\times$)	3.51 ms (2748 $\times$)	3.80 ms (2558 $\times$)
ConvDiff	Jacobi	>12.73 s ^{†32}	9.37 ms ($> 10^3 \times$)	4.95 ms ($> 10^3 \times$; $\tau=10$)	4.43 ms ($> 10^3 \times$)	4.81 ms ($> 10^3 \times$)	4.76 ms ($> 10^3 \times$)
	Jacobi (0.67)	>12.55 s ^{†32}	9.31 ms ($> 10^3 \times$)	5.68 ms ($> 10^3 \times$; $\tau=10$)	5.93 ms ($> 10^3 \times$)	5.94 ms ($> 10^3 \times$)	5.76 ms ($> 10^3 \times$)
	GS	30.05 s	31.0 ms (963 $\times$)	10.5 ms (2830 $\times$; $\tau=5$)	7.20 ms (4211 $\times$)	7.15 ms (4238 $\times$)	7.05 ms (4326 $\times$)
	SymGS	27.77 s	42.3 ms (653 $\times$)	10.9 ms (2530 $\times$; $\tau=5$)	6.93 ms (4053 $\times$)	6.89 ms (4083 $\times$)	6.96 ms (4026 $\times$)
	SOR (1.5)	8.46 s	46.1 ms (182 $\times$)	10.6 ms (785 $\times$; $\tau=5$)	5.90 ms (1416 $\times$)	5.74 ms (1462 $\times$)	5.80 ms (1444 $\times$)
AnisoDiff	Jacobi	>9.29 s ^{†16}	14.9 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$; $\tau=5$)	9.29 ms ($> 10^3 \times$)	9.42 ms ($> 10^3 \times$)	9.55 ms ($> 10^3 \times$)
	GS	>34.18 s ^{†16}	29.7 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$; $\tau=5$)	9.40 ms ($> 10^3 \times$)	9.32 ms ($> 10^3 \times$)	9.30 ms ($> 10^3 \times$)
	SOR (1.5)	>33.90 s ^{†16}	29.7 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$; $\tau=5$)	9.33 ms ($> 10^3 \times$)	9.27 ms ($> 10^3 \times$)	9.93 ms ($> 10^3 \times$)

Table 31: 256×256 : paired median speedup of the learned router over HINTS (32 instances; bold: $\geq 1.1 \times$ with one-sided Wilcoxon $p < 0.01$).

Equation	Solver	vs. HINTS ($\tau=25$)			vs. best fixed τ		
		$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$
Poisson	Jacobi	4.02$\times$	2.52$\times$	1.23$\times$	1.38$\times$ ($\tau=5$)	1.56$\times$ ($\tau=5$)	1.11$\times$ ($\tau=50$)
	Jacobi (0.67)	4.63$\times$	2.26$\times$	1.10 $\times$	1.55$\times$ ($\tau=5$)	1.73$\times$ ($\tau=5$)	1.10 $\times$ ($\tau=25$)
	GS	15.0$\times$	5.77$\times$	1.67$\times$	3.29$\times$ ($\tau=5$)	1.70$\times$ ($\tau=5$)	1.20$\times$ ($\tau=10$)
	SymGS	27.1$\times$	9.96$\times$	2.97$\times$	5.36$\times$ ($\tau=5$)	2.27$\times$ ($\tau=5$)	1.22$\times$ ($\tau=5$)
	SOR (1.5)	14.9$\times$	10.9$\times$	2.48$\times$	3.36$\times$ ($\tau=5$)	2.33$\times$ ($\tau=5$)	1.09 $\times$ ($\tau=10$)
ConvDiff	Jacobi	5.13$\times$	1.92$\times$	1.12$\times$	1.55$\times$ ($\tau=5$)	1.30$\times$ ($\tau=10$)	1.08 $\times$ ($\tau=50$)
	Jacobi (0.67)	5.64$\times$	1.80$\times$	1.06 $\times$	1.66$\times$ ($\tau=5$)	1.25$\times$ ($\tau=10$)	1.06 $\times$ ($\tau=25$)
	GS	15.0$\times$	5.52$\times$	1.95$\times$	3.31$\times$ ($\tau=5$)	1.53$\times$ ($\tau=5$)	1.17$\times$ ($\tau=10$)
	SymGS	25.3$\times$	6.26$\times$	2.63$\times$	4.99$\times$ ($\tau=5$)	1.69$\times$ ($\tau=5$)	1.18$\times$ ($\tau=5$)
	SOR (1.5)	14.8$\times$	7.99$\times$	2.50$\times$	3.24$\times$ ($\tau=5$)	1.96$\times$ ($\tau=5$)	1.19$\times$ ($\tau=5$)
AnisoDiff	Jacobi	1.59$\times$	1.58$\times$	1.93$\times$	1.09 $\times$ ($\tau=5$)	1.11$\times$ ($\tau=5$)	1.80$\times$ ($\tau=50$)
	GS	3.23$\times$	3.18$\times$	2.26$\times$	1.39$\times$ ($\tau=5$)	1.37$\times$ ($\tau=5$)	1.64$\times$ ($\tau=10$)
	SOR (1.5)	3.25$\times$	2.98$\times$	1.54$\times$	1.38$\times$ ($\tau=5$)	1.38$\times$ ($\tau=5$)	1.35$\times$ ($\tau=10$)

Table 32: 256×256 : median number of iterations and corrector calls to reach $\varepsilon = h^2$.

Equation	Solver	HINTS ($\tau=25$)		Cost-aware oracle		Learned router	
		iters	NO calls	iters	NO calls	iters	NO calls
Poisson	Jacobi	25	1	6	1	6	1
	Jacobi (0.67)	25	1	8	1	8	1
	GS	27	1	4	1	4	1
	SymGS	26	1	3	1	3	1
	SOR (1.5)	50	2	4	1	4	1
ConvDiff	Jacobi	25	1	10	1	10	1
	Jacobi (0.67)	25	1	15	1	15	1
	GS	33	1	7	1	7	1
	SymGS	27	1	4	1	4	1
	SOR (1.5)	50	2	6	1	6	1
AnisoDiff	Jacobi	25	1	1	1	1	1
	GS	25	1	1	1	1	1
	SOR (1.5)	25	1	1	1	1	1

Table 33: 256×256 : measured single-thread per-iteration costs and macro-action sizes.

Equation	Solver	classical iteration	corrector iteration	m_{solver}	m_{NO}
Poisson	Jacobi	167 μs	1004 μs	6	1
	Jacobi (0.67)	150 μs	971 μs	6	1
	GS	780 μs	1007 μs	1	1
	SymGS	1402 μs	931 μs	1	1
	SOR (1.5)	771 μs	960 μs	1	1
ConvDiff	Jacobi	286 μs	1130 μs	4	1
	Jacobi (0.67)	288 μs	1147 μs	4	1
	GS	855 μs	1077 μs	1	1
	SymGS	1551 μs	1114 μs	1	1
	SOR (1.5)	813 μs	1032 μs	1	1
AnisoDiff	Jacobi	235 μs	8886 μs	38	1
	GS	826 μs	8905 μs	11	1
	SOR (1.5)	777 μs	8803 μs	11	1

Table 34: 256×256 , Poisson: median wall-clock time to relative error ε for every policy and pairing (32 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	5.65 s	$>7.61 \text{ s}^{132}$	$>7.61 \text{ s}^{132}$	$>7.61 \text{ s}^{132}$	$>7.61 \text{ s}^{132}$	$>7.61 \text{ s}^{132}$
	HINTS ($\tau=25$)	5.90 ms (948 $\times$)	5.90 ms ($> 10^3 \times$)	5.95 ms ($> 10^3 \times$)	5.95 ms ($> 10^3 \times$)	6.50 ms ($> 10^3 \times$)	85.5 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	3.00 ms (1824 $\times$)	3.00 ms ($> 10^3 \times$)	3.31 ms ($> 10^3 \times$)	3.31 ms ($> 10^3 \times$)	6.06 ms ($> 10^3 \times$)	108.2 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	2.03 ms (2721 $\times$)	2.07 ms ($> 10^3 \times$)	2.47 ms ($> 10^3 \times$)	3.21 ms ($> 10^3 \times$)	8.72 ms ($> 10^3 \times$)	163.1 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	10.4 ms (536 $\times$)	10.4 ms ($> 10^3 \times$)	10.6 ms ($> 10^3 \times$)	10.6 ms ($> 10^3 \times$)	11.1 ms ($> 10^3 \times$)	72.9 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.27 ms (4345 $\times$)	1.28 ms ($> 10^3 \times$)	2.30 ms ($> 10^3 \times$)	2.68 ms ($> 10^3 \times$)	5.50 ms ($> 10^3 \times$)	67.2 ms ($> 10^3 \times$)
	Cost-aware oracle	1.42 ms (3889 $\times$)	1.42 ms ($> 10^3 \times$)	2.32 ms ($> 10^3 \times$)	2.74 ms ($> 10^3 \times$)	5.34 ms ($> 10^3 \times$)	67.0 ms ($> 10^3 \times$)
	Learned router (ours)	1.46 ms (3740 $\times$)	1.47 ms ($> 10^3 \times$)	2.38 ms ($> 10^3 \times$)	2.76 ms ($> 10^3 \times$)	5.18 ms ($> 10^3 \times$)	73.1 ms ($> 10^3 \times$)
Jacobi (0.67)	Solver only	$>7.74 \text{ s}^{131}$	$>7.74 \text{ s}^{132}$	$>7.74 \text{ s}^{132}$	$>7.74 \text{ s}^{132}$	$>7.74 \text{ s}^{132}$	$>7.74 \text{ s}^{132}$
	HINTS ($\tau=25$)	6.02 ms ($> 10^3 \times$)	6.02 ms ($> 10^3 \times$)	6.41 ms ($> 10^3 \times$)	6.45 ms ($> 10^3 \times$)	8.13 ms ($> 10^3 \times$)	16.8 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	3.06 ms ($> 10^3 \times$)	3.06 ms ($> 10^3 \times$)	3.54 ms ($> 10^3 \times$)	3.87 ms ($> 10^3 \times$)	8.78 ms ($> 10^3 \times$)	21.9 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	2.08 ms ($> 10^3 \times$)	2.10 ms ($> 10^3 \times$)	3.43 ms ($> 10^3 \times$)	4.50 ms ($> 10^3 \times$)	13.7 ms ($> 10^3 \times$)	33.6 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	10.6 ms ($> 10^3 \times$)	10.6 ms ($> 10^3 \times$)	10.6 ms ($> 10^3 \times$)	10.6 ms ($> 10^3 \times$)	11.0 ms ($> 10^3 \times$)	21.1 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.29 ms ($> 10^3 \times$)	1.30 ms ($> 10^3 \times$)	2.93 ms ($> 10^3 \times$)	3.59 ms ($> 10^3 \times$)	7.67 ms ($> 10^3 \times$)	16.2 ms ($> 10^3 \times$)
	Cost-aware oracle	1.29 ms ($> 10^3 \times$)	1.30 ms ($> 10^3 \times$)	2.77 ms ($> 10^3 \times$)	3.45 ms ($> 10^3 \times$)	7.19 ms ($> 10^3 \times$)	16.1 ms ($> 10^3 \times$)
	Learned router (ours)	1.30 ms ($> 10^3 \times$)	1.32 ms ($> 10^3 \times$)	2.83 ms ($> 10^3 \times$)	3.48 ms ($> 10^3 \times$)	7.56 ms ($> 10^3 \times$)	16.8 ms ($> 10^3 \times$)
GS	Solver only	11.78 s	17.80 s	28.83 s	29.94 s	$>31.70 \text{ s}^{132}$	$>31.70 \text{ s}^{132}$
	HINTS ($\tau=25$)	20.4 ms (574 $\times$)	20.4 ms (871 $\times$)	22.0 ms (1316 $\times$)	22.9 ms (1307 $\times$)	40.4 ms ($> 10^3 \times$)	43.5 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	8.34 ms (1398 $\times$)	8.34 ms (2120 $\times$)	9.92 ms (2899 $\times$)	10.9 ms (2723 $\times$)	17.3 ms ($> 10^3 \times$)	30.4 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	4.49 ms (2626 $\times$)	4.49 ms (3981 $\times$)	6.07 ms (4681 $\times$)	7.23 ms (4133 $\times$)	14.0 ms ($> 10^3 \times$)	31.1 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	39.9 ms (294 $\times$)	39.9 ms (445 $\times$)	41.4 ms (698 $\times$)	42.3 ms (703 $\times$)	79.5 ms ($> 10^3 \times$)	79.8 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.34 ms (8746 $\times$)	1.36 ms (13154 $\times$)	3.85 ms (7702 $\times$)	5.37 ms (5649 $\times$)	11.3 ms ($> 10^3 \times$)	25.4 ms ($> 10^3 \times$)
	Cost-aware oracle	1.31 ms (8992 $\times$)	1.33 ms (13490 $\times$)	3.78 ms (7797 $\times$)	5.28 ms (5826 $\times$)	11.4 ms ($> 10^3 \times$)	25.6 ms ($> 10^3 \times$)
	Learned router (ours)	1.33 ms (8750 $\times$)	1.36 ms (13075 $\times$)	3.92 ms (7591 $\times$)	5.56 ms (5508 $\times$)	12.2 ms ($> 10^3 \times$)	26.0 ms ($> 10^3 \times$)
SymGS	Solver only	10.82 s	16.41 s	26.53 s	27.56 s	33.16 s	44.30 s
	HINTS ($\tau=25$)	36.1 ms (302 $\times$)	36.1 ms (457 $\times$)	37.3 ms (712 $\times$)	37.6 ms (732 $\times$)	71.9 ms (462 $\times$)	71.9 ms (616 $\times$)
	HINTS ($\tau=10$)	14.3 ms (754 $\times$)	14.3 ms (1143 $\times$)	15.8 ms (1669 $\times$)	15.9 ms (1721 $\times$)	28.5 ms (1164 $\times$)	34.6 ms (1277 $\times$)
	HINTS ($\tau=5$)	7.18 ms (1523 $\times$)	7.18 ms (2301 $\times$)	8.71 ms (3052 $\times$)	8.77 ms (3153 $\times$)	15.5 ms (2137 $\times$)	29.8 ms (1493 $\times$)
	HINTS ($\tau=50$)	72.5 ms (149 $\times$)	72.5 ms (226 $\times$)	73.3 ms (359 $\times$)	74.3 ms (372 $\times$)	144.3 ms (229 $\times$)	144.4 ms (306 $\times$)
	Greedy oracle (Alg. 1)	1.35 ms (7971 $\times$)	1.36 ms (11909 $\times$)	4.18 ms (7002 $\times$)	4.90 ms (5659 $\times$)	10.7 ms (3086 $\times$)	24.3 ms (1817 $\times$)
	Cost-aware oracle	1.33 ms (8231 $\times$)	1.35 ms (12187 $\times$)	4.19 ms (7038 $\times$)	4.95 ms (5652 $\times$)	10.5 ms (3150 $\times$)	23.8 ms (1863 $\times$)
	Learned router (ours)	1.33 ms (8089 $\times$)	1.34 ms (12258 $\times$)	4.22 ms (7027 $\times$)	4.97 ms (5654 $\times$)	10.8 ms (3081 $\times$)	24.4 ms (1821 $\times$)
SOR (1.5)	Solver only	3.93 s	5.94 s	9.60 s	9.97 s	11.97 s	15.97 s
	HINTS ($\tau=25$)	20.5 ms (190 $\times$)	20.5 ms (287 $\times$)	40.7 ms (235 $\times$)	40.7 ms (244 $\times$)	40.7 ms (293 $\times$)	60.9 ms (261 $\times$)
	HINTS ($\tau=10$)	8.39 ms (461 $\times$)	8.39 ms (700 $\times$)	16.4 ms (577 $\times$)	16.5 ms (598 $\times$)	17.7 ms (683 $\times$)	27.2 ms (592 $\times$)
	HINTS ($\tau=5$)	4.57 ms (855 $\times$)	4.57 ms (1299 $\times$)	9.06 ms (1065 $\times$)	9.20 ms (1104 $\times$)	14.2 ms (832 $\times$)	27.4 ms (577 $\times$)
	HINTS ($\tau=50$)	39.4 ms (98.9 $\times$)	39.4 ms (151 $\times$)	78.5 ms (122 $\times$)	78.8 ms (126 $\times$)	78.8 ms (152 $\times$)	104.0 ms (154 $\times$)
	Greedy oracle (Alg. 1)	1.30 ms (3003 $\times$)	1.31 ms (4507 $\times$)	3.69 ms (2650 $\times$)	4.46 ms (2252 $\times$)	10.4 ms (1142 $\times$)	22.7 ms (698 $\times$)
	Cost-aware oracle	1.29 ms (3009 $\times$)	1.30 ms (4527 $\times$)	3.51 ms (2748 $\times$)	4.26 ms (2320 $\times$)	10.3 ms (1142 $\times$)	22.4 ms (702 $\times$)
	Learned router (ours)	1.34 ms (2896 $\times$)	1.36 ms (4339 $\times$)	3.80 ms (2558 $\times$)	4.64 ms (2126 $\times$)	10.9 ms (1083 $\times$)	24.2 ms (654 $\times$)

Table 35: 256×256 , convection–diffusion: median wall-clock time to relative error ε for every policy and pairing (32 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	8.83 s	$>12.73 \text{ s}^{126}$	$>12.73 \text{ s}^{132}$	$>12.73 \text{ s}^{132}$	$>12.73 \text{ s}^{132}$	$>12.73 \text{ s}^{132}$
	HINTS ($\tau=25$)	9.15 ms (967 $\times$)	9.15 ms ($> 10^3 \times$)	9.37 ms ($> 10^3 \times$)	9.46 ms ($> 10^3 \times$)	11.2 ms ($> 10^3 \times$)	500.8 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	4.33 ms (2010 $\times$)	4.37 ms ($> 10^3 \times$)	4.95 ms ($> 10^3 \times$)	5.24 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$)	631.6 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	2.73 ms (3051 $\times$)	2.80 ms ($> 10^3 \times$)	4.97 ms ($> 10^3 \times$)	6.76 ms ($> 10^3 \times$)	14.9 ms ($> 10^3 \times$)	869.7 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	17.0 ms (517 $\times$)	17.0 ms ($> 10^3 \times$)	17.3 ms ($> 10^3 \times$)	17.3 ms ($> 10^3 \times$)	18.5 ms ($> 10^3 \times$)	475.6 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.55 ms (5578 $\times$)	1.57 ms ($> 10^3 \times$)	4.43 ms ($> 10^3 \times$)	5.06 ms ($> 10^3 \times$)	9.42 ms ($> 10^3 \times$)	436.3 ms ($> 10^3 \times$)
	Cost-aware oracle	1.69 ms (5129 $\times$)	1.70 ms ($> 10^3 \times$)	4.81 ms ($> 10^3 \times$)	5.42 ms ($> 10^3 \times$)	9.60 ms ($> 10^3 \times$)	437.5 ms ($> 10^3 \times$)
	Learned router (ours)	1.73 ms (4987 $\times$)	1.77 ms ($> 10^3 \times$)	4.76 ms ($> 10^3 \times$)	5.42 ms ($> 10^3 \times$)	9.76 ms ($> 10^3 \times$)	435.1 ms ($> 10^3 \times$)
Jacobi (0.67)	Solver only	$>12.55 \text{ s}^{120}$	$>12.55 \text{ s}^{132}$	$>12.55 \text{ s}^{132}$	$>12.55 \text{ s}^{132}$	$>12.55 \text{ s}^{132}$	$>12.55 \text{ s}^{132}$
	HINTS ($\tau=25$)	9.03 ms ($> 10^3 \times$)	9.03 ms ($> 10^3 \times$)	9.31 ms ($> 10^3 \times$)	9.35 ms ($> 10^3 \times$)	14.9 ms ($> 10^3 \times$)	30.4 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	4.25 ms ($> 10^3 \times$)	4.26 ms ($> 10^3 \times$)	5.68 ms ($> 10^3 \times$)	7.16 ms ($> 10^3 \times$)	15.6 ms ($> 10^3 \times$)	34.5 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	2.68 ms ($> 10^3 \times$)	2.73 ms ($> 10^3 \times$)	7.87 ms ($> 10^3 \times$)	9.37 ms ($> 10^3 \times$)	21.7 ms ($> 10^3 \times$)	47.7 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	16.2 ms ($> 10^3 \times$)	16.2 ms ($> 10^3 \times$)	16.3 ms ($> 10^3 \times$)	16.4 ms ($> 10^3 \times$)	17.9 ms ($> 10^3 \times$)	33.8 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.53 ms ($> 10^3 \times$)	1.54 ms ($> 10^3 \times$)	5.93 ms ($> 10^3 \times$)	6.88 ms ($> 10^3 \times$)	13.1 ms ($> 10^3 \times$)	26.8 ms ($> 10^3 \times$)
	Cost-aware oracle	1.56 ms ($> 10^3 \times$)	1.58 ms ($> 10^3 \times$)	5.94 ms ($> 10^3 \times$)	6.88 ms ($> 10^3 \times$)	13.4 ms ($> 10^3 \times$)	27.5 ms ($> 10^3 \times$)
	Learned router (ours)	1.53 ms ($> 10^3 \times$)	1.61 ms ($> 10^3 \times$)	5.76 ms ($> 10^3 \times$)	6.65 ms ($> 10^3 \times$)	13.0 ms ($> 10^3 \times$)	26.9 ms ($> 10^3 \times$)
GS	Solver only	11.85 s	18.36 s	30.05 s	31.23 s	$>36.70 \text{ s}^{126}$	$>36.70 \text{ s}^{132}$
	HINTS ($\tau=25$)	23.5 ms (501 $\times$)	23.5 ms (773 $\times$)	31.0 ms (963 $\times$)	46.0 ms (666 $\times$)	47.0 ms ($> 10^3 \times$)	59.3 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	9.79 ms (1213 $\times$)	9.79 ms (1875 $\times$)	17.1 ms (1695 $\times$)	19.4 ms (1618 $\times$)	21.3 ms ($> 10^3 \times$)	36.7 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	5.27 ms (2247 $\times$)	5.29 ms (3468 $\times$)	10.5 ms (2830 $\times$)	11.4 ms (2674 $\times$)	19.4 ms ($> 10^3 \times$)	39.2 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	46.2 ms (257 $\times$)	46.2 ms (397 $\times$)	53.6 ms (565 $\times$)	71.9 ms (427 $\times$)	92.8 ms ($> 10^3 \times$)	94.2 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	1.55 ms (7777 $\times$)	1.57 ms (11258 $\times$)	7.20 ms (4211 $\times$)	8.15 ms (3852 $\times$)	16.2 ms ($> 10^3 \times$)	31.1 ms ($> 10^3 \times$)
	Cost-aware oracle	1.56 ms (7586 $\times$)	1.61 ms (10966 $\times$)	7.15 ms (4238 $\times$)	8.13 ms (3865 $\times$)	16.2 ms ($> 10^3 \times$)	31.0 ms ($> 10^3 \times$)
	Learned router (ours)	1.56 ms (7414 $\times$)	1.57 ms (10840 $\times$)	7.05 ms (4326 $\times$)	7.97 ms (3960 $\times$)	16.1 ms ($> 10^3 \times$)	31.4 ms ($> 10^3 \times$)
SymGS	Solver only	10.93 s	16.90 s	27.77 s	28.89 s	34.88 s	46.85 s
	HINTS ($\tau=25$)	39.2 ms (278 $\times$)	39.2 ms (429 $\times$)	42.3 ms (653 $\times$)	43.9 ms (651 $\times$)	78.0 ms (446 $\times$)	78.1 ms (594 $\times$)
	HINTS ($\tau=10$)	15.6 ms (697 $\times$)	15.6 ms (1076 $\times$)	18.9 ms (1464 $\times$)	20.4 ms (1401 $\times$)	30.9 ms (1118 $\times$)	41.4 ms (1143 $\times$)
	HINTS ($\tau=5$)	7.80 ms (1391 $\times$)	7.80 ms (2149 $\times$)	10.9 ms (2530 $\times$)	12.6 ms (2276 $\times$)	18.5 ms (1858 $\times$)	34.8 ms (1345 $\times$)
	HINTS ($\tau=50$)	78.3 ms (138 $\times$)	78.3 ms (213 $\times$)	81.4 ms (341 $\times$)	83.1 ms (345 $\times$)	156.3 ms (221 $\times$)	156.3 ms (298 $\times$)
	Greedy oracle (Alg. 1)	1.53 ms (6906 $\times$)	1.58 ms (10455 $\times$)	6.93 ms (4053 $\times$)	7.74 ms (3675 $\times$)	14.1 ms (2383 $\times$)	28.4 ms (1616 $\times$)
	Cost-aware oracle	1.52 ms (7162 $\times$)	1.56 ms (10557 $\times$)	6.89 ms (4083 $\times$)	7.71 ms (3756 $\times$)	14.1 ms (2414 $\times$)	28.4 ms (1633 $\times$)
	Learned router (ours)	1.54 ms (7007 $\times$)	1.55 ms (10689 $\times$)	6.96 ms (4026 $\times$)	7.92 ms (3739 $\times$)	14.8 ms (2372 $\times$)	29.6 ms (1569 $\times$)
SOR (1.5)	Solver only	3.31 s	5.14 s	8.46 s	8.79 s	10.59 s	14.22 s
	HINTS ($\tau=25$)	23.2 ms (141 $\times$)	23.2 ms (219 $\times$)	46.1 ms (182 $\times$)	46.1 ms (189 $\times$)	46.1 ms (228 $\times$)	69.0 ms (204 $\times$)
	HINTS ($\tau=10$)	9.72 ms (339 $\times$)	9.72 ms (523 $\times$)	19.1 ms (435 $\times$)	19.1 ms (453 $\times$)	23.2 ms (456 $\times$)	32.5 ms (443 $\times$)
	HINTS ($\tau=5$)	5.18 ms (622 $\times$)	5.22 ms (970 $\times$)	10.6 ms (785 $\times$)	11.9 ms (745 $\times$)	17.4 ms (609 $\times$)	31.2 ms (449 $\times$)
	HINTS ($\tau=50$)	45.7 ms (71.2 $\times$)	45.7 ms (110 $\times$)	91.0 ms (90.8 $\times$)	91.0 ms (94.4 $\times$)	91.0 ms (114 $\times$)	136.7 ms (102 $\times$)
	Greedy oracle (Alg. 1)	1.52 ms (2097 $\times$)	1.56 ms (3139 $\times$)	5.90 ms (1416 $\times$)	7.29 ms (1200 $\times$)	14.3 ms (733 $\times$)	28.1 ms (504 $\times$)
	Cost-aware oracle	1.52 ms (2172 $\times$)	1.57 ms (3151 $\times$)	5.74 ms (1462 $\times$)	7.13 ms (1230 $\times$)	14.0 ms (749 $\times$)	27.5 ms (516 $\times$)
	Learned router (ours)	1.53 ms (2088 $\times$)	1.57 ms (3051 $\times$)	5.80 ms (1444 $\times$)	7.21 ms (1218 $\times$)	14.2 ms (740 $\times$)	27.6 ms (510 $\times$)

Table 36: 256×256 , anisotropic diffusion: median wall-clock time to relative error ε for the Jacobi, GS and SOR pairings (16 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	$>9.29 \text{ s}^{116}$	$>9.29 \text{ s}^{116}$	$>9.29 \text{ s}^{116}$	$>9.29 \text{ s}^{116}$	$>9.29 \text{ s}^{116}$	$>9.29 \text{ s}^{116}$
	HINTS ($\tau=25$)	14.9 ms ($> 10^3 \times$)	14.9 ms ($> 10^3 \times$)	14.9 ms ($> 10^3 \times$)	14.9 ms ($> 10^3 \times$)	30.0 ms ($> 10^3 \times$)	127.4 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	11.3 ms ($> 10^3 \times$)	11.3 ms ($> 10^3 \times$)	11.3 ms ($> 10^3 \times$)	11.4 ms ($> 10^3 \times$)	22.7 ms ($> 10^3 \times$)	252.6 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	10.2 ms ($> 10^3 \times$)	10.2 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$)	10.3 ms ($> 10^3 \times$)	20.4 ms ($> 10^3 \times$)	507.5 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	20.7 ms ($> 10^3 \times$)	20.7 ms ($> 10^3 \times$)	20.7 ms ($> 10^3 \times$)	20.7 ms ($> 10^3 \times$)	40.8 ms ($> 10^3 \times$)	94.6 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	9.19 ms ($> 10^3 \times$)	9.19 ms ($> 10^3 \times$)	9.29 ms ($> 10^3 \times$)	9.29 ms ($> 10^3 \times$)	18.7 ms ($> 10^3 \times$)	65.1 ms ($> 10^3 \times$)
	Cost-aware oracle	9.26 ms ($> 10^3 \times$)	9.26 ms ($> 10^3 \times$)	9.42 ms ($> 10^3 \times$)	9.42 ms ($> 10^3 \times$)	23.8 ms ($> 10^3 \times$)	65.3 ms ($> 10^3 \times$)
	Learned router (ours)	9.35 ms ($> 10^3 \times$)	9.35 ms ($> 10^3 \times$)	9.55 ms ($> 10^3 \times$)	9.55 ms ($> 10^3 \times$)	23.0 ms ($> 10^3 \times$)	65.6 ms ($> 10^3 \times$)
GS	Solver only	$>34.18 \text{ s}^{116}$	$>34.18 \text{ s}^{116}$	$>34.18 \text{ s}^{116}$	$>34.18 \text{ s}^{116}$	$>34.18 \text{ s}^{116}$	$>34.18 \text{ s}^{116}$
	HINTS ($\tau=25$)	29.7 ms ($> 10^3 \times$)	29.7 ms ($> 10^3 \times$)	29.7 ms ($> 10^3 \times$)	29.7 ms ($> 10^3 \times$)	59.0 ms ($> 10^3 \times$)	64.5 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	16.8 ms ($> 10^3 \times$)	16.8 ms ($> 10^3 \times$)	16.8 ms ($> 10^3 \times$)	16.8 ms ($> 10^3 \times$)	33.6 ms ($> 10^3 \times$)	37.8 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	12.7 ms ($> 10^3 \times$)	12.7 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$)	25.6 ms ($> 10^3 \times$)	40.7 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	50.9 ms ($> 10^3 \times$)	50.9 ms ($> 10^3 \times$)	50.9 ms ($> 10^3 \times$)	50.9 ms ($> 10^3 \times$)	81.1 ms ($> 10^3 \times$)	102.5 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	9.24 ms ($> 10^3 \times$)	9.24 ms ($> 10^3 \times$)	9.40 ms ($> 10^3 \times$)	9.40 ms ($> 10^3 \times$)	20.3 ms ($> 10^3 \times$)	38.7 ms ($> 10^3 \times$)
	Cost-aware oracle	9.20 ms ($> 10^3 \times$)	9.20 ms ($> 10^3 \times$)	9.32 ms ($> 10^3 \times$)	9.32 ms ($> 10^3 \times$)	18.7 ms ($> 10^3 \times$)	35.1 ms ($> 10^3 \times$)
	Learned router (ours)	9.20 ms ($> 10^3 \times$)	9.20 ms ($> 10^3 \times$)	9.30 ms ($> 10^3 \times$)	9.30 ms ($> 10^3 \times$)	18.8 ms ($> 10^3 \times$)	35.9 ms ($> 10^3 \times$)
SOR (1.5)	Solver only	$>33.90 \text{ s}^{116}$	$>33.90 \text{ s}^{116}$	$>33.90 \text{ s}^{116}$	$>33.90 \text{ s}^{116}$	$>33.90 \text{ s}^{116}$	$>33.90 \text{ s}^{116}$
	HINTS ($\tau=25$)	29.6 ms ($> 10^3 \times$)	29.6 ms ($> 10^3 \times$)	29.7 ms ($> 10^3 \times$)	29.7 ms ($> 10^3 \times$)	58.9 ms ($> 10^3 \times$)	60.2 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	16.9 ms ($> 10^3 \times$)	16.9 ms ($> 10^3 \times$)	17.0 ms ($> 10^3 \times$)	17.0 ms ($> 10^3 \times$)	33.7 ms ($> 10^3 \times$)	51.7 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	12.7 ms ($> 10^3 \times$)	12.7 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$)	12.8 ms ($> 10^3 \times$)	25.4 ms ($> 10^3 \times$)	64.9 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	50.5 ms ($> 10^3 \times$)	50.5 ms ($> 10^3 \times$)	50.5 ms ($> 10^3 \times$)	50.7 ms ($> 10^3 \times$)	100.7 ms ($> 10^3 \times$)	101.3 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	9.21 ms ($> 10^3 \times$)	9.21 ms ($> 10^3 \times$)	9.33 ms ($> 10^3 \times$)	9.33 ms ($> 10^3 \times$)	21.4 ms ($> 10^3 \times$)	43.3 ms ($> 10^3 \times$)
	Cost-aware oracle	9.12 ms ($> 10^3 \times$)	9.12 ms ($> 10^3 \times$)	9.27 ms ($> 10^3 \times$)	9.27 ms ($> 10^3 \times$)	19.9 ms ($> 10^3 \times$)	34.2 ms ($> 10^3 \times$)
	Learned router (ours)	9.18 ms ($> 10^3 \times$)	9.18 ms ($> 10^3 \times$)	9.93 ms ($> 10^3 \times$)	9.93 ms ($> 10^3 \times$)	23.6 ms ($> 10^3 \times$)	44.1 ms ($> 10^3 \times$)

Table 37: 256×256 : strong classical baselines (32 instances; paired median speedup of the learned router with its best stationary pairing over each method in parentheses).

Equation	Method	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$	iters to h^2
Poisson	FFT direct solve	1.32 ms (0.58 $\times$)	1.32 ms (0.58 $\times$)	1.32 ms (0.58 $\times$)	1.32 ms (0.58 $\times$)	1
	Multigrid V(2,2) alone	9.27 ms (3.92 $\times$)	14.1 ms (6.70 $\times$)	18.5 ms (8.63 $\times$)	27.0 ms (12.4 $\times$)	3
	CG	58.0 ms (24.5 $\times$)	73.1 ms (32.2 $\times$)	88.4 ms (35.5 $\times$)	102.5 ms (43.6 $\times$)	348
	PCG (SymGS)	146.5 ms (64.5 $\times$)	254.7 ms (112 $\times$)	297.2 ms (127 $\times$)	365.0 ms (161 $\times$)	170
	PCG (multigrid)	10.5 ms (3.94 $\times$)	15.2 ms (6.30 $\times$)	19.6 ms (8.55 $\times$)	24.5 ms (10.7 $\times$)	3
	HINTS ($\tau=25$), Jacobi	5.90 ms (2.52 $\times$)	5.95 ms (2.52 $\times$)	6.50 ms (4.82 $\times$)	85.5 ms (34.1 $\times$)	25
	Learned router, {NO, Jacobi}	1.47 ms	2.38 ms	5.18 ms	73.1 ms	6
	Learned router, {NO, Multigrid}	1.32 ms	5.97 ms	10.8 ms	16.6 ms	2
ConvDiff	Cost-aware oracle, {NO, Multigrid}	1.31 ms	6.09 ms	10.8 ms	16.2 ms	2
	FFT direct solve	1.56 ms (0.39 $\times$)	1.56 ms (0.39 $\times$)	1.56 ms (0.39 $\times$)	1.56 ms (0.39 $\times$)	1
	Multigrid V(2,2) alone	12.3 ms (2.73 $\times$)	18.8 ms (4.60 $\times$)	24.8 ms (5.87 $\times$)	36.1 ms (7.77 $\times$)	3
	BiCGSTAB	377.7 ms (78.9 $\times$)	486.6 ms (102 $\times$)	545.8 ms (113 $\times$)	605.9 ms (130 $\times$)	731
	BiCGSTAB (multigrid)	12.5 ms (2.92 $\times$)	24.0 ms (5.06 $\times$)	24.5 ms (5.28 $\times$)	36.1 ms (7.53 $\times$)	2
	GMRES(20)	1.85 s (387 $\times$)	3.00 s (644 $\times$)	3.75 s (797 $\times$)	5.14 s (1089 $\times$)	5030
	HINTS ($\tau=25$), Jacobi	9.15 ms (1.92 $\times$)	9.37 ms (1.92 $\times$)	11.2 ms (5.96 $\times$)	500.8 ms (28.0 $\times$)	25
	Learned router, {NO, Jacobi}	1.77 ms	4.76 ms	9.76 ms	435.1 ms	10
	Learned router, {NO, Multigrid}	1.60 ms	7.26 ms	12.4 ms	20.2 ms	2
	Cost-aware oracle, {NO, Multigrid}	1.55 ms	7.23 ms	12.7 ms	19.9 ms	2

Table 38: 512×512 : median wall-clock time to $\varepsilon = h^2 = 3.8 \times 10^{-6}$ (16 instances; Jacobi, GS, SOR and multigrid pairings; solver-only runs capped at 8000 iterations).

Equation	Solver	Solver only	HINTS ($\tau=25$)	HINTS (best τ)	Greedy oracle	Cost-aware oracle	Learned router (ours)
Poisson	Jacobi	>9.85 s ^{†16}	87.0 ms (> $10^3 \times$)	87.0 ms (> $10^3 \times$; $\tau=25$)	78.2 ms (> $10^3 \times$)	74.2 ms (> $10^3 \times$)	78.8 ms (> $10^3 \times$) ^{†2}
	GS	>29.96 s ^{†16}	142.8 ms (> $10^3 \times$)	142.8 ms (> $10^3 \times$; $\tau=25$)	141.4 ms (> $10^3 \times$)	130.0 ms (> $10^3 \times$)	131.5 ms (> $10^3 \times$)
	SOR (1.5)	>29.74 s ^{†16}	182.7 ms (> $10^3 \times$)	74.4 ms (> $10^3 \times$; $\tau=5$)	61.1 ms (> $10^3 \times$)	60.3 ms (> $10^3 \times$)	60.7 ms (> $10^3 \times$)
ConvDiff	Jacobi	>13.71 s ^{†16}	153.8 ms (> $10^3 \times$)	140.6 ms (> $10^3 \times$; $\tau=50$)	148.0 ms (> $10^3 \times$)	165.2 ms (> $10^3 \times$)	148.5 ms (> $10^3 \times$) ^{†4}
	GS	>32.99 s ^{†16}	214.1 ms (> $10^3 \times$)	192.7 ms (> $10^3 \times$; $\tau=10$)	184.8 ms (> $10^3 \times$)	185.9 ms (> $10^3 \times$)	179.9 ms (> $10^3 \times$)
	SOR (1.5)	>32.91 s ^{†16}	206.6 ms (> $10^3 \times$)	89.3 ms (> $10^3 \times$; $\tau=5$)	83.3 ms (> $10^3 \times$)	85.0 ms (> $10^3 \times$)	83.6 ms (> $10^3 \times$)

Table 39: 512×512 : paired median speedup of the learned router over HINTS (16 instances; bold: $\geq 1.1 \times$ with $p < 0.01$).

Equation	Solver	vs. HINTS ($\tau=25$)			vs. best fixed τ		
		$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-8}$
Poisson	Jacobi	6.37$\times$	1.11$\times$	1.00 $\times$	1.85$\times$ ($\tau=5$)	1.11$\times$ ($\tau=25$)	1.00 $\times$ ($\tau=50$)
	GS	19.3$\times$	1.11$\times$	1.07 $\times$	4.07$\times$ ($\tau=5$)	1.11$\times$ ($\tau=25$)	1.07 $\times$ ($\tau=25$)
	SOR (1.5)	19.2$\times$	3.10$\times$	1.37$\times$	3.93$\times$ ($\tau=5$)	1.24$\times$ ($\tau=5$)	1.09 $\times$ ($\tau=10$)
ConvDiff	Jacobi	7.73$\times$	1.04 $\times$	1.00 $\times$	1.94$\times$ ($\tau=5$)	1.18$\times$ ($\tau=50$)	1.00 $\times$ ($\tau=50$)
	GS	17.2$\times$	1.05 $\times$	1.03 $\times$	3.68$\times$ ($\tau=5$)	1.09 $\times$ ($\tau=10$)	1.03 $\times$ ($\tau=25$)
	SOR (1.5)	18.4$\times$	2.48$\times$	1.41$\times$	3.72$\times$ ($\tau=5$)	1.18$\times$ ($\tau=5$)	1.08 $\times$ ($\tau=10$)

Table 40: 512×512 : median iterations and corrector calls to h^2 (left) and measured per-iteration costs (right).

Equation	Solver	HINTS ($\tau=25$)		Cost-aware oracle		Learned router	
		iters	NO calls	iters	NO calls	iters	NO calls
Poisson	Jacobi	65	2	64	1	64	1
	GS	38	1	35	1	35	1
	SOR (1.5)	50	2	17	1	17	2
ConvDiff	Jacobi	85	2	84	1	84	1
	GS	51	2	44	1	44	2
	SOR (1.5)	50	2	20	1	20	2

Equation	Solver	classical iteration	corrector iteration	m_{solver}	m_{NO}
Poisson	Jacobi	1072 μs	3363 μs	3	1
	GS	3497 μs	3300 μs	1	1
	SOR (1.5)	3412 μs	3195 μs	1	1
ConvDiff	Jacobi	1418 μs	3614 μs	3	1
	GS	3943 μs	3667 μs	1	1
	SOR (1.5)	3896 μs	3618 μs	1	1

Table 41: 512×512 , Poisson: median wall-clock time to relative error ε for every policy and pairing (16 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	$>9.85 \text{ s}^{116}$	$>9.85 \text{ s}^{116}$	$>9.85 \text{ s}^{116}$	$>9.85 \text{ s}^{116}$	$>9.85 \text{ s}^{116}$	$>9.85 \text{ s}^{116}$
	HINTS ($\tau=25$)	33.1 ms ($> 10^3 \times$)	33.1 ms ($> 10^3 \times$)	87.0 ms ($> 10^3 \times$)	56.4 ms ($> 10^3 \times$)	127.7 ms ($> 10^3 \times$) ¹²	309.2 ms ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=10$)	14.5 ms ($> 10^3 \times$)	15.2 ms ($> 10^3 \times$)	103.3 ms ($> 10^3 \times$)	67.0 ms ($> 10^3 \times$)	152.1 ms ($> 10^3 \times$) ¹²	363.9 ms ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=5$)	9.18 ms ($> 10^3 \times$)	9.27 ms ($> 10^3 \times$)	126.3 ms ($> 10^3 \times$)	82.0 ms ($> 10^3 \times$)	192.2 ms ($> 10^3 \times$) ¹²	460.4 ms ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=50$)	60.8 ms ($> 10^3 \times$)	60.8 ms ($> 10^3 \times$)	88.9 ms ($> 10^3 \times$)	75.1 ms ($> 10^3 \times$)	125.8 ms ($> 10^3 \times$) ¹²	293.9 ms ($> 10^3 \times$) ¹⁷
	Greedy oracle (Alg. 1)	4.57 ms ($> 10^3 \times$)	4.74 ms ($> 10^3 \times$)	78.2 ms ($> 10^3 \times$)	53.6 ms ($> 10^3 \times$)	116.6 ms ($> 10^3 \times$) ¹²	275.5 ms ($> 10^3 \times$) ¹⁷
	Cost-aware oracle	4.62 ms ($> 10^3 \times$)	4.74 ms ($> 10^3 \times$)	74.2 ms ($> 10^3 \times$)	47.2 ms ($> 10^3 \times$)	111.0 ms ($> 10^3 \times$) ¹²	261.5 ms ($> 10^3 \times$) ¹⁷
	Learned router (ours)	4.80 ms ($> 10^3 \times$)	4.86 ms ($> 10^3 \times$)	78.8 ms ($> 10^3 \times$) ¹²	52.6 ms ($> 10^3 \times$)	119.4 ms ($> 10^3 \times$) ¹⁴	282.9 ms ($> 10^3 \times$) ¹⁷
GS	Solver only	$>29.96 \text{ s}^{116}$	$>29.96 \text{ s}^{116}$	$>29.96 \text{ s}^{116}$	$>29.96 \text{ s}^{116}$	$>29.96 \text{ s}^{116}$	$>29.96 \text{ s}^{116}$
	HINTS ($\tau=25$)	92.5 ms ($> 10^3 \times$)	92.5 ms ($> 10^3 \times$)	142.8 ms ($> 10^3 \times$)	107.0 ms ($> 10^3 \times$)	210.1 ms ($> 10^3 \times$)	439.6 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	37.3 ms ($> 10^3 \times$)	37.3 ms ($> 10^3 \times$)	147.0 ms ($> 10^3 \times$)	96.0 ms ($> 10^3 \times$)	214.6 ms ($> 10^3 \times$)	465.9 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	19.5 ms ($> 10^3 \times$)	20.0 ms ($> 10^3 \times$)	162.5 ms ($> 10^3 \times$)	107.8 ms ($> 10^3 \times$)	238.2 ms ($> 10^3 \times$)	522.1 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	183.5 ms ($> 10^3 \times$)	183.5 ms ($> 10^3 \times$)	212.2 ms ($> 10^3 \times$)	190.8 ms ($> 10^3 \times$)	346.0 ms ($> 10^3 \times$)	478.7 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	4.59 ms ($> 10^3 \times$)	4.65 ms ($> 10^3 \times$)	141.4 ms ($> 10^3 \times$)	95.2 ms ($> 10^3 \times$)	221.0 ms ($> 10^3 \times$)	447.1 ms ($> 10^3 \times$)
	Cost-aware oracle	4.68 ms ($> 10^3 \times$)	4.87 ms ($> 10^3 \times$)	130.0 ms ($> 10^3 \times$)	88.6 ms ($> 10^3 \times$)	190.1 ms ($> 10^3 \times$)	433.0 ms ($> 10^3 \times$)
	Learned router (ours)	4.66 ms ($> 10^3 \times$)	4.70 ms ($> 10^3 \times$)	131.5 ms ($> 10^3 \times$)	89.1 ms ($> 10^3 \times$)	194.5 ms ($> 10^3 \times$)	417.8 ms ($> 10^3 \times$)
SOR (1.5)	Solver only	$>29.74 \text{ s}^{116}$	$>29.74 \text{ s}^{116}$	$>29.74 \text{ s}^{116}$	$>29.74 \text{ s}^{116}$	$>29.74 \text{ s}^{116}$	$>29.74 \text{ s}^{116}$
	HINTS ($\tau=25$)	93.2 ms ($> 10^3 \times$)	93.2 ms ($> 10^3 \times$)	182.7 ms ($> 10^3 \times$)	129.0 ms ($> 10^3 \times$)	187.1 ms ($> 10^3 \times$)	269.4 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	37.3 ms ($> 10^3 \times$)	37.7 ms ($> 10^3 \times$)	82.5 ms ($> 10^3 \times$)	78.6 ms ($> 10^3 \times$)	108.5 ms ($> 10^3 \times$)	214.9 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	18.7 ms ($> 10^3 \times$)	18.7 ms ($> 10^3 \times$)	74.4 ms ($> 10^3 \times$)	49.3 ms ($> 10^3 \times$)	107.4 ms ($> 10^3 \times$)	237.4 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	185.7 ms ($> 10^3 \times$)	185.7 ms ($> 10^3 \times$)	355.2 ms ($> 10^3 \times$)	209.1 ms ($> 10^3 \times$)	368.3 ms ($> 10^3 \times$)	376.4 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	4.55 ms ($> 10^3 \times$)	4.56 ms ($> 10^3 \times$)	61.1 ms ($> 10^3 \times$)	41.8 ms ($> 10^3 \times$)	88.7 ms ($> 10^3 \times$)	186.7 ms ($> 10^3 \times$)
	Cost-aware oracle	4.47 ms ($> 10^3 \times$)	4.62 ms ($> 10^3 \times$)	60.3 ms ($> 10^3 \times$)	41.1 ms ($> 10^3 \times$)	89.3 ms ($> 10^3 \times$)	185.9 ms ($> 10^3 \times$)
	Learned router (ours)	4.62 ms ($> 10^3 \times$)	4.69 ms ($> 10^3 \times$)	60.7 ms ($> 10^3 \times$)	41.5 ms ($> 10^3 \times$)	89.1 ms ($> 10^3 \times$)	188.5 ms ($> 10^3 \times$)

Table 42: 512×512 , convection–diffusion: median wall-clock time to relative error ε for every policy and pairing (16 instances).

Solver	Method	$\varepsilon=10^{-2}$	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-5}$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
Jacobi	Solver only	$>13.71 \text{ s}^{116}$	$>13.71 \text{ s}^{116}$	$>13.71 \text{ s}^{116}$	$>13.71 \text{ s}^{116}$	$>13.71 \text{ s}^{116}$	$>13.71 \text{ s}^{116}$
	HINTS ($\tau=25$)	45.3 ms ($> 10^3 \times$)	45.8 ms ($> 10^3 \times$)	153.8 ms ($> 10^3 \times$)	114.5 ms ($> 10^3 \times$)	214.1 ms ($> 10^3 \times$) ¹⁴	1.38 s ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=10$)	20.5 ms ($> 10^3 \times$)	20.8 ms ($> 10^3 \times$)	170.0 ms ($> 10^3 \times$)	126.6 ms ($> 10^3 \times$)	237.8 ms ($> 10^3 \times$) ¹⁴	1.57 s ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=5$)	12.5 ms ($> 10^3 \times$)	13.8 ms ($> 10^3 \times$)	250.2 ms ($> 10^3 \times$)	185.3 ms ($> 10^3 \times$)	335.6 ms ($> 10^3 \times$) ¹⁴	2.33 s ($> 10^3 \times$) ¹⁷
	HINTS ($\tau=50$)	93.0 ms ($> 10^3 \times$)	93.0 ms ($> 10^3 \times$)	140.6 ms ($> 10^3 \times$)	121.5 ms ($> 10^3 \times$)	195.2 ms ($> 10^3 \times$) ¹⁴	1.33 s ($> 10^3 \times$) ¹⁷
	Greedy oracle (Alg. 1)	5.84 ms ($> 10^3 \times$)	6.49 ms ($> 10^3 \times$)	148.0 ms ($> 10^3 \times$)	111.4 ms ($> 10^3 \times$)	202.0 ms ($> 10^3 \times$) ¹⁴	1.29 s ($> 10^3 \times$) ¹⁷
	Cost-aware oracle	5.85 ms ($> 10^3 \times$)	8.02 ms ($> 10^3 \times$)	165.2 ms ($> 10^3 \times$)	120.5 ms ($> 10^3 \times$)	232.9 ms ($> 10^3 \times$) ¹⁴	1.28 s ($> 10^3 \times$) ¹⁷
	Learned router (ours)	5.77 ms ($> 10^3 \times$)	6.18 ms ($> 10^3 \times$)	148.5 ms ($> 10^3 \times$) ¹⁴	109.8 ms ($> 10^3 \times$)	204.2 ms ($> 10^3 \times$) ¹⁴	1.36 s ($> 10^3 \times$) ¹⁷
GS	Solver only	$>32.99 \text{ s}^{116}$	$>32.99 \text{ s}^{116}$	$>32.99 \text{ s}^{116}$	$>32.99 \text{ s}^{116}$	$>32.99 \text{ s}^{116}$	$>32.99 \text{ s}^{116}$
	HINTS ($\tau=25$)	105.1 ms ($> 10^3 \times$)	105.1 ms ($> 10^3 \times$)	214.1 ms ($> 10^3 \times$)	151.5 ms ($> 10^3 \times$)	257.6 ms ($> 10^3 \times$)	504.5 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	42.5 ms ($> 10^3 \times$)	42.9 ms ($> 10^3 \times$)	192.7 ms ($> 10^3 \times$)	143.4 ms ($> 10^3 \times$)	265.1 ms ($> 10^3 \times$)	519.2 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	21.8 ms ($> 10^3 \times$)	22.0 ms ($> 10^3 \times$)	212.5 ms ($> 10^3 \times$)	157.4 ms ($> 10^3 \times$)	294.0 ms ($> 10^3 \times$)	582.6 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	209.5 ms ($> 10^3 \times$)	209.5 ms ($> 10^3 \times$)	295.3 ms ($> 10^3 \times$)	236.8 ms ($> 10^3 \times$)	414.1 ms ($> 10^3 \times$)	570.7 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	5.58 ms ($> 10^3 \times$)	5.70 ms ($> 10^3 \times$)	184.8 ms ($> 10^3 \times$)	134.4 ms ($> 10^3 \times$)	256.3 ms ($> 10^3 \times$)	493.9 ms ($> 10^3 \times$)
	Cost-aware oracle	5.69 ms ($> 10^3 \times$)	5.97 ms ($> 10^3 \times$)	185.9 ms ($> 10^3 \times$)	138.4 ms ($> 10^3 \times$)	254.8 ms ($> 10^3 \times$)	489.9 ms ($> 10^3 \times$)
	Learned router (ours)	5.65 ms ($> 10^3 \times$)	5.79 ms ($> 10^3 \times$)	179.9 ms ($> 10^3 \times$)	133.2 ms ($> 10^3 \times$)	247.3 ms ($> 10^3 \times$)	486.0 ms ($> 10^3 \times$)
SOR (1.5)	Solver only	$>32.91 \text{ s}^{116}$	$>32.91 \text{ s}^{116}$	$>32.91 \text{ s}^{116}$	$>32.91 \text{ s}^{116}$	$>32.91 \text{ s}^{116}$	$>32.91 \text{ s}^{116}$
	HINTS ($\tau=25$)	103.8 ms ($> 10^3 \times$)	103.8 ms ($> 10^3 \times$)	206.6 ms ($> 10^3 \times$)	206.6 ms ($> 10^3 \times$)	206.6 ms ($> 10^3 \times$)	308.3 ms ($> 10^3 \times$)
	HINTS ($\tau=10$)	41.1 ms ($> 10^3 \times$)	41.5 ms ($> 10^3 \times$)	91.5 ms ($> 10^3 \times$)	84.9 ms ($> 10^3 \times$)	121.2 ms ($> 10^3 \times$)	234.2 ms ($> 10^3 \times$)
	HINTS ($\tau=5$)	21.5 ms ($> 10^3 \times$)	22.1 ms ($> 10^3 \times$)	89.3 ms ($> 10^3 \times$)	66.2 ms ($> 10^3 \times$)	123.9 ms ($> 10^3 \times$)	242.4 ms ($> 10^3 \times$)
	HINTS ($\tau=50$)	203.2 ms ($> 10^3 \times$)	203.2 ms ($> 10^3 \times$)	406.5 ms ($> 10^3 \times$)	403.1 ms ($> 10^3 \times$)	406.5 ms ($> 10^3 \times$)	460.1 ms ($> 10^3 \times$)
	Greedy oracle (Alg. 1)	5.41 ms ($> 10^3 \times$)	6.82 ms ($> 10^3 \times$)	83.3 ms ($> 10^3 \times$)	60.8 ms ($> 10^3 \times$)	111.5 ms ($> 10^3 \times$)	212.8 ms ($> 10^3 \times$)
	Cost-aware oracle	5.22 ms ($> 10^3 \times$)	5.68 ms ($> 10^3 \times$)	85.0 ms ($> 10^3 \times$)	62.2 ms ($> 10^3 \times$)	114.0 ms ($> 10^3 \times$)	217.1 ms ($> 10^3 \times$)
	Learned router (ours)	5.48 ms ($> 10^3 \times$)	5.64 ms ($> 10^3 \times$)	83.6 ms ($> 10^3 \times$)	61.2 ms ($> 10^3 \times$)	112.5 ms ($> 10^3 \times$)	216.7 ms ($> 10^3 \times$)

Table 43: 512×512 : strong classical baselines (16 instances).

Equation	Method	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$	iters to h^2
Poisson	FFT direct solve	7.65 ms (0.13 $\times$)	7.65 ms (0.13 $\times$)	7.65 ms (0.13 $\times$)	7.65 ms (0.13 $\times$)	1
	Multigrid V(2,2) alone	42.8 ms (0.78 $\times$)	81.9 ms (1.49 $\times$)	90.2 ms (1.63 $\times$)	126.5 ms (2.21 $\times$)	4
	CG	689.5 ms (12.2 $\times$)	920.9 ms (16.9 $\times$)	1.04 s (18.1 $\times$)	1.18 s (20.7 $\times$)	776
	PCG (SymGS)	1.33 s (22.6 $\times$)	2.52 s (42.9 $\times$)	2.62 s (44.3 $\times$)	3.14 s (54.2 $\times$)	364
	PCG (multigrid)	44.2 ms (0.73 $\times$)	85.6 ms (1.64 $\times$)	85.5 ms (1.58 $\times$)	113.2 ms (1.87 $\times$)	4
	HINTS ($\tau=25$), SOR (1.5)	93.2 ms (1.61 $\times$)	182.7 ms (3.10 $\times$)	187.1 ms (3.10 $\times$)	269.4 ms (4.35 $\times$)	50
	Learned router, {NO, SOR (1.5)}	4.69 ms	60.7 ms	89.1 ms	188.5 ms	17
	Learned router, {NO, Multigrid}	4.76 ms	38.7 ms	52.5 ms	99.5 ms	3
	Cost-aware oracle, {NO, Multigrid}	4.86 ms	38.6 ms	50.6 ms	97.0 ms	2
ConvDiff	FFT direct solve	8.06 ms (0.10 $\times$)	8.06 ms (0.10 $\times$)	8.06 ms (0.10 $\times$)	8.06 ms (0.10 $\times$)	1
	Multigrid V(2,2) alone	52.4 ms (0.63 $\times$)	99.7 ms (1.23 $\times$)	107.7 ms (1.36 $\times$)	150.7 ms (1.87 $\times$)	4
	BiCGSTAB	3.79 s (43.1 $\times$)	5.56 s (68.3 $\times$)	5.83 s (70.6 $\times$) ¹¹	6.72 s (80.6 $\times$) ¹¹	1695
	BiCGSTAB (multigrid)	46.1 ms (0.82 $\times$)	93.5 ms (1.18 $\times$)	93.7 ms (1.12 $\times$)	144.8 ms (1.69 $\times$)	2
	HINTS ($\tau=25$), SOR (1.5)	103.8 ms (1.24 $\times$)	206.6 ms (2.48 $\times$)	206.6 ms (2.48 $\times$)	308.3 ms (3.75 $\times$)	50
	Learned router, {NO, SOR (1.5)}	5.64 ms	83.6 ms	112.5 ms	216.7 ms	20
	Learned router, {NO, Multigrid}	5.63 ms	53.7 ms	61.3 ms	102.6 ms	3
	Cost-aware oracle, {NO, Multigrid}	5.74 ms	50.9 ms	62.3 ms	104.1 ms	3

1067 E.6 Statistical significance and multi-trial robustness

1068 All wall-clock comparisons are paired: every policy solves the same 64 test instances, so we test
1069 the per-instance log time ratios with a one-sided Wilcoxon signed-rank test (alternative: the router
1070 is faster) and, as a parametric complement, a one-sided paired t -test on log times; instances that a
1071 baseline fails to bring below the tolerance within the iteration cap are counted against the router
1072 (ratio 0). Table 44 reports the p -values of the pairwise comparisons of Tables 2 and 17, together with
1073 a multi-trial analysis: the router of every pairing was retrained from scratch with five independent
1074 seeds (different training instances, exploration and initialisation; the corrector and the test set are
1075 fixed) and re-run on the test set. Because these trials ran in separate sessions, whose live timings
1076 are not comparable with the main run’s, they are compared in *work units* (measured per-iteration
1077 costs times executed operations, plus $5 \mu\text{s}$ per decision; the main run’s HINTS work units serve as the
1078 reference), which are determined by the decision sequence alone. We report the mean and standard
1079 deviation over seeds of the median work-unit time to h^2 , in parentheses the fraction of test instances
1080 on which the seed router’s iteration count to h^2 coincides with the main router’s (identical decision
1081 sequences up to the crossing), the range over seeds of the paired speedup over HINTS ($\tau=25$), and
1082 the number of seeds for which the router is significantly faster ($p < 0.01$) than both HINTS ($\tau=25$)
1083 and the best fixed period. The learned policy turns out to be insensitive to the training seed: the seed
1084 routers reproduce the main router’s decision sequence up to the h^2 crossing on essentially every test
1085 instance, so their work-unit times and speedups coincide and every seed passes both significance
1086 tests. Table 45 gives the corresponding tests for the ensembles (including the tests in the opposite
1087 direction, to show where an ensemble router is significantly *slower* than the best pairwise router or
1088 than its oracle), and Table 46 for the strong classical baselines of Section E.4. The iteration-based
1089 table (Table 18) already carries paired t -tests on the AUC.

Table 44: Significance of the pairwise comparisons on the 128×128 grid (64 paired instances; one-sided Wilcoxon / paired t -test p -values on live times, “ 10^{-k} ” denotes $p < 10^{-k}$; a baseline run that does not reach the tolerance within the iteration cap enters at the time it spent up to the cap, a lower bound that makes the test conservative) and robustness over five independent router-training seeds (work-unit times; the percentage is the share of instances whose decision sequence to h^2 is identical to the main router’s).

Equation	Solver	$\varepsilon = h^2$; p (Wilcoxon / t)			$\varepsilon = 10^{-8}$; p		Router over 5 training seeds		
		vs. solver only	vs. HINTS-25	vs. best τ	vs. HINTS-25	vs. best τ	time to h^2 (decisions as main router)	speedup vs. HINTS-25	seeds with $p < 0.01$
Poisson	Jacobi	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-8} / 10^{-7}$	$< 10^{-10} / < 10^{-10}$	$10^{-9} / 10^{-8}$	$0.74 \pm 0.00 \text{ ms (100\%)}$	$2.59\text{--}2.59\times$	5/5
	Jacobi (0.67)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-9} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-4} / 10^{-5}$	$0.75 \pm 0.00 \text{ ms (100\%)}$	$2.58\text{--}2.58\times$	5/5
	GS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-10} / < 10^{-10}$	$0.90 \pm 0.00 \text{ ms (100\%)}$	$6.30\text{--}6.30\times$	5/5
	SymGS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$1.06 \pm 0.00 \text{ ms (100\%)}$	$9.29\text{--}9.29\times$	5/5
	SOR (1.5)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-9} / 10^{-10}$	$0.89 \pm 0.00 \text{ ms (100\%)}$	$10.45\text{--}10.45\times$	5/5
ConvDiff	Jacobi	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / 10^{-7}$	$0.84 \pm 0.00 \text{ ms (100\%)}$	$2.91\text{--}2.91\times$	5/5
	Jacobi (0.67)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-5} / 10^{-6}$	$0.91 \pm 0.00 \text{ ms (100\%)}$	$2.67\text{--}2.67\times$	5/5
	GS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-10} / < 10^{-10}$	$1.11 \pm 0.00 \text{ ms (100\%)}$	$7.22\text{--}7.22\times$	5/5
	SymGS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$1.12 \pm 0.00 \text{ ms (100\%)}$	$9.10\text{--}9.10\times$	5/5
	SOR (1.5)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$1.30 \pm 0.00 \text{ ms (100\%)}$	$9.41\text{--}9.41\times$	5/5
AnisoDiff	Jacobi	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-9} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$0.76 \pm 0.00 \text{ ms (100\%)}$	$2.97\text{--}2.97\times$	5/5
	Jacobi (0.67)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$10^{-9} / 10^{-10}$	$0.76 \pm 0.00 \text{ ms (100\%)}$	$3.01\text{--}3.01\times$	5/5
	GS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$0.78 \pm 0.00 \text{ ms (100\%)}$	$7.63\text{--}7.63\times$	5/5
	SymGS	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$0.78 \pm 0.00 \text{ ms (100\%)}$	$12.77\text{--}12.77\times$	5/5
	SOR (1.5)	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$< 10^{-10} / < 10^{-10}$	$0.80 \pm 0.00 \text{ ms (100\%)}$	$7.58\text{--}7.58\times$	5/5

Table 45: Ensembles: one-sided Wilcoxon p -values at $\varepsilon = h^2$ (64 paired instances) for the ensemble router against the fastest member solver alone (censored runs at their lower bound, as in Table 44), against the best pairwise router in both directions, and against the cost-aware oracle over the same ensemble (alternative: the ensemble router is slower).

Equation	$\mathcal{W}$	vs. best solver only	vs. best pairwise router (faster)	vs. best pairwise router (slower)	vs. oracle($\text{NO} \cup \mathcal{W}$) (slower)
Poisson	$\{Jacobi, GS\}$	$< 10^{-10}$	10^{-7}	1.000	0.040
	$\{Jacobi, GS, SymGS\}$	$< 10^{-10}$	0.001	0.999	0.001
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	$< 10^{-10}$	0.856	0.144	10^{-4}
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	$< 10^{-10}$	1.000	10^{-8}	10^{-8}
ConvDiff	$\{Jacobi, GS\}$	$< 10^{-10}$	10^{-4}	1.000	10^{-5}
	$\{Jacobi, GS, SymGS\}$	$< 10^{-10}$	10^{-10}	1.000	0.492
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	$< 10^{-10}$	0.783	0.217	0.001
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	$< 10^{-10}$	0.047	0.953	0.566
AnisoDiff	$\{Jacobi, GS\}$	$< 10^{-10}$	0.315	0.685	0.014
	$\{Jacobi, GS, SymGS\}$	$< 10^{-10}$	0.945	0.055	10^{-4}
	$\{Jacobi, GS, SymGS, Jacobi(0.67)\}$	$< 10^{-10}$	0.551	0.449	0.064
	$\{Jacobi, GS, SymGS, Jacobi(0.67), SOR(1.5)\}$	$< 10^{-10}$	0.633	0.367	0.012

Table 46: Strong baselines: one-sided Wilcoxon p -values that the learned router (best stationary pairing) is faster than each method at each tolerance; where the baseline’s median is lower the entry gives the p -value that the router is slower.

Equation	Method	$\varepsilon=10^{-3}$	$\varepsilon=h^2$	$\varepsilon=10^{-6}$	$\varepsilon=10^{-8}$
ConvDiff 128 ²	FFT direct solve	slower: $< 10^{-10}$	slower: $< 10^{-10}$	slower: $< 10^{-10}$	slower: $< 10^{-10}$
	Multigrid V(2,2) alone	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	BiCGSTAB	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	BiCGSTAB (multigrid)	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	GMRES(20)	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
ConvDiff 256 ²	FFT direct solve	slower: 10^{-4}	slower: 10^{-6}	slower: 10^{-10}	slower: 10^{-10}
	Multigrid V(2,2) alone	10^{-10}	0.006	0.533	slower: 10^{-4}
	BiCGSTAB	10^{-10}	10^{-9}	10^{-6}	0.105
	BiCGSTAB (multigrid)	10^{-10}	0.005	0.562	slower: 10^{-4}
	GMRES(20)	10^{-10}	10^{-10}	10^{-10}	10^{-5}
ConvDiff 512 ²	FFT direct solve	0.768	slower: 10^{-5}	slower: 10^{-5}	slower: 10^{-5}
	Multigrid V(2,2) alone	10^{-5}	0.065	slower: 0.768	slower: 0.008
	BiCGSTAB	10^{-5}	10^{-5}	10^{-5}	10^{-5}
	BiCGSTAB (multigrid)	10^{-5}	0.088	slower: 0.490	slower: 0.009
Poisson 128 ²	FFT direct solve	slower: $< 10^{-10}$	slower: $< 10^{-10}$	slower: $< 10^{-10}$	slower: $< 10^{-10}$
	Multigrid V(2,2) alone	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	CG	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	PCG (SymGS)	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
	PCG (multigrid)	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$	$< 10^{-10}$
Poisson 256 ²	FFT direct solve	slower: 10^{-5}	slower: 10^{-6}	slower: 10^{-9}	slower: 10^{-10}
	Multigrid V(2,2) alone	10^{-10}	10^{-5}	0.165	slower: 0.009
	CG	10^{-10}	10^{-9}	10^{-5}	0.789
	PCG (SymGS)	10^{-10}	10^{-10}	10^{-7}	0.014
	PCG (multigrid)	10^{-10}	10^{-5}	0.195	slower: 0.007
Poisson 512 ²	FFT direct solve	0.353	slower: 10^{-4}	slower: 10^{-5}	slower: 10^{-5}
	Multigrid V(2,2) alone	10^{-5}	0.052	0.217	slower: 0.005
	CG	10^{-5}	10^{-5}	10^{-5}	10^{-5}
	PCG (SymGS)	10^{-5}	10^{-5}	10^{-5}	10^{-5}
	PCG (multigrid)	10^{-5}	0.037	slower: 0.702	slower: 0.005

F Limitations

A primary limitation of our approach is the higher on-time computational cost requires to learn the greedy routing strategy. In contrast, the simpler baselines such as HINTS incur no training overhead. While this cost is amortized during inference time and leads to improved solution quality, it may be prohibitive in settings with limited computational resources or when rapid deployment is required. On the deployment side, Section E shows that the gains survive cost accounting: with a corrector that covers the smooth half of the spectrum and a classical smoother that covers the rest, one or two corrector calls suffice and the router’s advantage over fixed schedules is uniform across solvers, equations and tolerances. The corrector we use is specific to the constant-coefficient periodic setting (a fixed Fourier trunk basis on a coarse sensor grid), its cost grows with the fourth power of the sensor-grid size, and undamped Jacobi remains limited far below truncation level by the near-Nyquist modes that neither component damps. Our wall-clock study is restricted to single-thread CPU execution and stationary smoothers; comparisons against optimal-complexity solvers such as multigrid, and hybrid ensembles that include them, are left to future work.

Additionally, the effectiveness of the learned router depends on the neural operator exhibiting heterogeneous performance across samples. When the neural operators produces predictions of relatively uniform quality, the benefit of adaptive routing diminishes and it is more practical to set a fixed schedule based on the quality of the neural operator. In practice, achieving such heterogeneity can be sensitive to training choices, and neural operators themselves can be finicky to train.

G LLM Usage

LLMs, specifically ChatGPT and Gemini, supported the writing process in an iterative manner. We drafted paragraphs and asked the models for feedback on grammar and clarity. We then incorporated selected suggestions into the writing and repeated this process until we were satisfied with the writing.

The code developed for the experiments was written by the authors with the help of occasional code completions. The central components (e.g., the hybrid solver implementation and the greedy-router training pipelines) were implemented exclusively by the authors.

1116 All substantive intellectual contributions, which include ideas, theorems, and analyses, are our own.
1117 LLMs were occasionally used to verify the correctness of proofs, but all proof strategies originated
1118 from the authors and relevant literature.

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